SAFe® Release Train Engineer

Facilitating Agile Release Trains

6.0

Workbook



SAFe®
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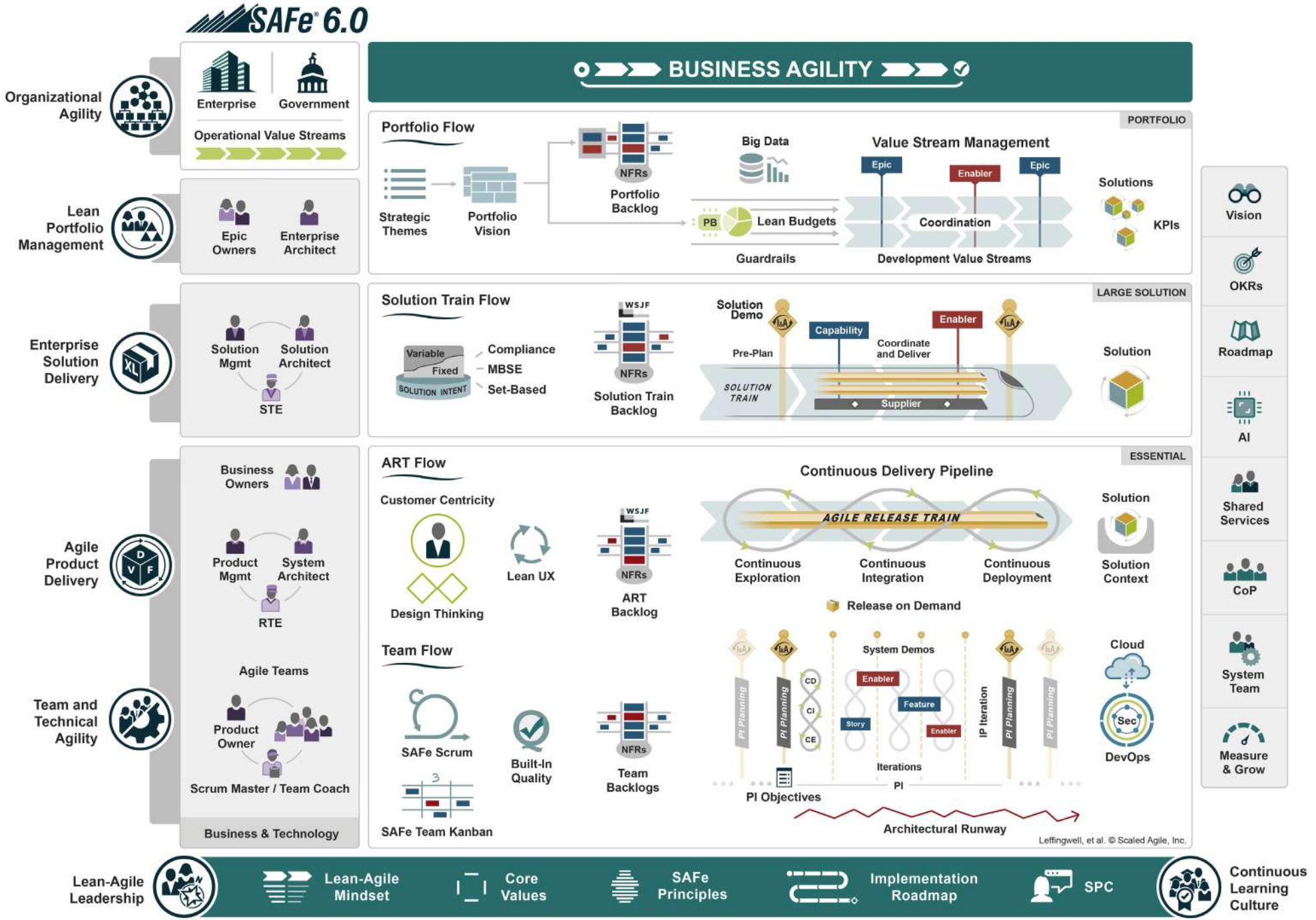
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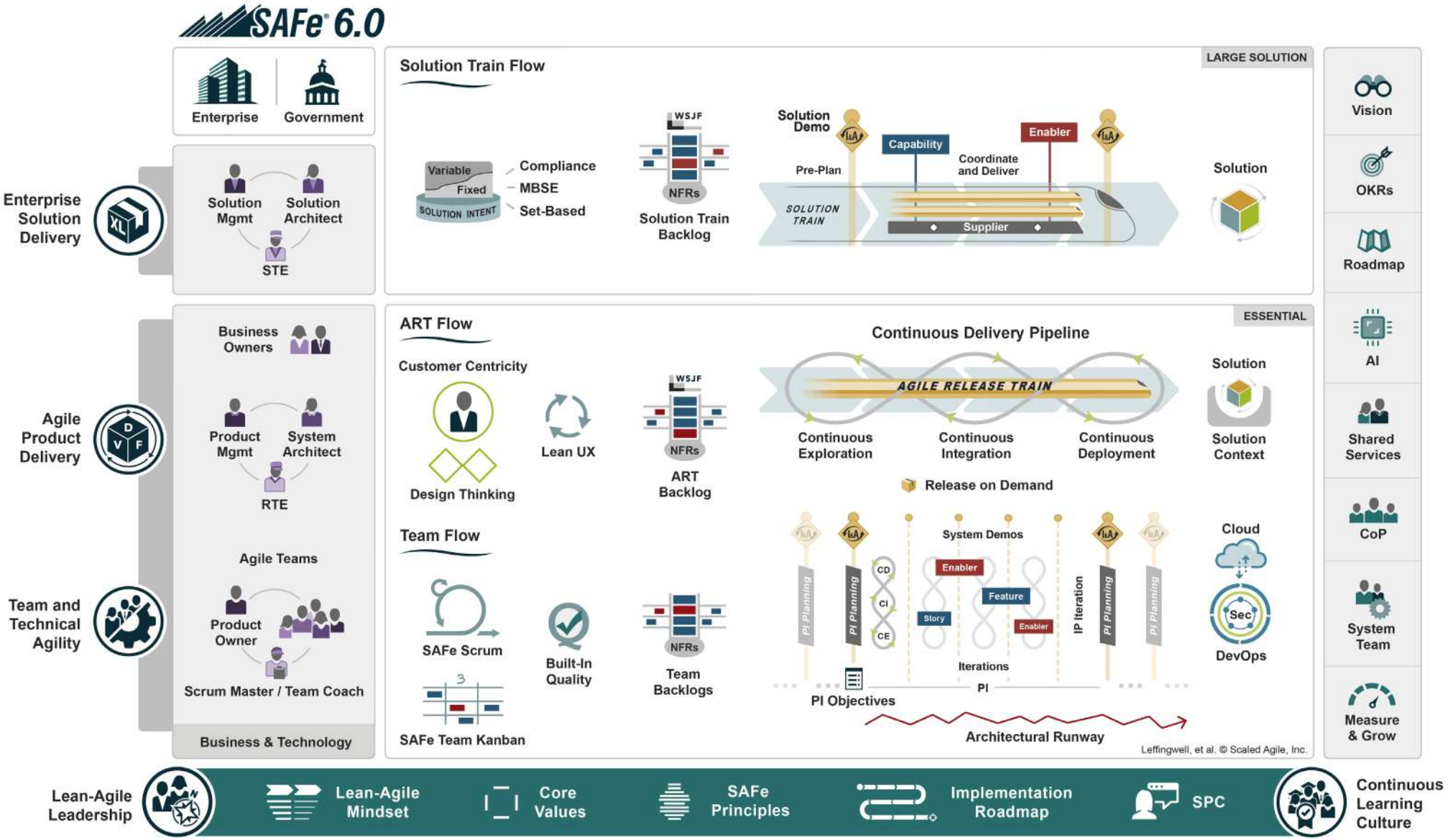
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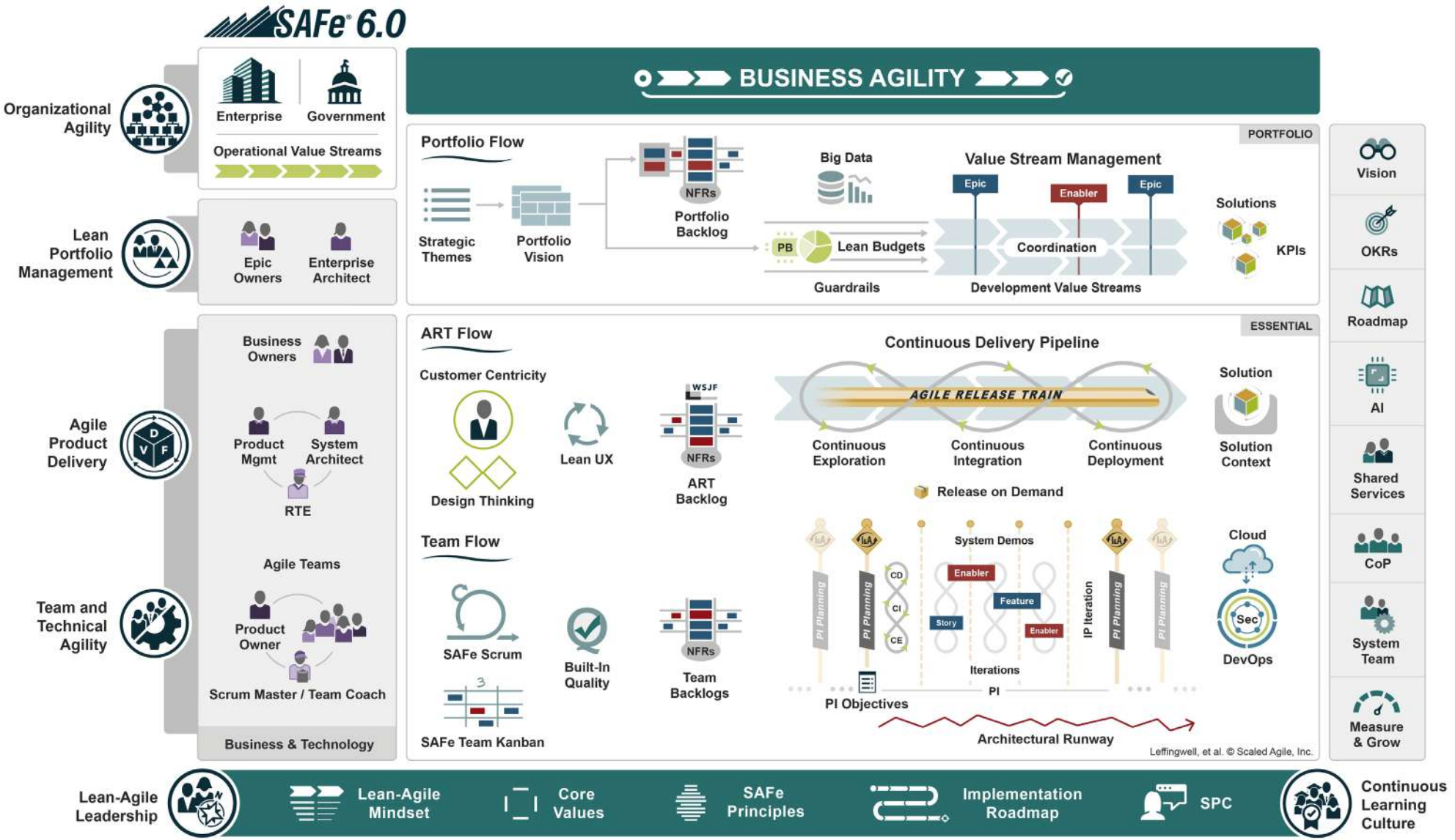


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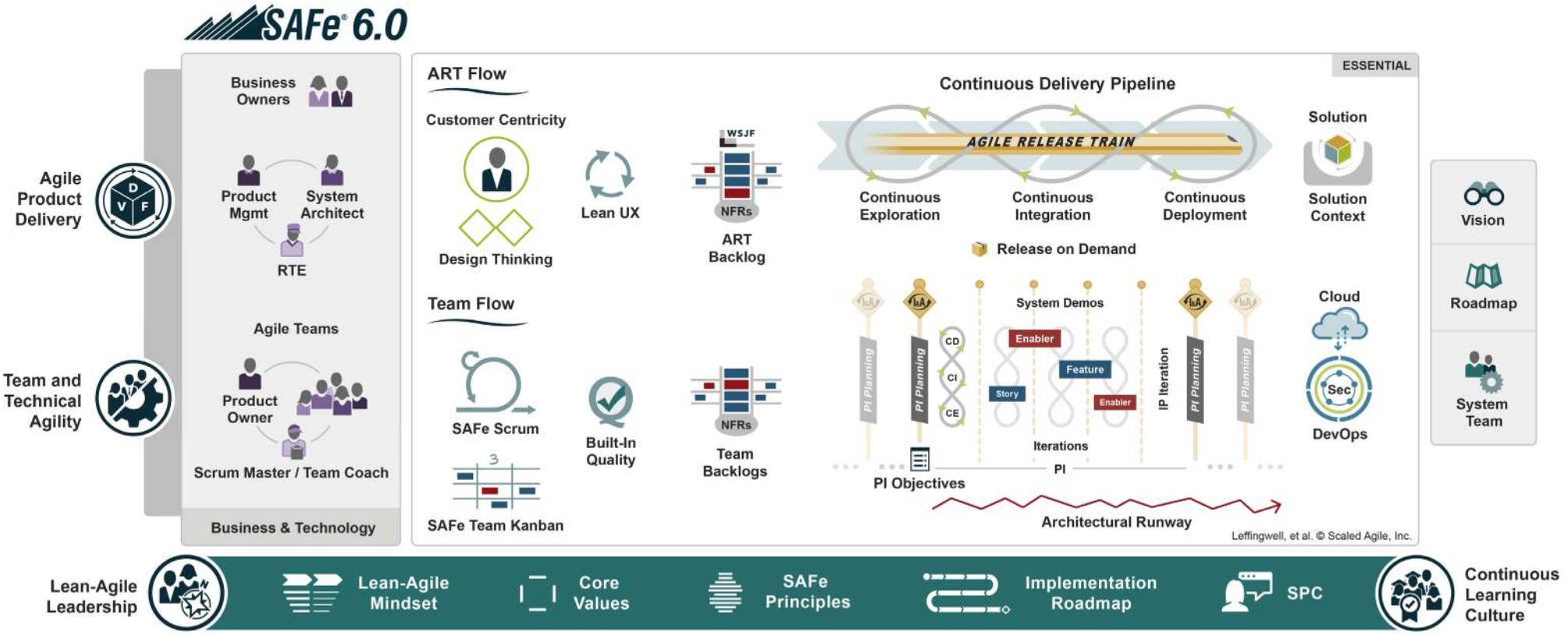


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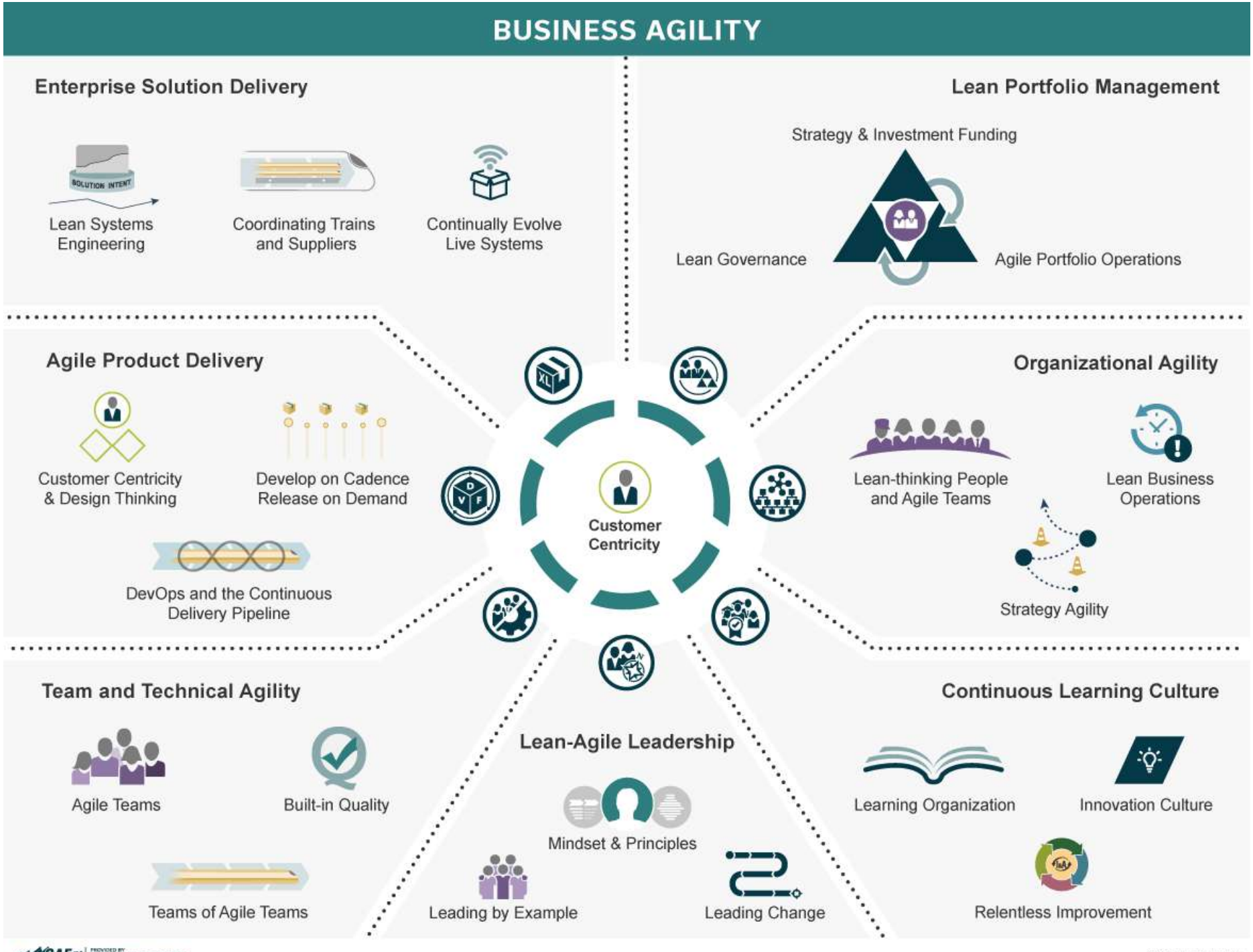
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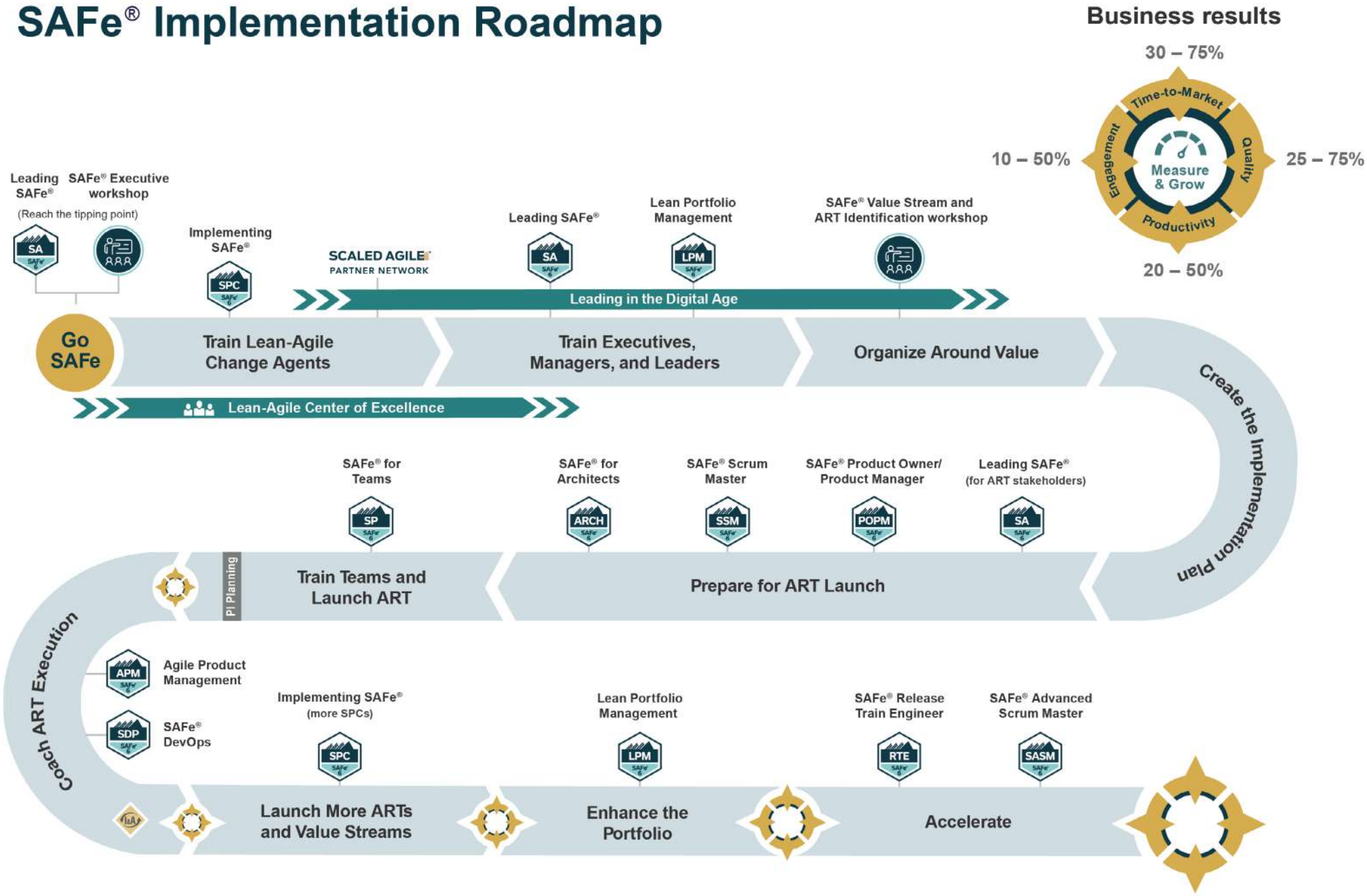


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SAFe® Implementation Roadmap



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SAFe®
Release Train Engineer

Facilitating Agile Release Trains



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6.0

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Logistics

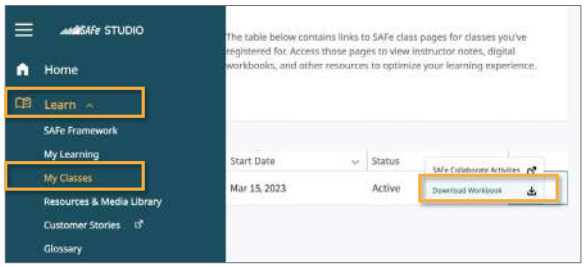
- Course meeting times
- Breaks
- Facilities
- Technology requirements
- Working agreements

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Activity: Access the Class Page

Duration: 5 min

- **Step 1**: Navigate to the Class Page on the SAFe Studio
- **Step 2**: Select Learn, then My Classes, then SAFe Release Train Engineer
- **Step 3**: Click on the link to Download the SAFe Release Train Engineer (6.0) workbook (PDF)



SAFe STUDIO™

Visit the SAFe Release Train Engineer Class Page to download the workbook

https://bit.ly/Studio-MyClasses

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Course outline

- Lesson 1: Exploring the RTE Role and Responsibilities
- Lesson 2: Applying SAFe Principles
- Lesson 3: Organizing the ART
- Lesson 4: Planning a PI
- Lesson 5: Executing a PI
- Lesson 6: Fostering Relentless Improvement
- Lesson 7: Serving the ART
- Lesson 8: Continuing Your Learning Journey
- Lesson 9: Practicing SAFe

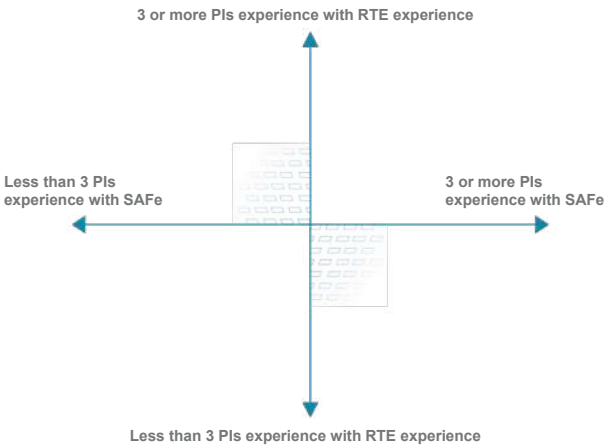
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Activity: RTE and SAFe experience

Duration: 5 min

- **Step 1:** Identify where you are on the RTE and SAFe experience matrix regarding your experience as an RTE and your experience with SAFe
- **Step 2:** Identify the most important thing you want to get out of this course
- **Step 3:** Place your name and your learning goal on the matrix



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Lesson 1

Exploring the RTE Role and Responsibilities

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Lesson Topics

1.1 Connect the RTE role to SAFe

1.2 Examine the responsibilities of the RTE role

1.3 Identify effective RTE behaviors



Learning objectives

At the end of this lesson, you should be able to:

- Define the RTE role in SAFe
- Describe the responsibilities of the RTE role
- Apply effective RTE behaviors in your role

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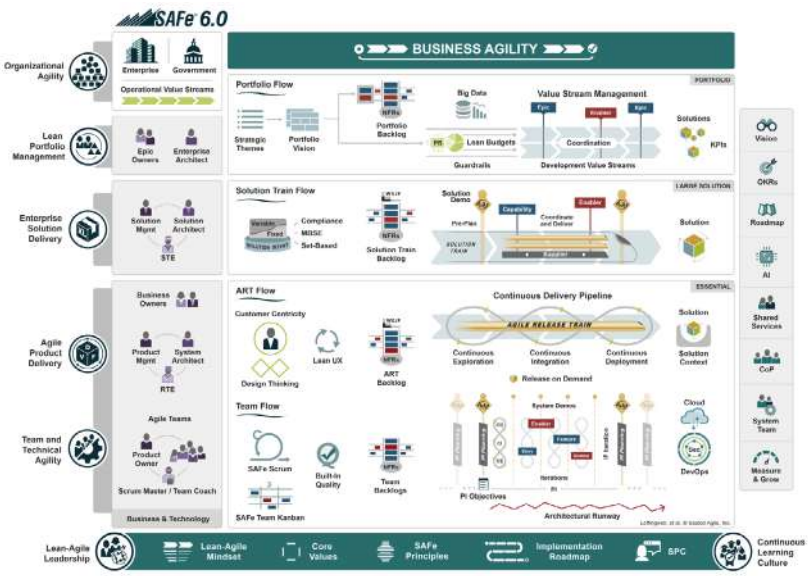
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1.1 Connect the RTE role to SAFe

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Achieving Business Agility with the Scaled Agile Framework® (SAFe)

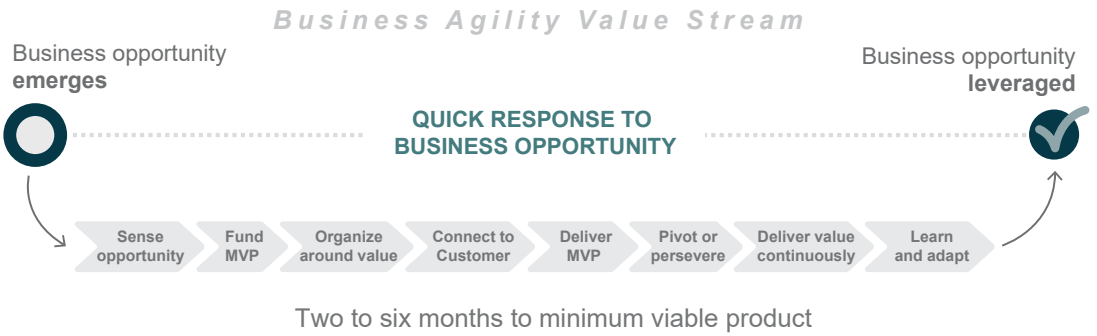


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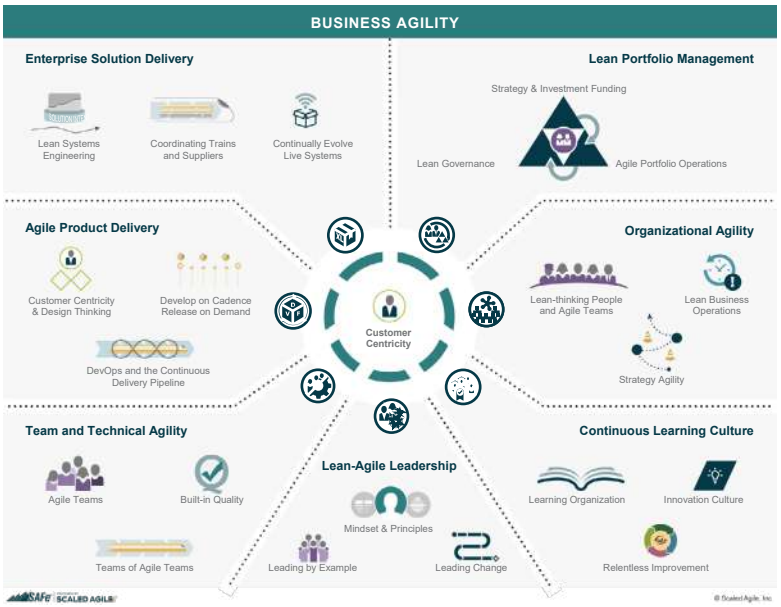
Business Agility Value Stream

Business Agility is the ability to compete and thrive in the digital age by quickly responding to market changes and emerging opportunities with innovative, digitally enabled business solutions.



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The Seven Core Competencies of Business Agility



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Team and Technical Agility

- High-performing, cross-functional Agile Teams
- Teams of business and technical teams build Solutions
- Quality business Solutions delight Customers

Agile Teams



Teams of Agile Teams



Built-in Quality



Agile Product Delivery

- The Customer is the center of the product strategy
- Decouple the release of value from the development cadence
- Continuously explore, integrate, deploy, and release

Customer Centricity and Design Thinking



Develop on Cadence, Release on Demand



DevOps and the Continuous Delivery Pipeline



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Enterprise Solution Delivery

- Apply Lean systems engineering practices to build really big systems
- Coordinate and align the full supply chain
- Continue to enhance value after release

Lean Systems Engineering



Coordinate Trains and Suppliers



Continually Evolve Live Systems

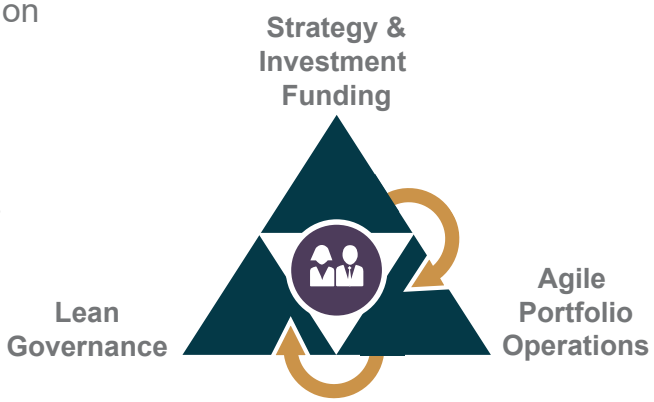


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Lean Portfolio Management

- Align strategy, funding, and execution
- Optimize operations across the portfolio
- Lightweight governance empowers decentralized decision-making



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Organizational Agility

- Create an enterprise-wide Lean-Agile mindset
- Map and continuously improve business processes
- Respond quickly to opportunities and threats

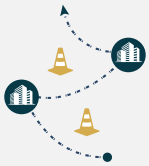
Lean-Thinking People and Agile Teams



Lean Business Operations



Strategy Agility



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Continuous Learning Culture

- Everyone in the organization learns and grows together
- Exploration and creativity are part of the organization's DNA
- Continuously improving Solutions, services, and processes is everyone's responsibility

Learning Organization



Innovation Culture



Relentless Improvement



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Lean-Agile Leadership

- Inspire others by modeling desired behaviors
- Align mindset, words, and actions to Lean-Agile values and principles
- Actively lead the change and guide others to the new way of working

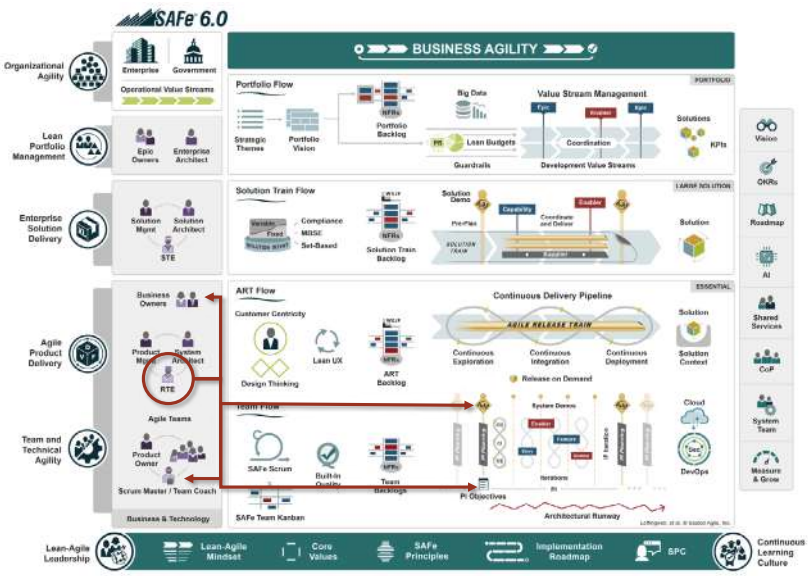
Leading by Example

Mindset & Principles

Leading Change

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Example of connecting the RTE role in SAFe



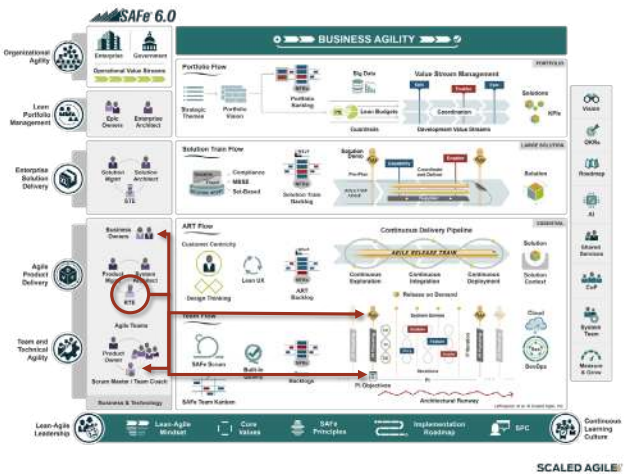
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Activity: Identify RTE connections

Prepare 7 min | Share 3 min

- **Step 1:** Working in your groups, use the Big Picture graphic to identify connections between the RTE and other Framework elements based on communication, collaboration, problem-solving, and inputs/outputs. Identify at least one connection of each type.
- **Step 2:** Be ready to present and discuss the connections you identify.



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1.2 Examine the responsibilities of the RTE role

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The RTE is the servant leader and coach for the ART

- Guides people in problem identification and decision-making
- Creates an environment of mutual influence
- Empathizes with others
- Encourages the personal development of teams
- Persuades rather than uses authority
- Applies systems thinking
- Supports the commitments made by the teams





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The RTE acts as the servant leader and coach for the ART

Responsibilities of the RTE include:

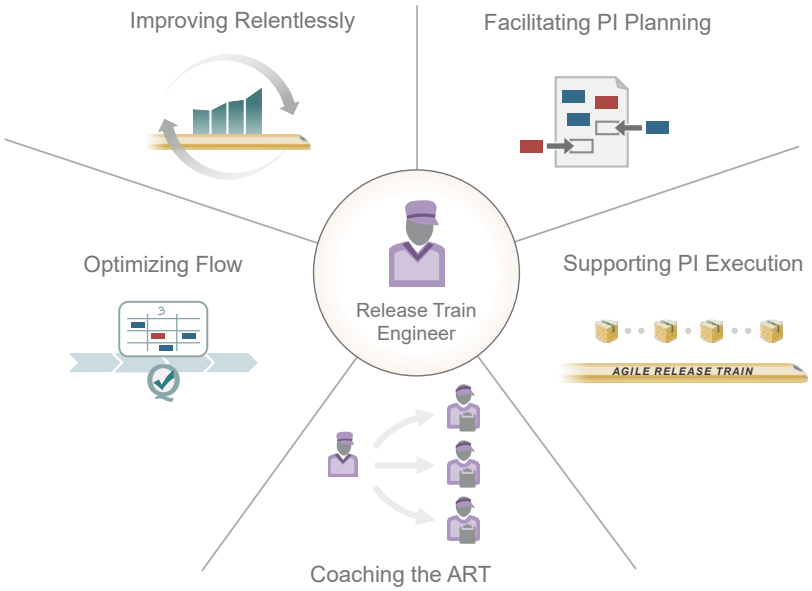
- Managing and optimizing the flow of value through the ART
- Fostering collaboration between teams and ART stakeholders
- Facilitating PI (Planning Interval) Planning readiness and the event itself
- Tracking and communicating key ART execution Metrics
- Escalating and tracking ART impediments
- Facilitating relentless improvement for the ART



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Responsibility areas of the RTE



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Activity: Describe RTE responsibilities

Prepare 7 min | Share 10 min

- **Step 1:** Working in your groups, select one key responsibility area of the RTE to explore.
- **Step 2:** Read the section of the RTE article that pertains to the chosen responsibility area. Identify key activities, actions, or events, alongside the key stakeholders and collaborators the RTE must work with to fulfill that responsibility.
  - Article Link: https://scaledagileframework.com/release-train-engineer/
- **Step 3:** Create a 2-minute presentation capturing the key information that other RTEs need to know to fulfill the responsibility area.

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1.3 Identify effective RTE behaviors

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Moving from a traditional mindset to a new one

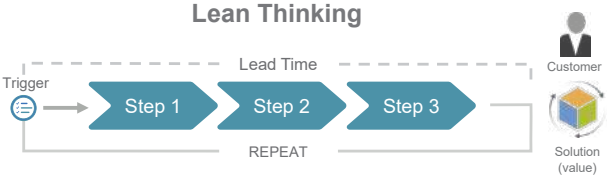
From…	To…
Coordinating team contributions	Coaching the teams to collaborate
Setting deadlines	Aligning to objectives
Driving toward specific outcomes	Being invested in the ART's overall performance
Knowing the answer	Asking the teams for the answer
Directing	Letting the teams self-organize and hit their stride
From fixing problems…	***…to helping others fix them!***

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SAFe Lean-Agile Mindset

Lean Thinking



Principles

1. Precisely specify value by product
2. Identify the Value Stream for each product
3. Make value flow without interruptions
4. Let the Customer pull value from the producer
5. Pursue perfection

Agile Values

We are uncovering better ways of developing software by doing it and helping others do it.

Through this work we have come to value:

- **Individuals and interactions** over processes and tools
- **Working software** over comprehensive documentation
- **Customer collaboration** over contract negotiation
- **Responding to change** over following a plan

That is, while there is value in the items on the right, we value the items on the left more.

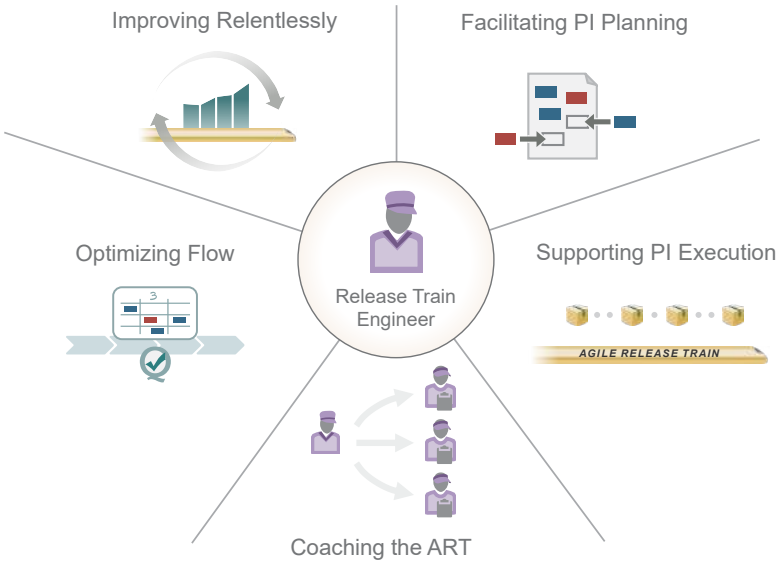
https://agilemanifesto.org/

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Discussion: Identify effective RTE behaviors

Duration: 10 min

- **Step 1:** As a class, identify behaviors that RTEs must exhibit to support each of their key responsibility areas
- **Step 2:** Brainstorm anti-pattern behaviors to avoid for each responsibility area



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Action Plan: Exploring the RTE role and responsibilities

Prepare 5 min | Share 2 min

- **Step 1:** Locate the Action Plan in your workbooks.
- **Step 2:** Identify which of the RTE's responsibility are will be the most challenging to fulfill. Brainstorm 1–3 actions you can take to prepare yourself to succeed that area.
- **Step 3:** Share one of your insights with the class.



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Lesson review

In this lesson you:

- Defined the RTE role in SAFe
- Described the responsibilities of the RTE role
- Applied effective RTE behaviors in your role

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Action Plan

Exploring the RTE role and Responsibilities

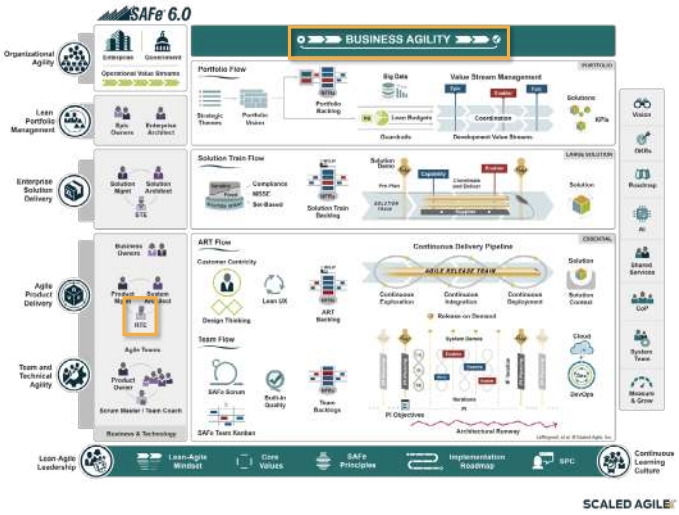
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Articles used in this lesson

Read these Framework articles to learn more about topics covered in this lesson

- "Release Train Engineer" https://www.scaledagileframework.com/release-train-engineer/
- "Business Agility" https://www.scaledagileframework.com/business-agility/



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1-33

Continue your SAFe journey with the following resources:

Watch this three-minute video, *Navigating the Big Picture*, to understand how to use the SAFe Big Picture. **https://bit.ly/Video-NavigatingtheBigPicture**	Watch this four-minute video, *Lean-Agile Mindset*, to learn why a Lean-Agile Mindset is an important Enabler for Business Agility. **https://bit.ly/Video-LeanAgileMindset**
Find "RTE Essentials" in your learning plan and enroll to gain the confidence needed to perform the responsibilities of the RTE role. The modules will take you through the critical knowledge and provide tips, techniques, and tools to improve your RTE skills. **https://bit.ly/Community-RTE-Essentials**	

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Lesson notes

Enter your notes below. If using a digital workbook, save your PDF often so you don't lose any of your notes.

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Lesson 2

Applying SAFe® Principles

SAFe® Course - Attending this course gives learners access to the SAFe Release Train Engineer exam and related preparation materials.

Lesson Topics

2.1 Explore the SAFe Principles



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Learning objectives

At the end of this lesson, you should be able to:

- Identify why the SAFe Principles are important to the RTE

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2.1 Explore the SAFe Principles

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SAFe RTE stories: Tamara Nation

Duration 2 min



https://bit.ly/Video-RTEStories

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2-5

SAFe Lean-Agile Principles

#1 Take an economic view

#2 Apply systems thinking

#3 Assume variability; preserve options

#4 Build incrementally with fast, integrated learning cycles

#5 Base milestones on objective evaluation of working systems

#6 Make value flow without interruptions

#7 Apply cadence, synchronize with cross-domain planning

#8 Unlock the intrinsic motivation of knowledge workers

#9 Decentralize decision-making

#10 Organize around value

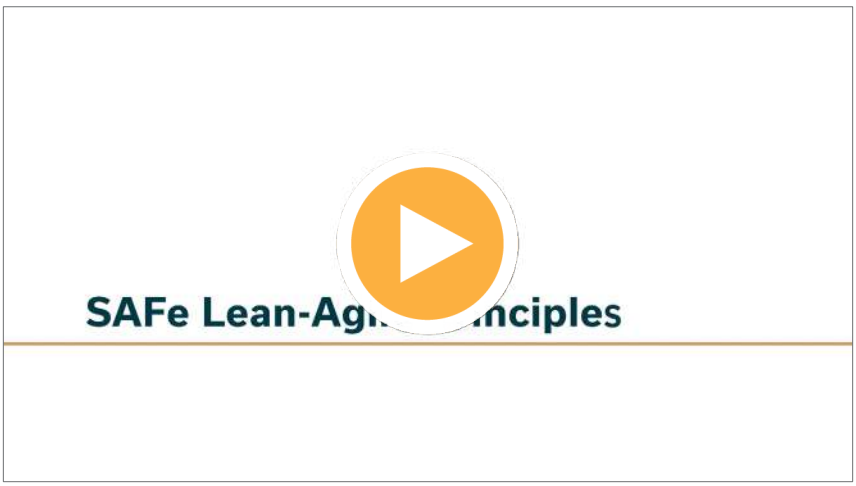
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2-6

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Video: SAFe Lean-Agile Principles

Duration 15 min



https://bit.ly/Video-SAFePrinciples-6

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Why do RTEs need the SAFe Principles?

- RTEs must understand why practices work; it's part of 'knowing what they must do'
- If a practice needs to change, understanding the Principles ensures that the change moves the ART in the right direction
- A shared understanding of the Principles helps decentralize decision-making within the teams and roles on the ART



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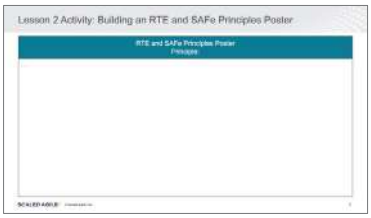
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Activity: Building an RTE and SAFe Principles poster

Prepare 10 min

Share 10 min

- **Step 1:** Working in your groups, select a SAFe Principle to explore.
- **Step 2:** Build a poster that explores how your Principle can be applied in the ART. Consider which areas of the ART the Principle applies to and how it might connect to the RTE.
- **Step 3:** Be prepared to share with the class.





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Without a shared understanding of the SAFe Principles…

- There is no systematic way to adapt practices to local context
- Business outcomes suffer
- Once-beneficial decisions become problematic
- The Lean-Agile Mindset is unachievable
- Conflict on processes and practices are difficult to resolve



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The RTE and applying SAFe Principles

Applying all 10 SAFe Principles in the ART is a key responsibility of the RTE.

- Reconnect to the Principles when discussing existing practices or potential improvements for the ART
- Build a shared understanding with stakeholders and the ART on why the Principles-based practices work
- Periodically ask the train to self-assess how well the Principles are applied



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RTE Action Plan

Duration 5 min

- **Step 1:** Locate the RTE Action Plan section in your workbooks
- **Step 2:** Begin adding tools to the Action Plan by brainstorming the following:
  - Identify one action you can take to ensure you maintain a regular connection to the SAFe Principles in your daily activities as an RTE
- **Step 3:** Share one of your insights with the class



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Action Plan

Applying SAFe Principles

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Lesson review

In this lesson you:

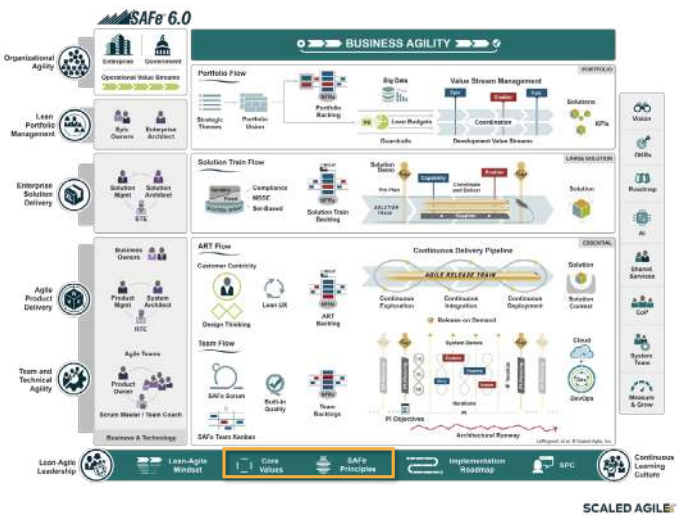
- Identified why the SAFe Principles are important to the RTE

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Articles used in this lesson

Read these Framework articles to learn more about topics covered in this lesson

- "SAFe Lean-Agile Principles" https://www.scaledagileframework.com/safe-lean-agile-principles/
- "SAFe Core Values," https://www.scaledagileframework.com/safe-core-values/



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Continue your SAFe journey with the following resources:

Explore each SAFe Principle in depth and learn how organizations have applied them in their own Agile practices in the 40-minute online learning “SAFe Lean-Agile Principles.” https://bit.ly/Community-GettingStarted	Watch the fifteen-minute video playlist “SAFe Lean-Agile Principles (6.0)” to reinforce your understanding of the SAFe Principles. https://bit.ly/Playlist-Principles6

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Lesson notes

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Lesson 3

Organizing the ART

SAFe® Course - Attending this course gives learners access to the SAFe Release Train Engineer exam and related preparation materials.

Lesson Topics

3.1 Identify the characteristics of an effective ART organization

3.2 Examine key ART roles and responsibilities

3.3 Identify the responsibilities of the System Team



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Learning objectives

At the end of this lesson, you should be able to:

- Define the characteristics of an effective ART organization
- Describe the key ART roles and responsibilities
- Summarize the responsibilities of the System Team

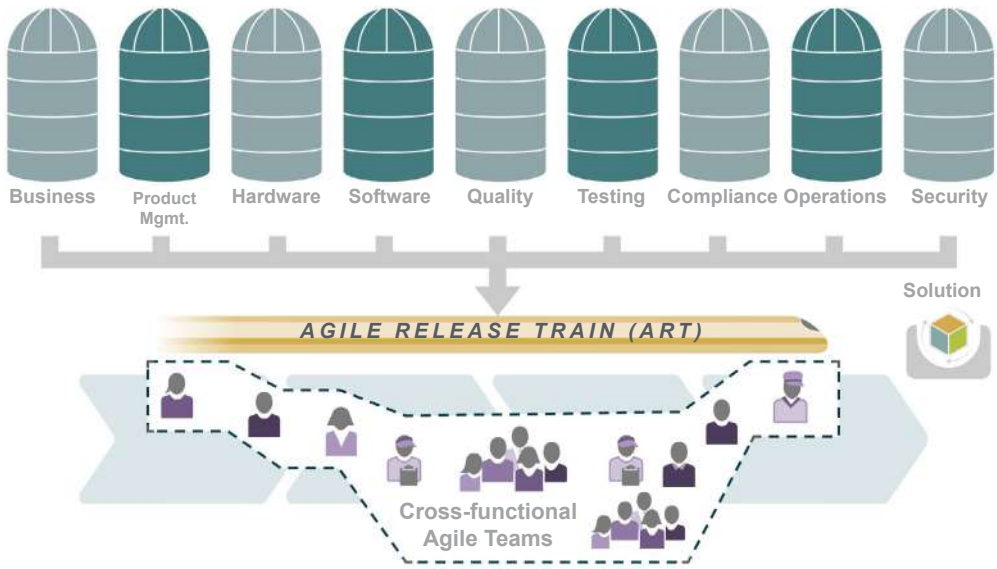
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3.1 Identify the characteristics of an effective ART organization

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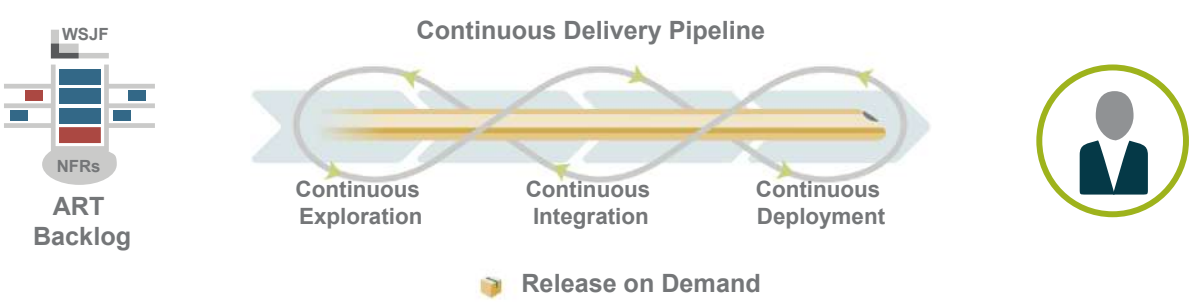
ARTs are cross-functional



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ARTs

- A virtual organization of 5–12 teams (50–125+ individuals)
- Synchronized on a common cadence (a PI)
- Aligned to a common mission via a single ART Backlog



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> [T]here is a cognitive limit to the number of individuals with whom any one person can maintain stable relationships, that this limit is a direct function of relative neocortical size, and that this in turn limits group size."
>
> —Robin Dunbar, "Coevolution of Neocortical Size, Group Size and Language in Humans"

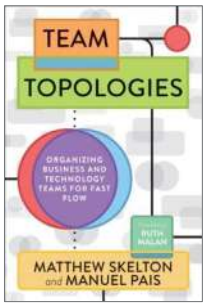


Portrait of Robin Dunbar. Creative Commons image.

3-7

Teams on the ART are organized for flow

- **Stream-aligned team** – Organized around the flow of work and can deliver value directly to the Customer or end user.
- **Complicated subsystem team** – Organized around specific subsystems that require deep specialty skills and expertise.
- **Platform team** – Organized around the development and support of platforms that provide services to other teams.
- **Enabling team** – Organized to assist other teams with specialized capabilities and help them become proficient in new technologies.



Team Topologies by Matthew Skelton and Manuel Pais

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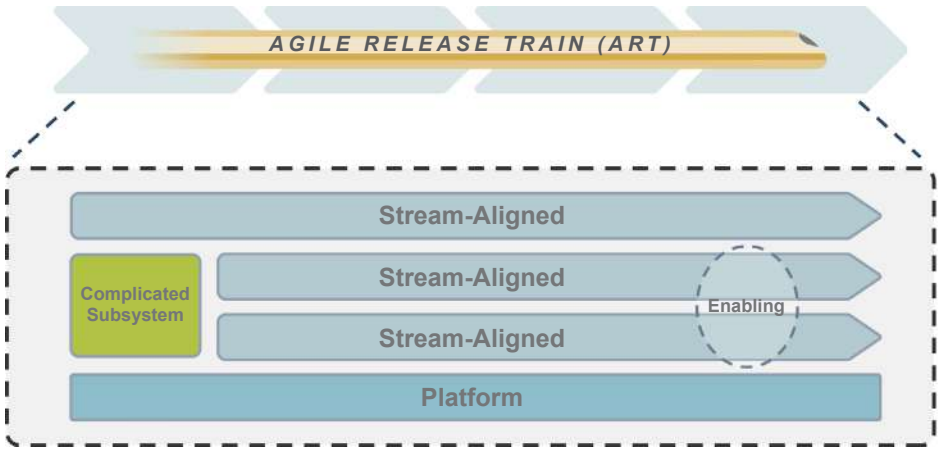
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ARTs are organized to deliver value continuously

Consider the necessary interactions between the teams and organize to maximize flow.



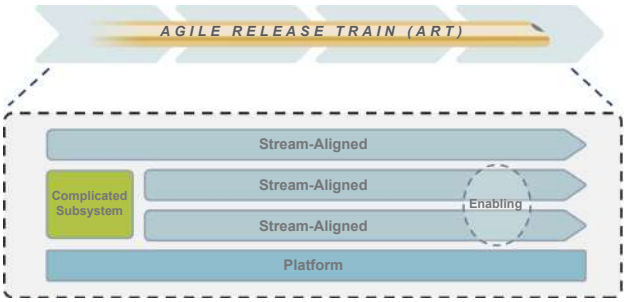
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Activity: Establish Agile Teams with team topologies

Prepare 7 min

Share 3 min

- **Step 1:** Consider your context and discuss with your group:
  - How might you organize stream-aligned teams so they could deliver end-to-end value?
  - When would you need complicated subsystem teams?
  - When would it be beneficial to create a platform team?
  - In addition to the System Team, what other enabling team might be beneficial?
- **Step 2:** Be prepared to share with the class

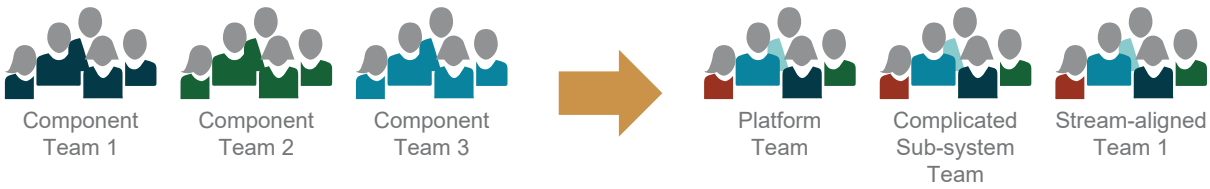


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Moving to Continuous Deployment and Release on Demand

- Organizations often migrate from component teams to other team types (usually stream-aligned teams) as the ART matures
- Over time, organizations build the cross-functional skills required for stream-aligned teams through cross-training and pair programming
- As skills advance, carefully rotate team members between different component teams to spread knowledge
- Create Enablers to architect for deploying and releasing more frequently



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3.2 Examine key ART roles and responsibilities

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Roles on the ART

Release Train Engineer is a coach for the ART.

Product Management owns, defines, and prioritizes the ART Backlog.

System Architect provides architectural guidance and technical enablement to the teams on the train.

The **System Team** provides processes and tools to integrate and evaluate assets early and often.

Business Owners are key stakeholders on the ART.

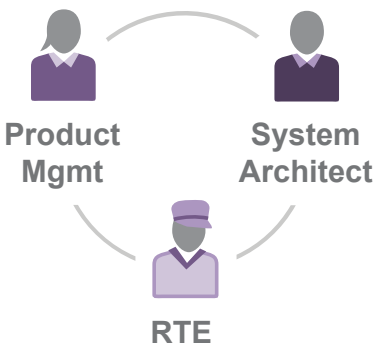
AGILE RELEASE TRAIN

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Key ART roles

ART leadership is collaboratively handled by the three key ART roles, each focusing on one area.

- ***ART Execution:*** The RTE facilitates optimizing the flow of value through the ART.
- ***Content Management:*** Product Management is the internal voice of the customer on the ART.
- ***Design Authority:*** The System Architect defines the overall architecture for the system.



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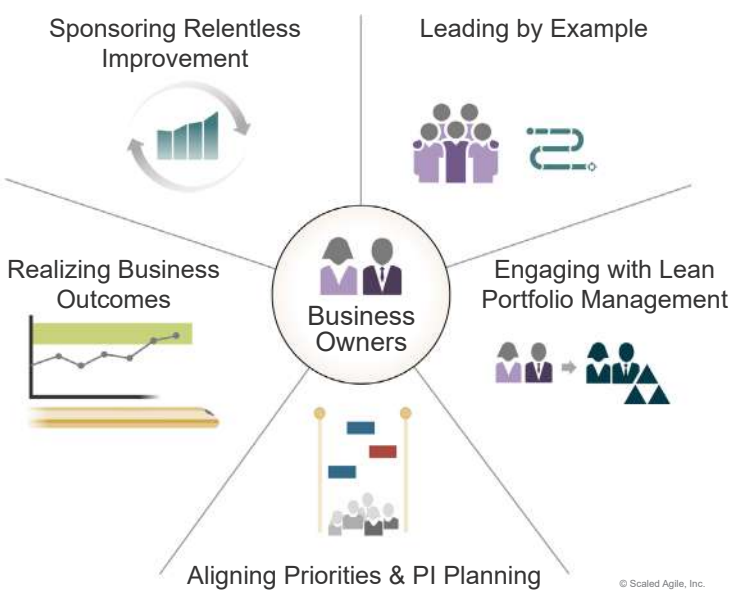
Discussion: Key ART collaborations

Duration: 8 min

- **Step 1:** As a class, discuss how the RTE facilitates connections between the three principal roles on the train (Product Management, System Architect, and Release Train Engineer). Consider the following:
  - What possible anti-patterns can arise in this group?
  - How can an RTE mitigate these anti-patterns?

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Business Owners responsibility areas



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Identifying Business Owners

The Business Owners can be identified by answering the following questions:

- Who can steer this train in the right direction, participate in planning and execution, and help eliminate impediments?
- Who can speak on behalf of development, the business, and the Customer?
- Who can approve and defend a set of PI plans, knowing that they will never satisfy everyone?



Active participation of Business Owners is a critical success factor on the ART!

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Video: Business Owners in SAFe

Duration 3 min



https://bit.ly/Video-BusinessOwnersSAFe

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Activity: Who are the Business Owners?

Duration: 5 min

- Step 1: As a class, discuss the following:
  - In your current context, who are the Business Owners for the ART?
  - Who else in your organization could fulfill the role, and why would they be the right person(s)?

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3.3 Identify the responsibilities of the System Team

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System Team

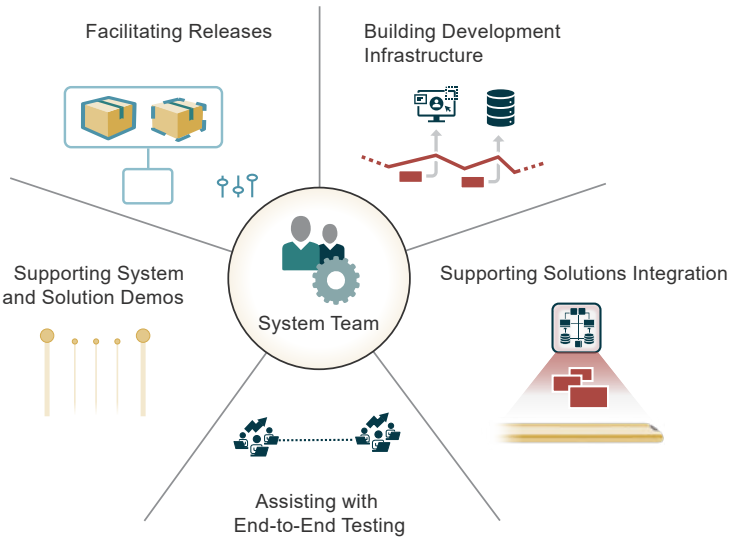
The teams on the ART are collectively responsible for delivering larger system and Solution value. To support this work, a System Team is often formed to:

- Create and maintain infrastructure, including Continuous Integration, automated builds, and automated build verification testing
- Perform end-to-end system integration and Solution performance testing
- Assist in staging System Demos
- Support DevOps and CDP activities, including Release on Demand



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System Team responsibility areas



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Discussion: The System Team

Prepare 5 min | Share 3 min

- **Step 1:** Working in your groups, discuss the following:
  - What might happen if an ART doesn't have a System Team?
  - When can the System Team turn into a bottleneck or impediment?
  - What is the role of the RTE in relation to the System Team?
- **Step 2:** Be prepared to share with the class

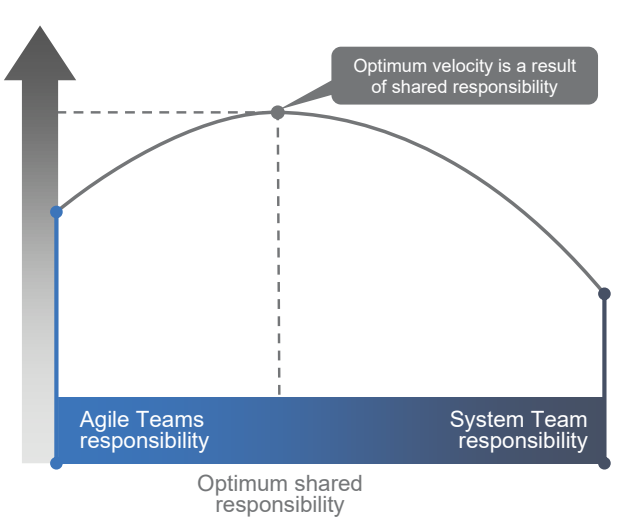
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Balancing responsibilities

The System Team can never be the entire solution to the integration challenge.

Maximizing ART velocity requires a sense of balance between Agile Teams and the System Team. Best practices include:

- Shared non-functional requirement (NFR) testing across other teams and the System Team
- Shared understanding of infrastructure and tooling across the teams on the ART
- Shared necessity to avoid turning the System Team into a bottleneck



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RTE Action Plan

Prepare 5 min

Share 2 min

- **Step 1:** Locate the RTE Action Plan section in your workbooks
- **Step 2:** Begin adding tools to the Action Plan by brainstorming the following:
  - How will you bridge connections between yourself and the other key roles on the ART?
- **Step 3:** Share one of your insights with the class



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3-25

Lesson Review

In this lesson you:

- Defined the characteristics of an effective ART organization
- Described the key ART roles and responsibilities
- Summarized the responsibilities of the System Team

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Action Plan

Organizing the ART

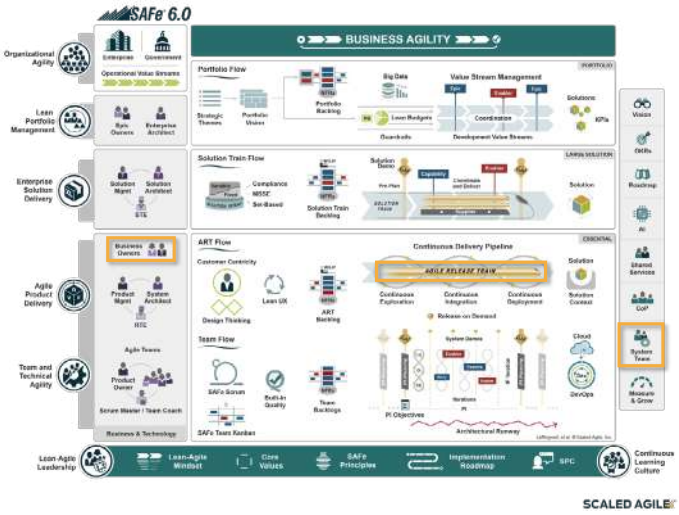
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Articles used in this lesson

Read these Framework articles to learn more about topics covered in this lesson

- "Organizing Agile Teams" https://www.scaledagileframework.com/organizing-agile-teams-and-arts-team-topologies-at-scale/
- "Agile Release Train" https://www.scaledagileframework.com/agile-release-train/
- "Business Owners" https://www.scaledagileframework.com/business-owners/
- "System Team" https://www.scaledagileframework.com/system-team/



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Continue your SAFe journey with the following resources:

Access the collaboration template "ART Team Organization" to evaluate the trade-offs that can happen when organizing teams on the train. **https://bit.ly/Template-ARTTeamOrg**	Access the collaboration template "The RTE and ART Events" to develop strong practices related to effectively facilitating ART events. **https://bit.ly/Template-RTEandARTEvents**
Access the "SAFe ART and Team Events" page on SAFe Studio to explore guidance and support for SAFe events. **https://bit.ly/Community-SAFeARTandTeamEvents**	Download and share the "SAFe Team Formation Toolkit" with your SM/TCs to help them align their teams on team purpose, skills, and working agreements. **https://bit.ly/Community-ToolkitsandTemplates**
Watch this 45-minute video, *Self-Selection: a SAFe approach to forming ARTs,* to hear about one company's experience organizing their ART and to learn about the benefits of allowing people to self-select their teams. **https://bit.ly/Video-SelfSelection**	

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References

Dunbar, R. I. M. “Coevolution of Neocortical Size, Group Size and Language in Humans.” *Behavioral and Brain Sciences* 16, no. 4. 1993. 691.

Skelton, Matthew and Manuel Pais. *Team Topologies: Organizing Business and Technology Teams for Fast Flow.* IT Revolution: Portland, 2019. Kindle Edition.

Wikipedia. 2023. "Portrait of Robin Dunbar." Wikimedia Foundation. Last modified April 13, 2023. https://en.wikipedia.org/wiki/Robin_Dunbar.

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Lesson notes

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Lesson 4

Planning a PI

SAFe® Course - Attending this course gives learners access to the SAFe Release Train Engineer exam and related preparation materials.

Lesson Topics

4.1 Identify preparation activities for the PI Planning event

4.2 Facilitate Day 1 activities for the PI Planning event

4.3 Facilitate final PI plan development and commitment

4.4 Explore facilitation of PI Planning across multiple locations



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Learning objectives

At the end of this lesson, you should be able to:

- Define the preparation activities for the PI Planning event
- Demonstrate how to facilitate Day 1 activities for the PI Planning event
- Demonstrate how to facilitate the final PI plan development and commitment
- Demonstrate how to facilitate PI Planning across multiple locations

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4.1 Identify preparation activities for the PI Planning event

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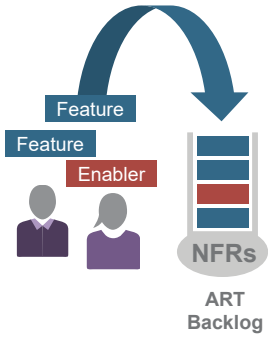
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Preparing the ART Backlog for PI Planning

In the weeks before PI Planning, Product Management does the final ART Backlog preparation with System Architects and ART stakeholders.

- RTEs will be more active in coaching a new ART when preparing the backlog
- Over time, the Product Management and the System Architect should be able to handle most of this preparation work
- The RTE will be one of the backlog stakeholders, often focused on infrastructure Enablers and improvement items



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Features represent the work for the Agile Release Train

- Feature benefit hypothesis justifies development costs and provides a business perspective for decision-making
- Acceptance criteria are typically defined during ART Backlog refinement
- Reflect functional and non-functional requirements
- Typically sized to fit into a single PI

Multi-factor authentication

Benefit hypothesis
Enhance user security via both password and a device

Acceptance criteria
1. USB tokens as a first layer
2. Password authentication second layer
3. Multiple tokens on a single device
4. User activity log reflecting both authentication factors

Example Feature

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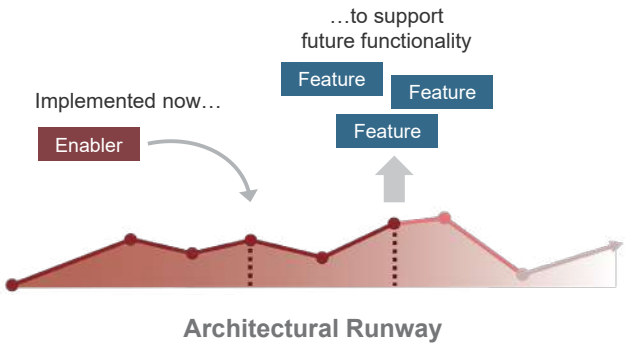
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Enablers build the Architectural Runway

Enablers define the existing code, hardware components, marketing guidelines, and other variables that enable near-term business Features.

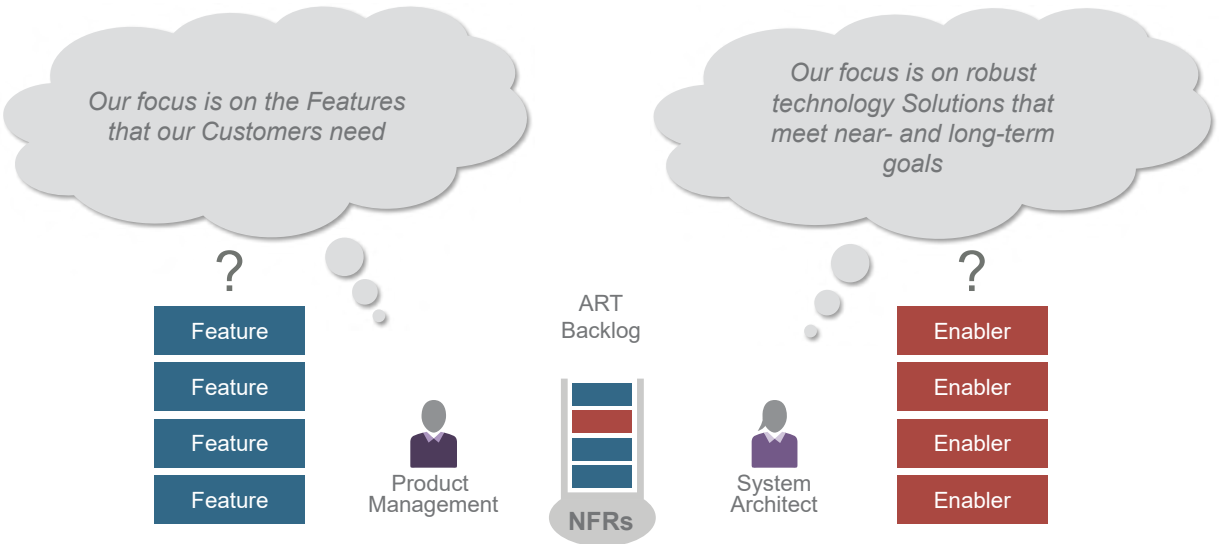
- Enablers also build up the runway, including exploration, architecture, infrastructure, and compliance
- Features consume the runway
- The Architectural Runway must be continuously maintained
- Capacity allocation (a percentage of the train's overall capacity in a PI) is used for Enablers that extend the runway

Example:
A single sign-on mechanism will enable sign-on in multiple applications.



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Using capacity allocation to create balance



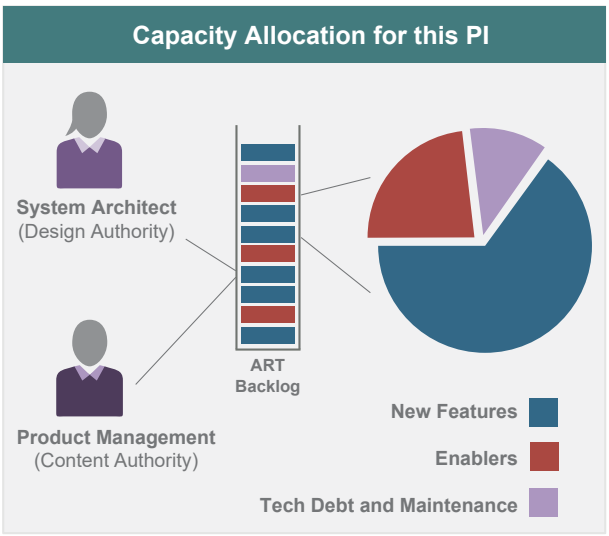
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Product integrity through capacity allocation

- At each PI boundary, agree on a healthy balance between each work type
- Establish policies to determine how work is performed
- The System Architect has design authority and prioritizes architectural work
- Product Management prioritizes business Features
- Agree to jointly prioritize work based on economics
- Agree to collaborate to sequence work to maximize Customer value.

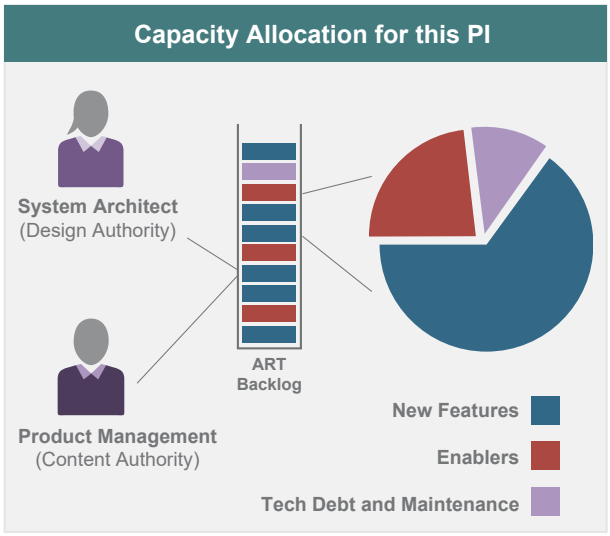


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Discussion: What can capacity allocation be used for?

Discuss 5 min

- **Step 1:** As a class, discuss the following:
  - Other than allocating dedicated capacity for Enablers, what other work types can leverage capacity allocation when preparing for PI Planning?
  - What about during PI Planning?

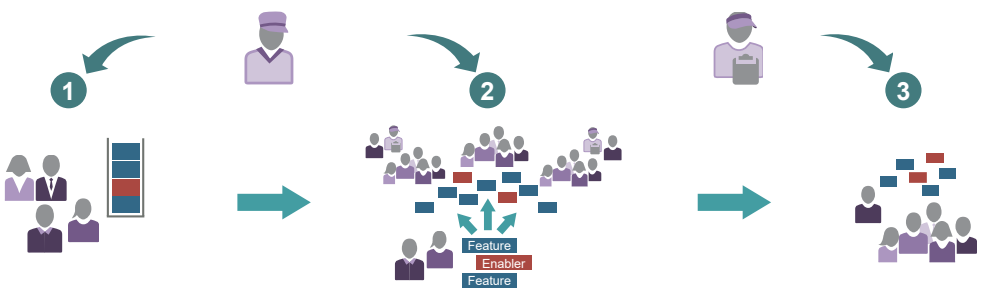


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Socializing Features and Enablers

Socializing Features and Enablers solves many problems later during PI Planning.



Product Management and other stakeholders refine the ART Backlog.

Features are socialized with the teams on the train, and each team identifies big/initial Stories.

Each team works to prepare a rough backlog for the PI Planning looking for feedback, big unknowns, etc.

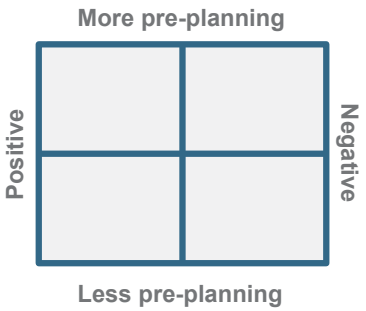
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Activity: How much preparation is enough?

Prepare 5 min

Share 5 min

- **Step 1:** Working in your groups, explore the benefits and drawbacks of doing more or less preparation before the PI Planning event.
- **Step 2:** Fill out the diagram. List the positives and negatives of each alternative in the quadrants.
- **Step 3:** Be prepared to share with the class.



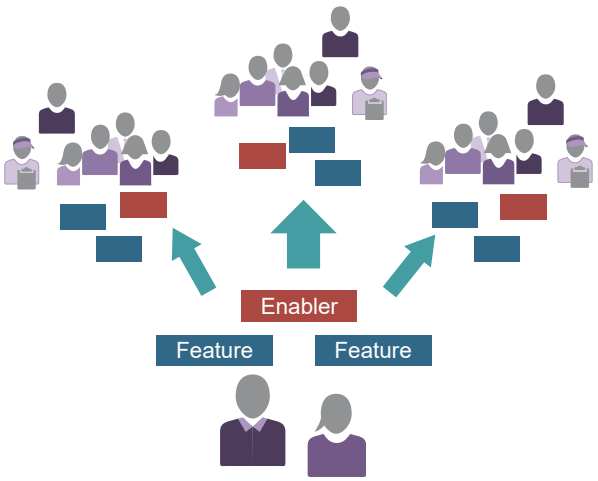
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How much preparation is enough?

Both too much and too little preparation can cause problems. The RTE helps the teams and stakeholders find a good trade-off.

- Do more pre-planning if you create new architecture or functionality
- Remind attendees that too much pre-planning minimizes exploration, interaction, and emergent designs/solutions
- Do enough preparation to maximize the amount of interaction in the event



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PI Planning content readiness – Day 1 presentations

Business context	8:00 – 9:00	• State of the business
Product/Solution Vision	9:00 – 10:30	• Vision and prioritized Features
Architecture Vision and development practices	10:30 – 11:30	• Architecture, common frameworks, etc. • Agile tooling, engineering practices, etc.
Planning context and lunch	11:30 – 1:00	• Facilitator explains the planning process
Team breakouts	1:00 – 4:00	• Teams develop draft plans and identify risks and impediments • Architects and Product Managers circulate
Draft plan review	4:00 – 5:00	• Teams present draft plans, risks, and impediments
Management review and problem solving	5:00 – 6:00	• Adjustments made based on challenges, risks, and impediments

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ART readiness checklist

Item	Considerations
Planning scope and context	Is the scope (product, system, technology domains) of the planning process understood? Have we identified our Value Stream(s) and Agile Release Train(s)?
RTE	Have we identified the RTE? Do they understand the scope of the role in preparing the organization and preparing for the PI Planning event?
Planning timeframe, Iteration and PI cadence	Have we identified the PI Planning dates, the Iteration cadence, and the PI cadence?
Agile Teams (SM/TCs, POs)	Does each team have an identified SM/TC and Product Owner (PO)?
Team makeup/commitment	Have we identified cross-functional Agile Teams? Are there dedicated team members on every team?

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ART readiness checklist (continued)

Item	Considerations
Agile Team attendance	Are all team members attending PI Planning, whether in person or involved remotely?
Executive, Business Owner	Do we know who will set and present the business context and strategy? Do we know who will assign business value and accept the plans?
Business alignment	Is there reasonable agreement on priorities amongst the Business Owners and Product Management?
Architecture Vision and Development Practices	Do we know who will present the Architecture Vision and required development practice?
Product/Solution Vision and ART Backlog	Is there a clear vision of what we are building, at least over the next few PIs? Have we identified the top 10 or so Features that are the subject of the first PI?

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ART readiness checklist (continued)

Item	Considerations
System Team	Has the System Team been identified and formed?
Shared Services	Have the Shared Services (User Experiences, Architecture, etc.) been identified?
Other Attendees	Do we know what other key business and technology stakeholders should attend?
Agile Lifecycle Management (ALM) tooling	Do we know how and where Iterations, PIs, Features, Stories, measures, etc., will be maintained?
Development infrastructure	Do we have a strategy for using and improving the Continuous Delivery Pipeline?
Quality practices	Is there a strategy for applying Built-in Quality practices?

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Activity: ART readiness preparation





- **Step 1:** Determine which items on the ART readiness checklist must be in place in order to launch your ART
- **Step 2:** In your workbook, mark each item with a Y, N, or D to determine if it is necessary for launching the ART
  - **(Y)** Yes, must be in place
  - **(N)** No, doesn't need to be in place
  - **(D)** Depends on context
- **Step 3:** Be prepared to share with the class

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ART readiness preparation

Instructions: Determine which items on the ART readiness checklist must be in place in order to launch your ART. Mark each item with a Y, N, or D to determine if it is a necessity for launching the ART: (Y) Yes, must be in place; (N) No, does not need to be in place; (D) Depends on context.

	Area	Questions	Y	N	D
1	Planning scope and context	Is the scope (product, system, technology domains) of the planning process understood? Have we identified our Value Stream(s) and Agile Release Train(s)?	○	○	○
2	RTE	Have we identified the RTE? Do they understand the scope of the role in preparing the organization and preparing for the PI Planning event?	○	○	○
3	Planning timeframe, Iteration and PI cadence	Have we identified the PI Planning dates, the Iteration cadence, and the PI cadence?	○	○	○
4	Agile Teams (SM/TCs, POs)	Does each team have an identified SM/TC and Product Owner (PO)?	○	○	○
5	Team makeup/ commitment	Have we identified cross-functional Agile Teams? Are there dedicated team members on every team?	○	○	○
6	Agile Team attendance	Are all team members attending PI Planning, whether in person or involved remotely?	○	○	○
7	Executive, Business Owner	Do we know who will set and present the business context and strategy? Do we know who will assign business value and accept the plans?	○	○	○
8	Business alignment	Is there reasonable agreement on priorities amongst the Business Owners and Product Management?	○	○	○
9	Architecture Vision and Development Practices	Do we know who will present the Architecture Vision and required development practice?	○	○	○
10	Product/Solution Vision and ART Backlog	Is there a clear vision of what we are building, at least over the next few PIs? Have we identified the top 10 or so Features that are the subject of the first PI?	○	○	○
11	System Team	Has the System Team been identified and formed?	○	○	○
12	Shared Services	Have the Shared Services (User Experiences, Architecture, etc.) been identified?	○	○	○
13	Other attendees	Do we know what other key business and technology stakeholders should attend?	○	○	○
14	Agile Lifecycle Management (ALM) tooling	Do we know how and where Iterations, PIs, Features, Stories, measures, etc., will be maintained?	○	○	○
15	Development infrastructure	Do we have a strategy for using and improving the Continuous Delivery Pipeline?	○	○	○
16	Quality practices	Is there a strategy for applying Built-in Quality practices?	○	○	○

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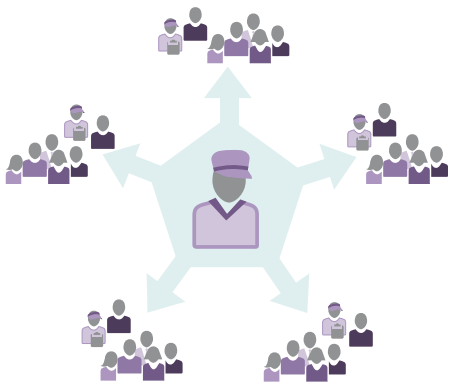
4.2 Facilitate Day 1 activities for the PI Planning event

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The responsibilities of the RTE in PI Planning

RTEs are the heartbeat of PI Planning and are responsible for:

- Helping the ART prepare for PI Planning
  - Planning scope and context
  - Content readiness
  - Logistics
- Facilitating Days 1 and 2 of PI Planning
  - Manage agenda
  - Run Coach Syncs
  - Facilitate draft and final plan readouts
  - Facilitate Management Review and problem-solving meeting



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Presentations at the start of Day 1

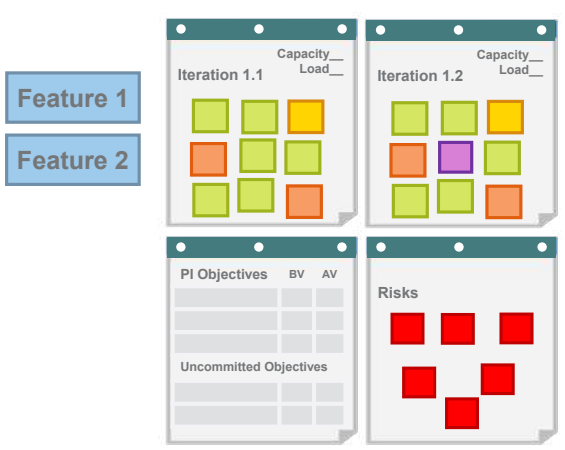
The day starts with the content presentations, creating a shared understanding of the business context and the boundaries that the teams are planning the PI inside. The RTE should prepare by:

- Double- and triple-checking audio/video equipment and other technological needs.
- Having one slideshow deck and one computer for presentations. Avoid switching between decks and devices to keep it flowing smoothly.
- Planning room for a short Q&A after each presentation and managing the timebox by pushing discussions to meet-afters as needed.

Team breakout #1

The teams create draft plans for each Iteration, write draft PI Objectives, and identify ART PI Risks and impediments.

- Teams should use visual radiators for planning to create transparency and collaboration
- Make sure teams are using common tools and standards (such as color-coded sticky notes)
- Ask SM/TCs to assist in preparing each team's physical/virtual spaces for the event

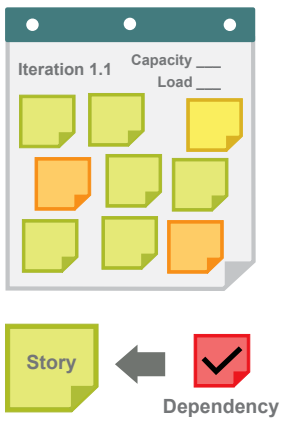


G User Story | P Maintenance | Y Exploration Enabler | O Infrastructure Enabler | R Risks or dependencies

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Team planning radiators in detail: Iterations

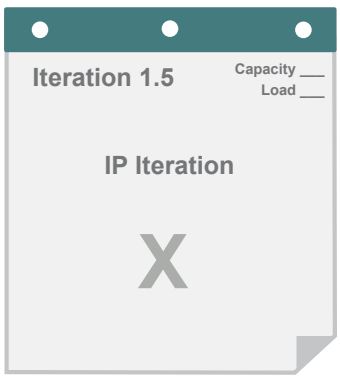
- Teams write and sequence User Stories and Enabler Stories across the PI
- Teams identify risks and escalate them if they cannot be managed within the team
- Stories are estimated so that teams can determine their capacity and load
- Teams should allocate a percentage of capacity for unplanned activities, such as maintenance and production support.



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Team planning radiators in detail: IP Iteration

- The last Iteration will be used for Innovation and Planning (IP)
- Teams should estimate their capacity but not a load on the IP Iteration
- Teams should not plan any Stories during the IP Iteration



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Team planning radiators in detail: PI Objectives

- PI Objectives should be written so that they can be understood by someone outside the team
- Uncommitted objectives are in the team plan and fit in the capacity, but the team is unsure if they can deliver
- SAFe suggests targeting 7-10 PI Objectives, with up to 3 uncommitted objectives



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Team planning radiators in detail: risks

- ART PI Risks are those that are outside of the team's control and need to be escalated. They will be captured and assessed after the final plan review.
- Team risks are those under the team's control. They won't be presented and are handled by the team themselves.



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Team breakout: Hourly Coach Sync checkpoints

The Coach Sync checkpoints help keep teams on track and facilitate early identification of risks.

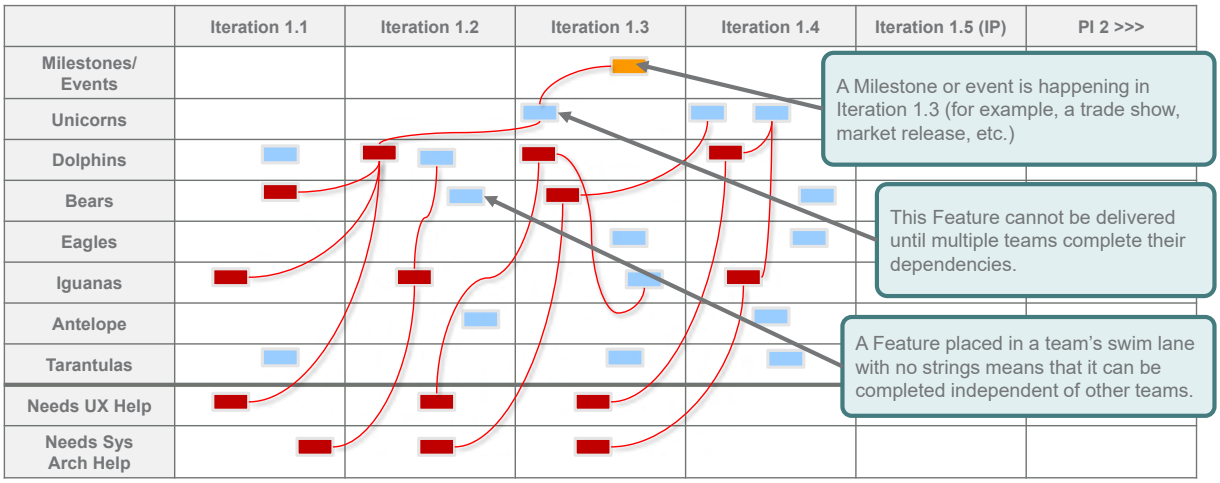
DAY 1 CHECK-IN 1: GETTING STARTED	1	2	3	4	5	6	7	8	9	10
Do you understand the planning requirements?										
Do you know who your team is for the whole PI?										
Is your working space setup?										
Do you have a Product Owner & Scrum Master / Team Coach?										
Do you have access to the team members and stakeholders you need?										
Do you understand (and can you find) the vision that drives your backlog?										
Have you identified the capacity for each Iteration in your PI?										
Do you understand the architectural context, and who to go to for questions?										
Do you understand which resources are shared (e.g., UX, Training, and Documentation) and who to go to for questions?										
Do you understand the role of the System Team and DevOps, and who to go to for questions?										
Is there anything you need to discuss with other Scrum Masters/Team Coaches? If so, stay for the "Meet After".										

Simple planning "radiators"

Day 1	
1:00 pm	Getting started
2:00 pm	Iteration Planning progress
3:00 pm	Team Objectives progress
3:45 pm	Draft plan readiness
Day 2	
10:30 am	Progress check-in
11:30 am	Final plan readiness

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ART Planning Board: Feature delivery, dependencies, and Milestones



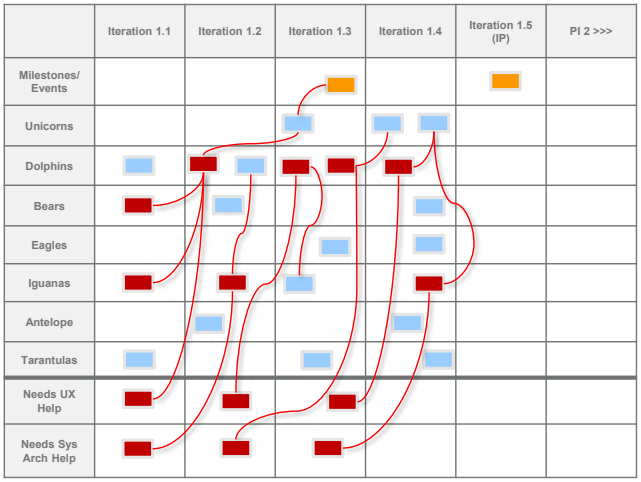
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Using the ART Planning Board

The ART Planning Board is a critical artifact during and after PI Planning

- Visually shows the sequence of inter-team dependencies that lead to Feature delivery
- Clearly indicates when Features are planned to be delivered
- Red sticky notes are agreed-upon and planned actions that will resolve a dependency and supports Feature delivery



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Activity: Reading the ART Planning Board in PI Planning

Prepare 7 min

Share 7 min

- **Step 1:** With your group, look at the three different ART Planning Board examples
- **Step 2:** Capture the answers to the following questions for each scenario:
  - What problems do you see?
  - How would you, the RTE, act if you could see these on the ART Planning Board at the end of Day 1 in PI Planning?
  - What could be a potential solution for the specific problems?
- **Step 3:** Be prepared to share with the class

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Example ART Planning Board #1

	Iteration 1.1	Iteration 1.2	Iteration 1.3	Iteration 1.4	Iteration 1.5 (IP)	PI 2 >>>
Milestones/ Events				M1		
Unicorns			F1			
Dolphins	D1	D2				
Bears			F2			
Eagles				F6		
Iguanas		F3				
Antelope				F5		
Tarantulas	D3	D4	F4			
Needs UX Help						
Needs Sys Arch Help						

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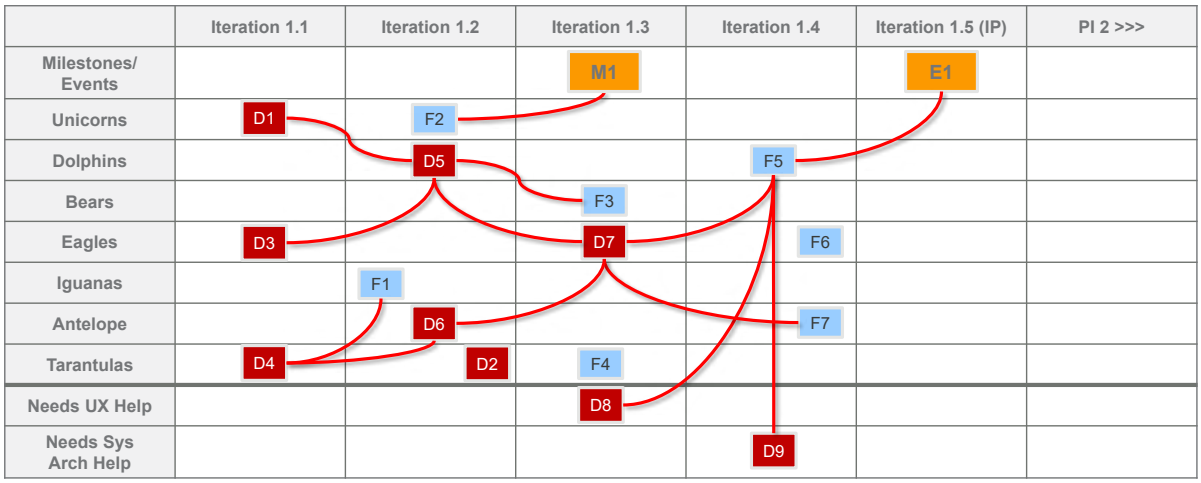
Example ART board #2

	Iteration 1.1	Iteration 1.2	Iteration 1.3	Iteration 1.4	Iteration 1.5 (IP)	PI 2 >>>
Milestones/ Events			M1			
Unicorns	F6	F1	F3	F5		
Dolphins						
Bears		D2				
Eagles	D1		F4	D7		
Iguanas		D4				
Antelope		F2		D6		
Tarantulas		D3	D5	F7		
Needs UX Help						
Needs Sys Arch Help						

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Example ART board #3



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Writing draft PI and uncommitted objectives

Draft objectives create transparency in PI Planning and help engage the Business Owners and stakeholders in the planning process.

- Draft versions are presented at the end of Day 1
- Targeting 7-10 committed objectives and up to 2-3 uncommitted objectives seems to work best
- Remind everyone to move team objectives to uncommitted if it is planned, but team confidence is not 100%

Objectives for PI 1	BV	ABV
1. Show routing calculations between the 5 most frequent destinations		
2. Navigate autonomously from distribution center the most frequent destination		
3. Parallel park for a delivery		
4. Return to distribution center after delivery		
5. Include traffic data in route planning		
6. Recall a delivery that is already in progress		
7. Reduce GPS signal loss by 25%		
Uncommitted Objectives		
8. Demonstrate real-time rerouting to avoid delays (for example, accident, construction)		

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Draft plan review

At the end of Day 1, the RTE facilitates the session where each team presents their draft plan to the ART.

- The main purpose is to get everyone in the room thinking about dependencies, problems, and opportunities for tomorrow
- Make sure everyone follows the agenda when presenting and make the agenda visible to everyone

Draft plan review agenda:

1. Current capacity and load for each Iteration
2. Draft PI Objectives
3. Identify ART PI Risks and impediments

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Management review and problem-solving

At the end of Day 1, management meets to make adjustments to scope and objectives based on the day's planning.

Common questions:

- What did we just learn?
- Where do we need to adjust Vision, scope, team assignments, or anything else?
- Where are the bottlenecks?
- What Features must be de-scoped?
- What decisions must we make between now and tomorrow to address these issues?



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Facilitating the management review and problem-solving meeting

The RTE is the facilitator of the management review and problem-solving meeting and is responsible for making sure that the session produces an output.

- Have a clear idea of **how** to facilitate the meeting so that it is effective
- Work with the key ART roles, teams, and stakeholders throughout Day 1 to identify potential inputs for the meeting
- Invite selected team members that can help clarify problems and find solutions
- Make sure to document inputs and outputs from the meeting
- Over time, evolve how you facilitate the meeting



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Activity: Management review and problem-solving meeting

Prepare 7 min | Share 5 min

- **Step 1:** With your group, consider the management review and problem-solving meeting and discuss the following:
  - Which roles or functions must attend?
  - What common challenges tend to arise during this meeting?
  - What techniques can RTEs use to address common challenges?
- **Step 2:** Based on your discussion, identify three best practices that can help RTEs prepare for and facilitate the management review and problem-solving meeting
- **Step 3:** Be prepared to share with the class

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4.3 Facilitate final PI plan development and commitment

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Make planning adjustments

- Based on the previous day's management review and problem-solving meeting, adjustments are discussed
- Possible changes include:
  - Business priorities
  - Adjustment to Vision
  - Changes to scope
  - Realignment of work and teams





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Team breakout #2

Based on new knowledge (and a good night's sleep), teams now work to create their final plans.

- Focus on getting all Feature deliveries and dependencies visualized on the ART Planning Board – the RTE should assist the teams and SM/TCs to get items placed
- Help the teams write readable, understandable, and S.M.A.R.T. PI Objectives
- Be available to facilitate discussions that might happen when assigning business value to PI Objectives

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PI Objectives should be S.M.A.R.T.

During the second team breakout, assist the teams in writing their PI Objectives using the S.M.A.R.T. format.

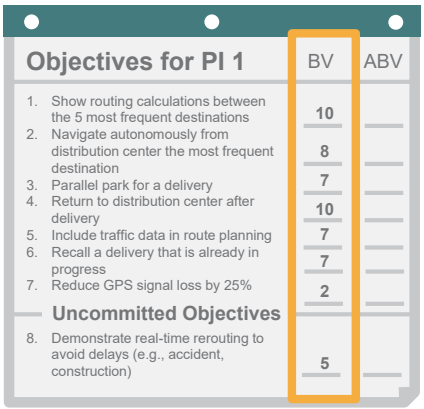
S	**Specific**	State the intended outcome as simply, concisely, and explicitly as possible. (Hint: Try starting with an action verb)
M	**Measurable**	It should be clear what a team needs to do to achieve the objective. The measures may be descriptive, yes/no, quantitative, or provide a range.
A	**Achievable**	Achieving the objective should be within the team's control and influence.
R	**Realistic**	Recognize factors that cannot be controlled. (Hint: Avoid making assumptions)
T	**Time-bound**	The time period for achievement must be within the PI. Therefore, all objectives must be scoped appropriately.

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Assigning business value to PI Objectives

- As the PI Objectives are finalized, each objective is assigned business value by the ART Business Owners on a scale of 1–10
- In a direct conversation with the Agile Teams, the ART Business Owners assign business value to each of the teams' individual PI Objectives
- Business value communicates the potential business impact of achieving an objective, which helps teams on the train prioritize work
- The RTE might need to be available to facilitate this conversation, especially during the first PIs



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4-43

Video: Assigning business value during PI Planning

Duration 3 min



https://bit.ly/Video-AssigningBVPIPlanning

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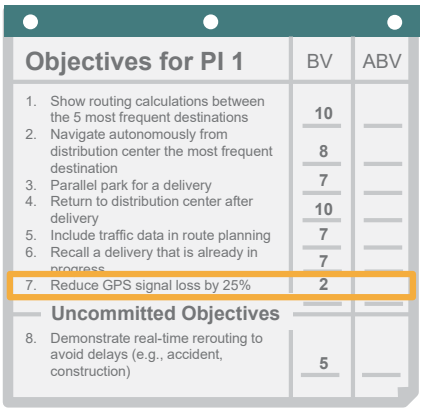
4-44

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Activity: Conflicting views of value

Prepare 7 min | Share 3 min

- **Step 1:** Consider the following scenario:
  - Business Owners have assigned a value of 2 to the PI Objective “Reduce GPS signal loss by 25%,” and they have requested that the team move the objective to uncommitted. The team believes this function provides a critical architectural enablement for the entire ART. The team is clearly disappointed and concerned that an important technical item is ranked so low.
    “We just don’t see much business value in it now,” said the Vice President of Product.
- **Step 2:** Working in your group, discuss what your steps should be for solving this problem
- **Step 3:** Be prepared to share with the class



Objectives for PI 1	BV	ABV
1. Show routing calculations between the 5 most frequent destinations	10	
2. Navigate autonomously from distribution center the most frequent destination	8	
3. Parallel park for a delivery	7	
4. Return to distribution center after delivery	10	
5. Include traffic data in route planning	7	
6. Recall a delivery that is already in progress	7	
7. Reduce GPS signal loss by 25%	2	
Uncommitted Objectives		
8. Demonstrate real-time rerouting to avoid delays (e.g., accident, construction)	5	

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Final plan review

The RTE facilitates the session where everyone in the room peer-reviews the final plans for all teams, and the Business Owners accept the plan.

- Make sure everyone follows the agenda when presenting. Make it visible in the room.
- Validate that presented risks and impediments are outside of the teams’ control before accepting them as ART PI Risks.
- Manage the timebox for the Q&A.
- After each review, ask the Business Owners in the room if they accept the plan.

Final plan review agenda:

1. Changes to capacity and load for each Iteration
2. Final PI Objectives with Business Values
3. ART PI risks and impediments
4. Q&A

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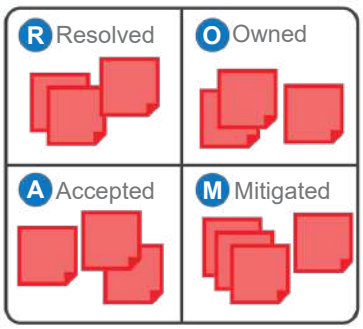
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Addressing ART PI Risks

After all plans have been presented, the remaining ART PI Risks and impediments are discussed and categorized.

ROAM stands for:

- **R**esolved - Addressed; the risk is no longer a concern
- **O**wned - Someone has taken responsibility
- **A**ccepted - Nothing more can be done, and if the risk occurs, the release may be compromised
- **M**itigated – The team has a plan to adjust as necessary



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Activity: ROAM ART PI risks

Prepare 7 min | Share 3 min

- **Step 1:** Working in your groups, read through the six risks on the next slide.
- **Step 2:** Develop a clear plan for how you, the RTE, would facilitate ROAMing the risks. Try to address as many as you can in the 7-minute timebox.
- **Step 3:** Document the approach for each risk. Indicate what ROAM you would suggest and why. Focus on **how** you would facilitate finding the solution during PI Planning, not **what** the solution would be.
- **Step 4:** Be prepared to share with the class.

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ART PI Risks

Team member X is a bottleneck. They are the only one who can handle critical parts of the implementation for an important Feature that is scheduled for delivery in the Iteration.

No System Team is available yet. The integration infrastructure is in bad shape and could put the integration and/or System Demo at risk.

Two teams have a shared Product Owner. This might have a negative impact on the team's Iteration Planning and demos.

The QA team is pointing out that application performance might be at risk during Iteration 3, as there will be an influx of new users. They are unsure how this might impact the ART.

In Iteration 3, an important supplier might be delayed with their crucial delivery. They have been late before, and this time it might impact several teams.

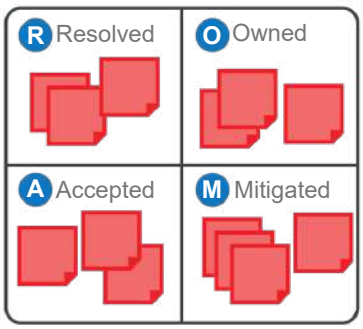
There is a holiday mid-PI, and many people will be on vacation.

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The risk ROAMing session

The risk ROAMing session can be challenging if you are new to the train as the RTE

- Make sure to continuously look at risks as they appear in the planning session so you can develop a plan for how to ROAM them.
- Ask the SM/TCs to be proactive and come to you if their teams find big risks during planning.
- Be the facilitator when ROAMing. Avoid trying to find the solution yourself and ask the room for help.
- Product Management and the System Architect should be an active part of the ROAMing process.



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Confidence vote: Teams and ART

Once ART PI Risks have been addressed, a confidence vote is taken by each team and the ART.

A commitment with two parts:

1. Teams agree to do everything in their power to meet the agreed-to objectives
2. If objectives become not achievable, teams agree to escalate immediately so that corrective action can be taken



No confidence	Little confidence	Good confidence	High confidence	Very high confidence

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Discussion: Confidence not high? Plan rework.

Duration 7 min

- **Step 1:** As a class, discuss the role of the RTE in facilitating the confidence vote. Consider the following:
  - How can the RTE ensure that attendees feel safe to voice their concerns?
  - How would you facilitate reworking the plan if there is little to no confidence (teams vote a 1 or 2)?
  - What agreements would you put in place to rework the plan?

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PI Planning retrospective

PI Planning will evolve over time. Ending it with a quick and focused retrospective will help it continuously improve.

- By this time, everyone in the room is exhausted and wants to go home. Make it as focused and quick as possible.
- Keep the focus on collecting opinions, ideas, and data; you can sort and prioritize later.
- Make sure to change how you run the retrospective over time to keep it fun and energizing.

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After the PI Planning event

Following PI Planning, the RTE needs to take care of many key outputs and artifacts:

- If appropriate, make sure the work is tracked in an Agile project management tool
- Move the ART Planning Board to a dedicated location
- Capture the decisions made during the risk ROAMing session
- Create a set of integrated ART PI Objectives
- Set up meetings for recurring ART events
- Start planning the next PI Planning event

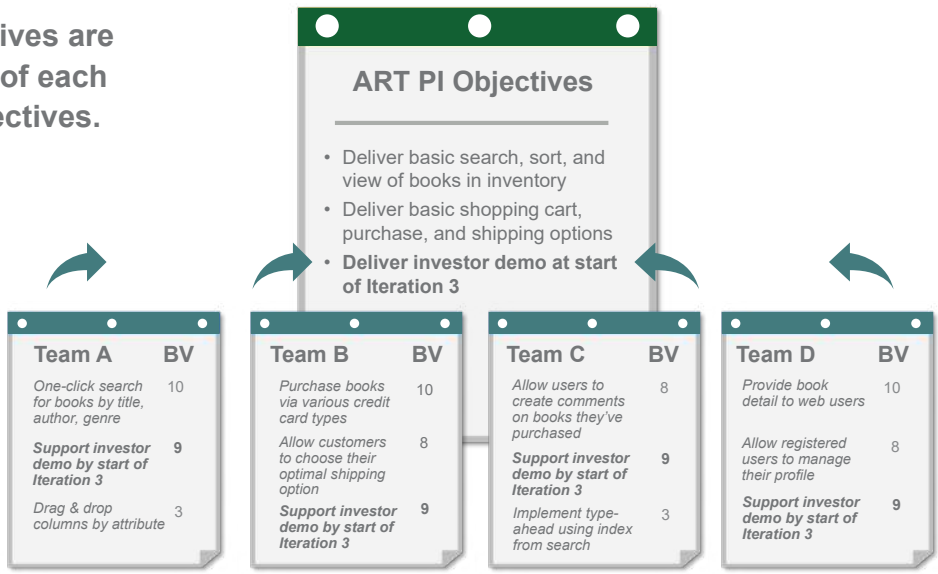


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RTE takeaway: integrated ART PI Objectives

ART PI Objectives are the synthesis of each team's PI Objectives.



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For large solutions, ARTs are combined into a Solution Train

Solution Trains are used to build large solutions that require coordination across multiple ARTs and suppliers.

- Requires additional roles to coordinate delivery, including Solution Train Engineer (STE), Solution Management, and Solution Architect
- Requires additional events to manage delivery across ARTs, including RTE Sync, Architect Sync, and Product Management Sync

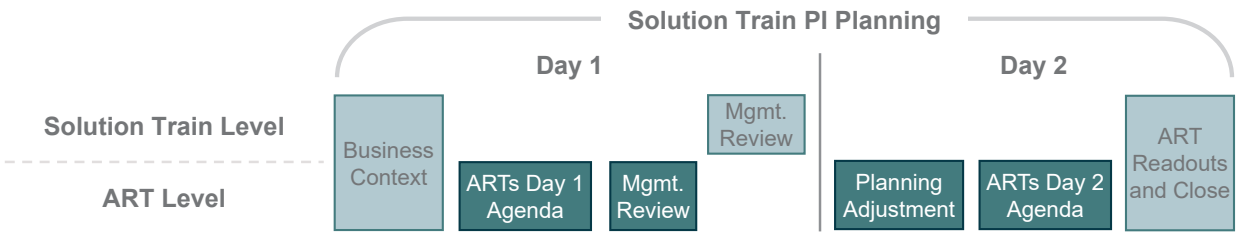


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The role of the RTE in planning the Solution Train PI

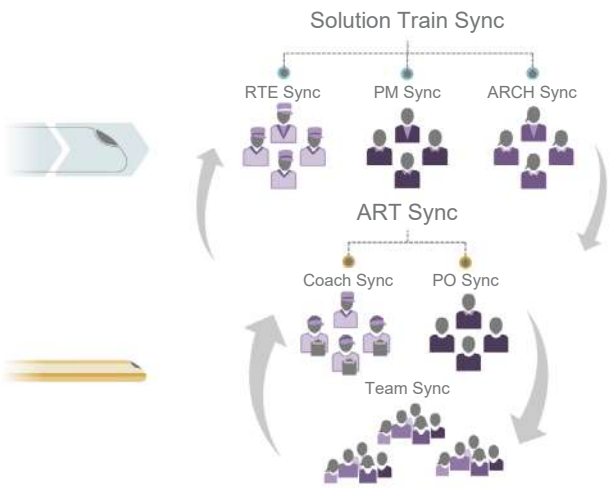
- Collaborate with the STE to build the PI Planning schedule and agenda
- Integrate the Solution Vision into the Business Context portion of PI Planning
- Provide input for the Solution Train Management Review and problem-solving meeting
- Contribute ART PI Objectives to the aggregated Solution Train PI Objectives



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Synchronize routinely throughout the PI

- Communicate progress toward objectives and track dependent work at RTE Sync or Solution Train Sync
- Coordinate large releases with STE and other Solution Train stakeholders



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4.4 Explore facilitation of PI Planning across multiple locations

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Video: SAFe at Travelport: PI Planning with distributed teams

Duration 2 min



https://bit.ly/Video-PIDistributed

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Distributed planning events

Distributed planning events require significantly more preparation and facilitation.

- Have a dedicated RTE proxy and tech support person at each location
- Test audio, video, and presentation-sharing connectivity multiple times
- Have a common understanding of how plans will be shared (video, intranet, collaborative ALM tool, emailing PPT, and so on)
- Establish team-based audio/video communication for breakout sessions



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Respect for people and culture

- Have clear working agreements to be respectful and compromise when it comes to time zone differences
- Follow the Lean principle to respect people and cultures by avoiding asking teams to stay up all night
- Avoid asking teams to commit to their PI Objectives in a sleep-deprived state
- Consider extending the PI Planning agenda to accommodate multiple time zones.



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Activity: Facilitating a distributed PI Planning

Prepare 15 min | Share 5 min

- **Step 1:** In your group, create an agenda for a distributed PI Planning event with large time zone differences
- **Step 2:** List ideas and tools regarding what you, the RTE, would do to effectively prepare for and facilitate this PI Planning event
- **Step 3:** Be prepared to share with the class



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Action Plan: RTE Action Plan

Prepare 5 min | Share 2 min

- **Step 1:** Locate the RTE Action Plan section in your workbooks
- **Step 2:** Add ideas to the Action Plan by brainstorming the following:
  - Review and select one of the facilitation responsibilities for an RTE during PI Planning and share how you will prepare to manage this part of the event successfully
- **Step 3:** Share one of your insights with the class



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Action Plan

Planning a PI

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Lesson review

In this lesson you:

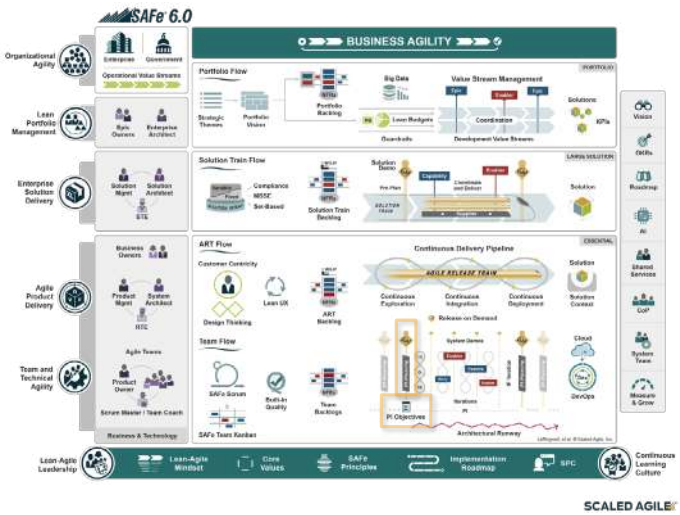
- Defined the preparation activities for the PI Planning event
- Demonstrated how to facilitate Day 1 activities for the PI Planning event
- Demonstrated how to facilitate the final PI plan development and commitment
- Demonstrated how to facilitate PI Planning across multiple locations

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Articles used in this lesson

Read these Framework articles to learn more about topics covered in this lesson

- "PI Planning" https://www.scaledagileframework.com/pi-planning/
- "Distributed PI Planning" https://www.scaledagileframework.com/distributed-pi-planning/
- "PI Objectives" https://www.scaledagileframework.com/pi-objectives/



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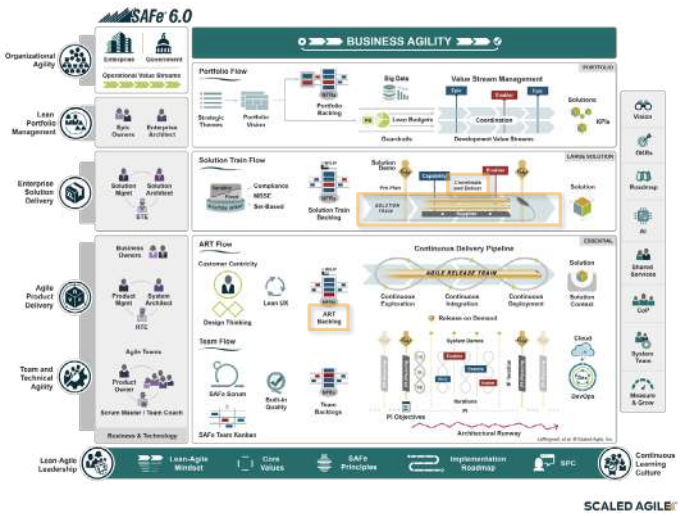
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Articles used in this lesson

Read these Framework articles to learn more about topics covered in this lesson

- "ART and Solution Train Backlogs" https://scaledagileframework.com/art-and-solution-train-backlogs/
- "Solution Train" https://scaledagileframework.com/supplier/
- "Coordinate and Deliver" https://scaledagileframework.com/coordinate-and-deliver/



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piplanning.io is now a part of SAFe®



A modern way to conduct PI Planning

With **piplanning.io**, collaborating virtually feels like being together in front of the same board. Empower RTEs to lead remote PI Planning events that ensure alignment and focus across all teams.

- **RTE Cockpit** streamlines PI Planning prep and includes a strong team breakout experience
- **Dependency Management** visualizes your requirements with easy-to-follow link options
- **Capacity Allocation** balances new Features versus technical debt with one click

Try it today at scaledagile.com/piplanning

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Continue your SAFe journey with the following resources:

Download the “SAFe PI Planning Toolkit” for resources to support preparation, coordination, and communication to guide an ART through its PI Planning event. https://bit.ly/Community-ToolkitsandTemplates	Watch this six-minute video, *Preparation for PI Planning*, to learn about three areas an ART should focus on when preparing for PI Planning. https://bit.ly/Video-PIPlanningPrep
Use the “ART Readiness Workbook” from the “SAFe PI Planning Toolkit” to create the implementation plan. The team and ART rosters help document the new organizational structure. https://bit.ly/Community-ToolkitsandTemplates	Watch the four-video playlist, *Introduction to PI Planning,* to review the PI Planning process. https://bit.ly/Playlist-PIPlanning
Use the SAFe Collaborate template, “Retrospective Plus, Minus, Delta,” to run an effective retrospective with your team. https://bit.ly/Template-PlusMinusDelta	Organize and manage ROAMing risks by using the “SAFe ROAM Board” Collaborate Template. https://bit.ly/Template-ROAMBoard

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Lesson notes

Enter your notes below. If using a digital workbook, save your PDF often so you don't lose any of your notes.

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Lesson 5

Executing a PI

SAFe® Course - Attending this course gives learners access to the SAFe Release Train Engineer exam and related preparation materials.

Lesson Topics

5.1 Using events to keep the ART on track

5.2 Executing the PI with Built-in Quality

5.3 Visualizing and measuring the flow of value

5.4 Prioritizing time to innovate and plan

5.5 Building a Continuous Delivery Pipeline with DevOps



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Learning objectives

At the end of this lesson, you should be able to:

- Identify key team and ART events
- Summarize basic Agile quality practices
- Describe the tools needed to visualize and measure flow
- Prepare and facilitate IP Iteration activities
- Summarize the Continuous Delivery Pipeline with DevOps

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5.1 Using events to keep the ART on track

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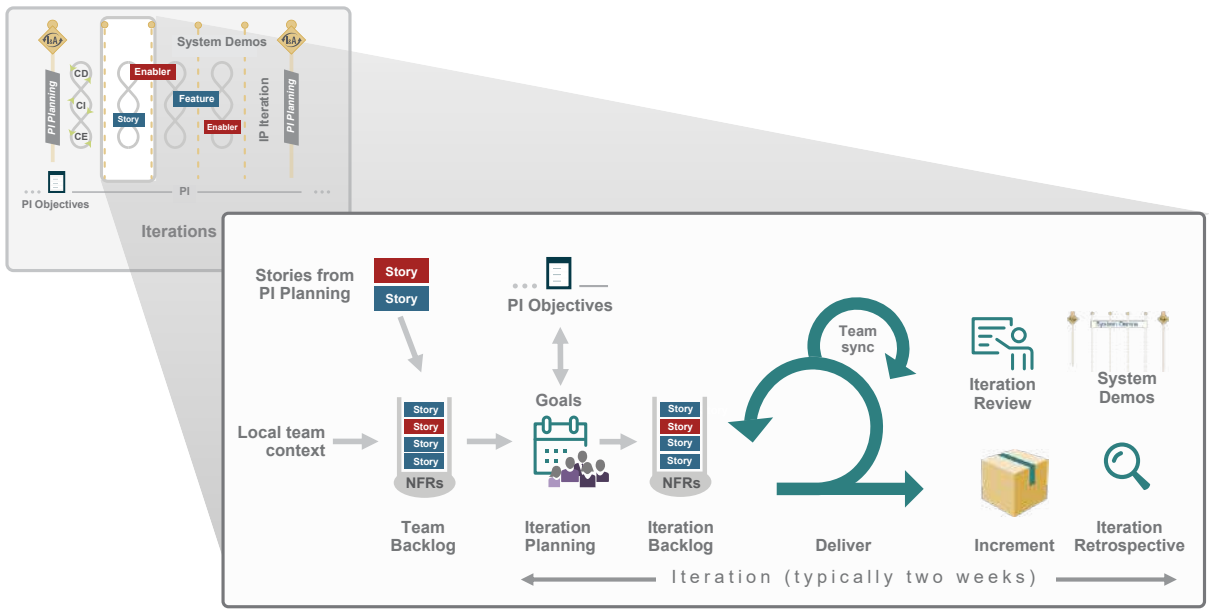
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Responsibilities of Agile Teams and ARTs



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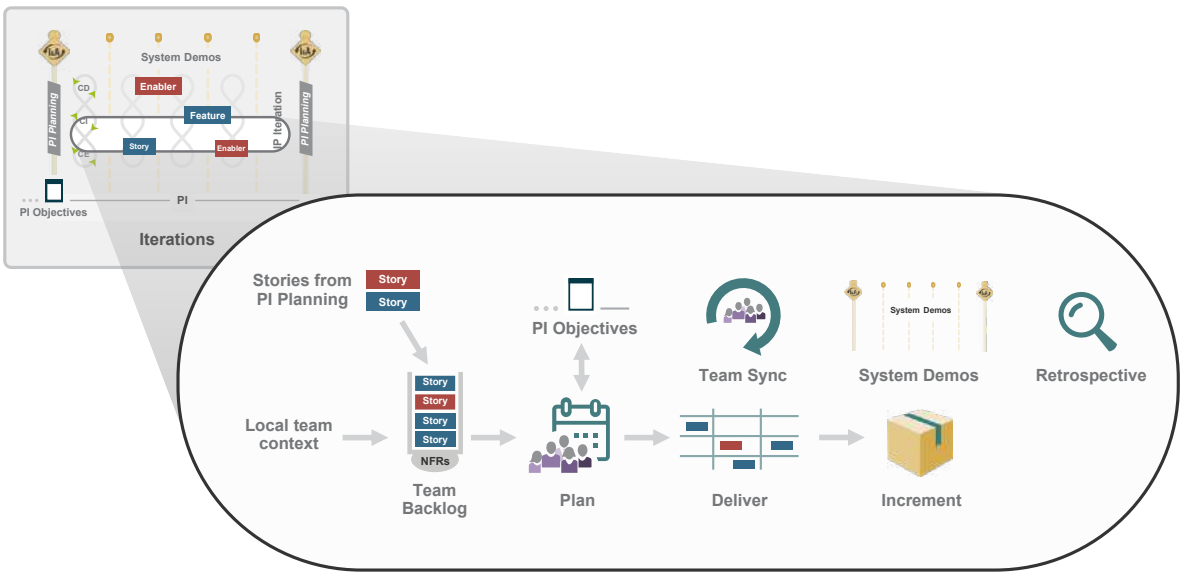
SAFe Scrum overview



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SAFe Team Kanban overview



Discussion: Selecting team methods

Prepare 7 min Share 7 min

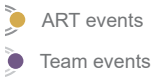
- **Step 1:** Working in your groups, consider SAFe Scrum and SAFe Team Kanban and your own context. Discuss the following:
  - Which approach is best suited for various teams on your ART?
  - What does the RTE need to consider when coaching ARTs that have teams using both approaches?
  - How can the RTE support teams combining models to work for their team needs?
- **Step 2:** Be prepared to share with the class

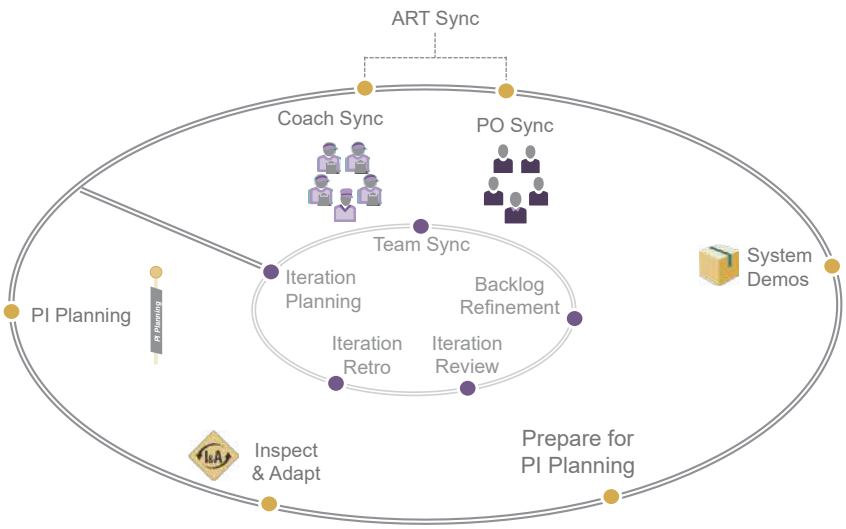
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ART execution events

ART events create a closed loop system to keep the train on the tracks.

Key PI execution events:
1. Coach Sync
2. PO sync
3. ART sync
4. System Demo

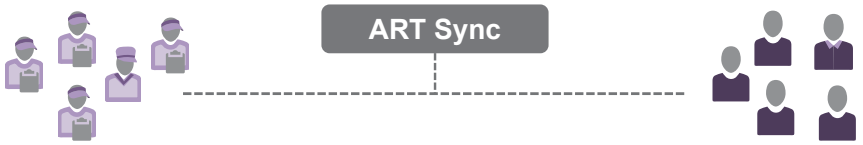




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5-9

ART Sync is used to coordinate progress



Coach Sync

- Visibility into progress and impediments
- Facilitated by the RTE
- Participants include SM/TCs, other select team members, SMEs if necessary
- Weekly or more frequently, 30–60 minutes
- Timeboxed and followed by a meet-after

PO Sync

- Visibility into progress, scope, and priority adjustments
- Facilitated by RTE or PM
- Participants include PM, POs, other stakeholders, and SMEs as necessary
- Weekly or more frequently, 30–60 minutes
- Timeboxed and followed by a meet-after

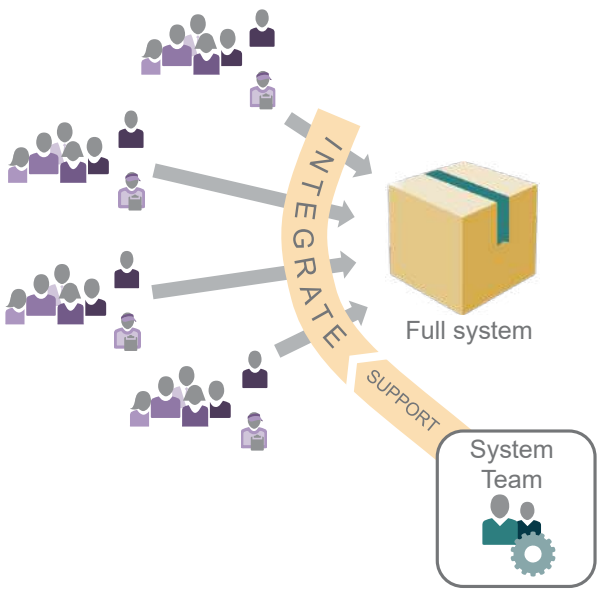
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Demo the full system increment every two weeks

- Features are functionally complete or toggled so as not to disrupt demonstrable functionality
- New Features work together and with existing functionality
- Happens after the Iteration Review (may lag by as much as one Iteration maximum)
- Demo from a staging environment that resembles production as much as possible



Activity: The RTE and the ART events

Prepare 10 min | Share 12 min

- **Step 1:** Working in your groups, prepare a three-minute presentation about an ART event facilitated by the RTE. Use the event descriptions as a guide and draw from your own experiences. Be sure to include:
  - A clear statement about the purpose of the event and why it's important for the RTE role
  - A list of facilitators and attendees, plus a suggested agenda for the event with timing and cadence
  - The key inputs and outputs of the event
  - Possible event anti-patterns, like a discussion of what would happen if this event was canceled
  - Success criteria for the RTE
  - General tips

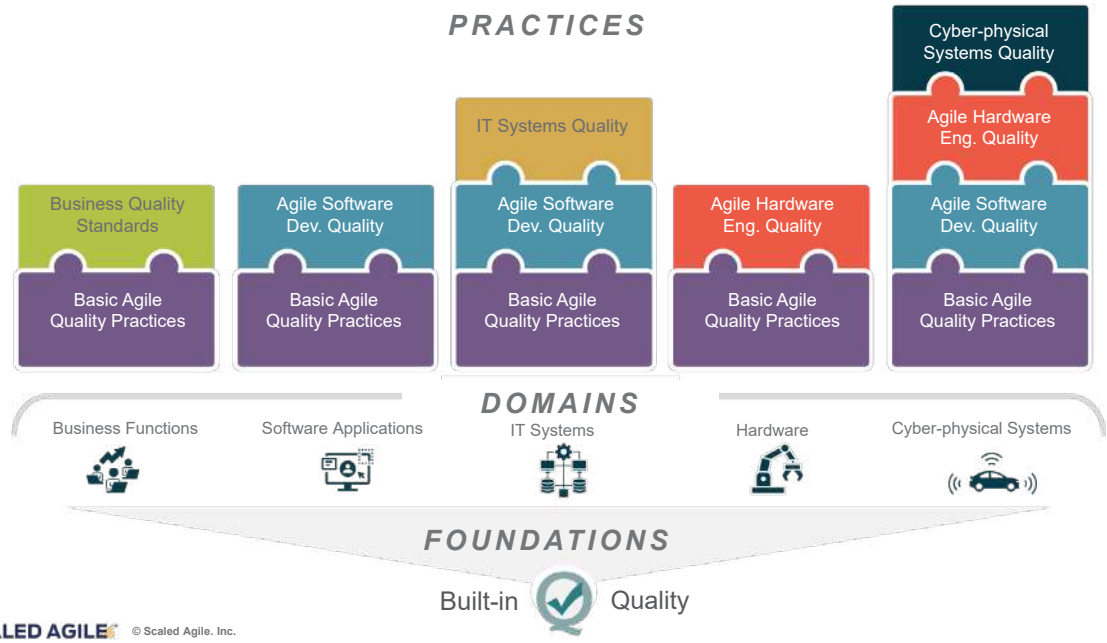
Step 2: Be prepared to share with the class.

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5.2 Executing the PI with Built-in Quality

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Built-in Quality



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Teams incorporate basic Agile quality practices

Agile quality practices apply to every team, whether business or technology.

- **Shift learning left** – Reveal problems sooner, take corrective action with minimum impact
- **Pairing and peer review** – Multiple viewpoints enhance work quality and grow knowledge
- **Collective ownership and T-shaped skills** – Reduce bottlenecks and increase flow
- **Artifact standards and definition of done** – Ensure consistent quality for each work product
- **Workflow automation** – Enable small batches and reduce errors



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5-15

Scalable definition of done

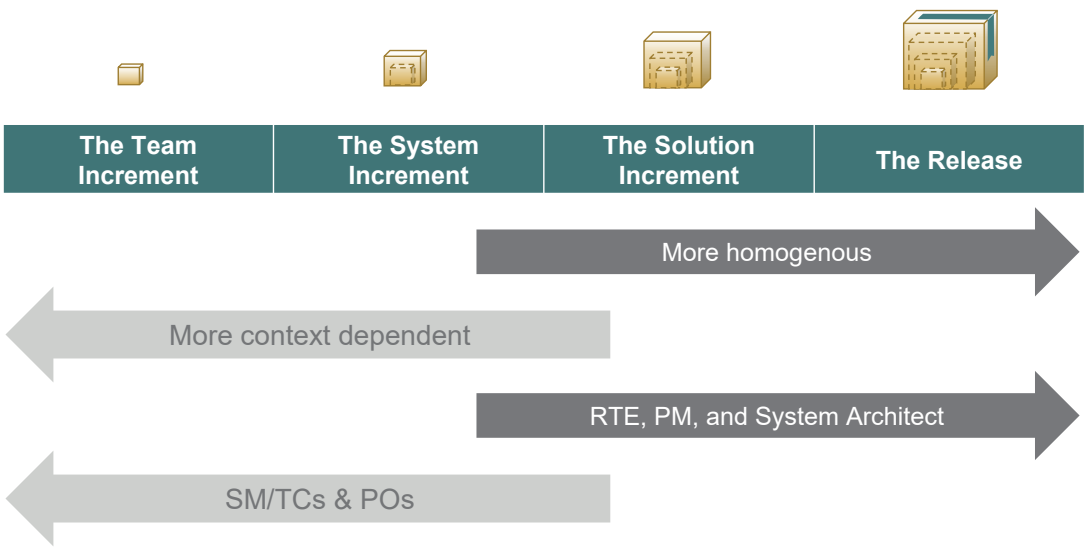


Team Increment	System Increment	Solution Increment	Release
• Stories satisfy acceptance criteria • Acceptance tests passed (automated where practical) • Unit tests • Cumulative unit tests passed • Assets under version control • Engineering standards followed • NFRs met • No must-fix defects • Stories accepted by PO	• Stories completed by all teams in the ART and integrated • Completed features meet acceptance criteria • NFRs met • No must-fix defects • Verification and validation of key scenarios • Included in build definition and deployment process • Increment demonstrated, feedback achieved • Accepted by Product Management	• Capabilities completed by all ARTs and meet acceptance criteria • Deployed/installed in the staging environment • NFRs met • System end-to-end integration, verification and validation done • No must-fix defects • Included in build definition and deployment/transition process • Documentation updated • Solution demonstrated, feedback achieved • Accepted by Solution Management	• All capabilities done and meet acceptance criteria • End-to-end integration and solutions V&V done • Regression testing done • NFRs met • No must-fix defects • Release documentation complete • All standards met • Approved by Solution and Release Management

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ART definition of done



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Discussion: Built-in Quality

Prepare 7 min

Share 3 min

- **Step 1:** Working in your groups, discuss how the RTE should coach the ART toward basic Agile quality practices:
  - What events can the RTE leverage?
  - What roles can the RTE collaborate with?
  - What tools can the RTE use?
- **Step 2:** Be prepared to share with the class

Basic Agile Quality Practices

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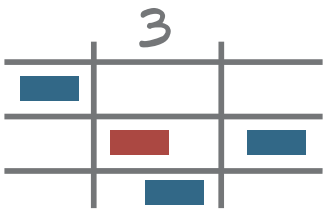
5.3 Visualizing and measuring the flow of value

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Visualize the workflow using an ART Kanban

Working with Kanban boards is lightweight and disciplined.

- All work is visualized, and the progress of individual items is continuously tracked
- Kanban board contains defined states that work moves through
- Stakeholders agree on work-in-process (WIP) limits for selected states and adjust as necessary to improve flow
- Kanban board has explicit policies detailing how work is managed in each state
- Flow through the Kanban board is measured, which informs future improvements



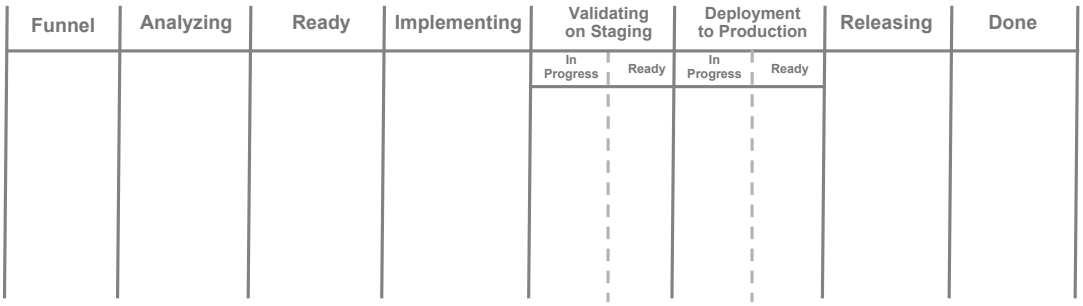
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The Kanban board columns

The columns define the individual states in the identified workflow that the Kanban board visualizes.

- Each column represents one state in the defined workflow
- Columns can be split into work area and buffer if more detailed visualization is needed

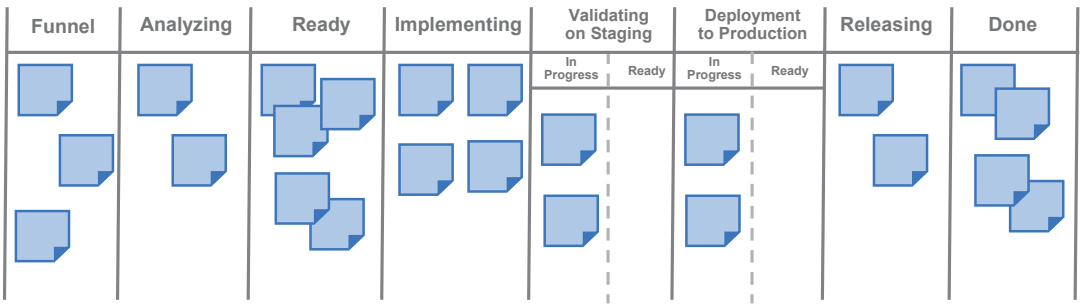


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The Kanban board cards

The Kanban board visualizes the flow of items from left to right. Each item on the board, represented as a card, is a unit of value that can be created and delivered on the board.

- Each card can be individually created, validated, and delivered
- Similarly-sized pieces of work enable smoother flow through the board



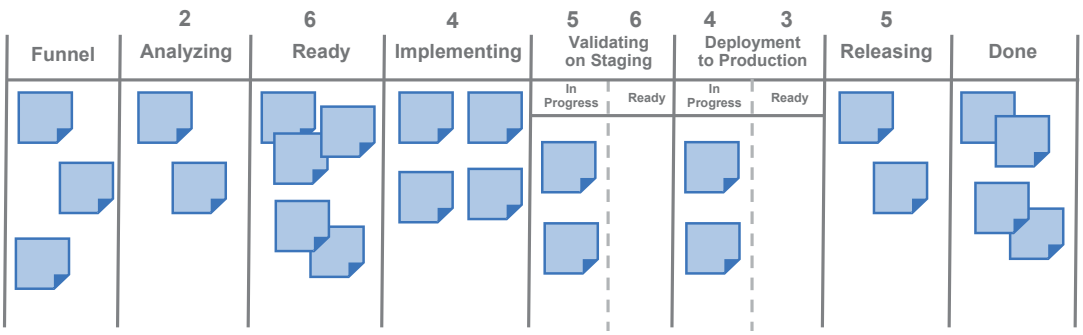
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The Kanban board WIP limits

The agreed-upon WIP limits are displayed above each column. These are the limits for how many cards each column can contain at any time.

- WIP limits adjust demand to capacity at bottlenecks and help flow through the board.
- WIP limits are initially a hypothesis. They will evolve over time as you learn.



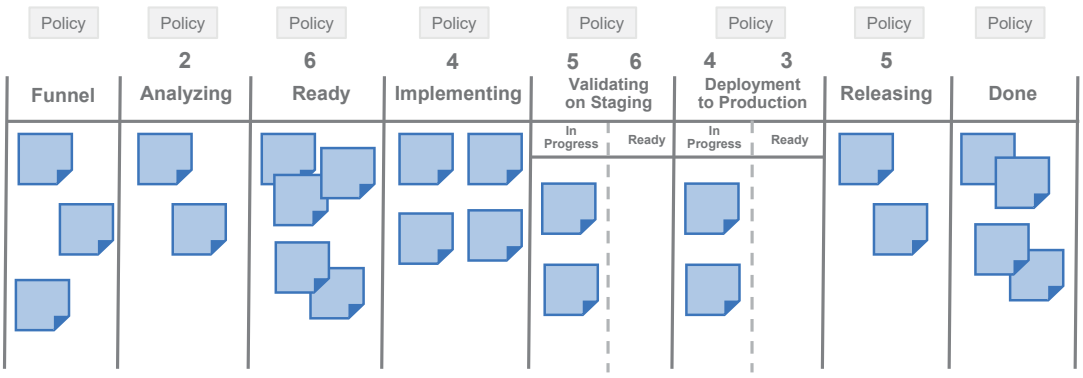
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The Kanban board explicit policies

The agreed-upon exit policies for each step in the workflow are displayed above each state on the board.

- These are the things that need to be done for the card to move to the next column
- The visualization of these policies encourages collaboration and removes uncertainty

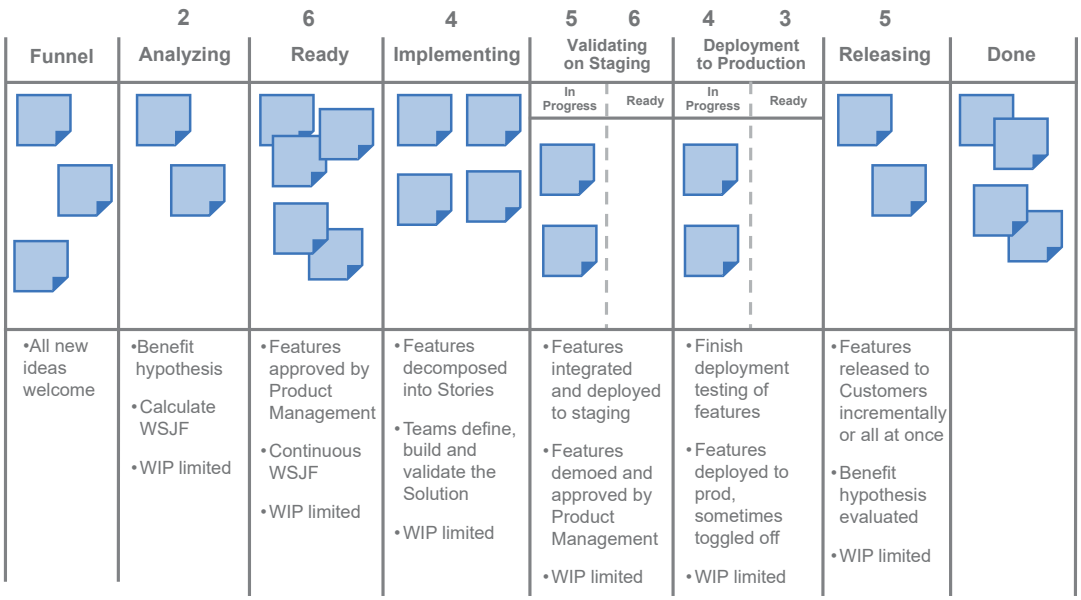


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Example ART Kanban Board



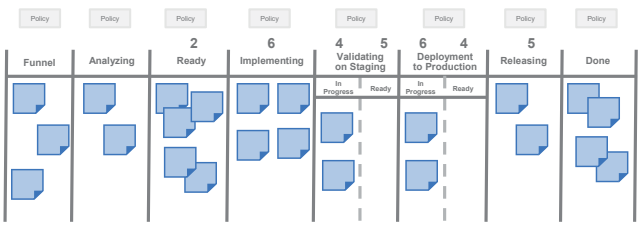
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Activity: Build your own ART Kanban Board

Prepare 15 min

Share 10 min

- **Step 1:** In your group, select one person's context.
- **Step 2:** Together, build an ART Kanban Board using the tools described in this lesson topic. Be sure the group:
  - Understands the actual workflow of Features
  - Decides how to map it to the board (columns, policies, and so on)
  - Determines initial WIP limits
- **Step 3:** Be prepared to share with the class.



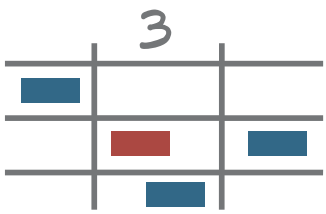
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The ART Kanban board

An ART Kanban board is a powerful tool to visualize the flow of Features and Enablers through the ART.

- Helps visualize PI Planning readiness and PI execution progress
- Useful tool for Coach Sync, PO Sync, ART Sync, and problem-solving meetings
- Accessible by RTE and stakeholders for ART Flow measurements
- Makes process policies and decisions visible to the entire ART



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Measuring the flow of value

How efficient is the organization at delivering value to the Customer?

- Flow Metrics provide insights into the efficiency of the delivery pipeline
- Highlights opportunities for removing delays and increasing our speed and agility

Metric
Flow Distribution
Flow Velocity
Flow Time
Flow Load
Flow Efficiency
Flow Predictability

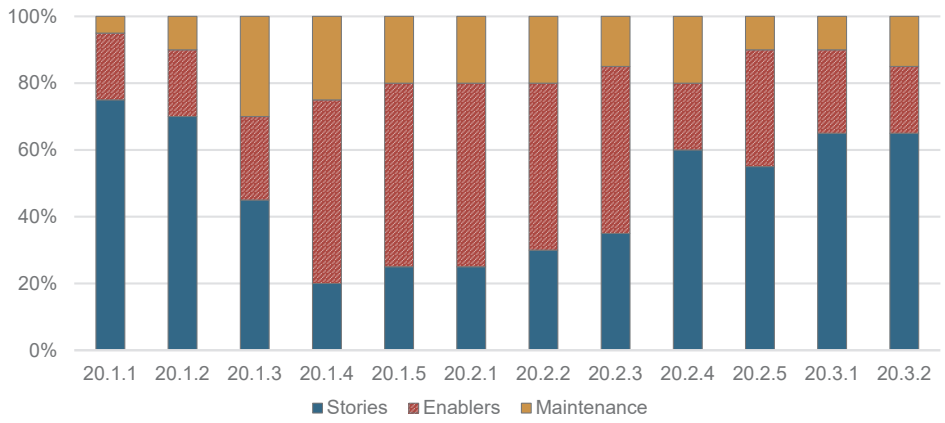
Reference: Adapted from Kersten (2018)

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Flow distribution

Flow distribution measures the proportion of work items by type in a system.

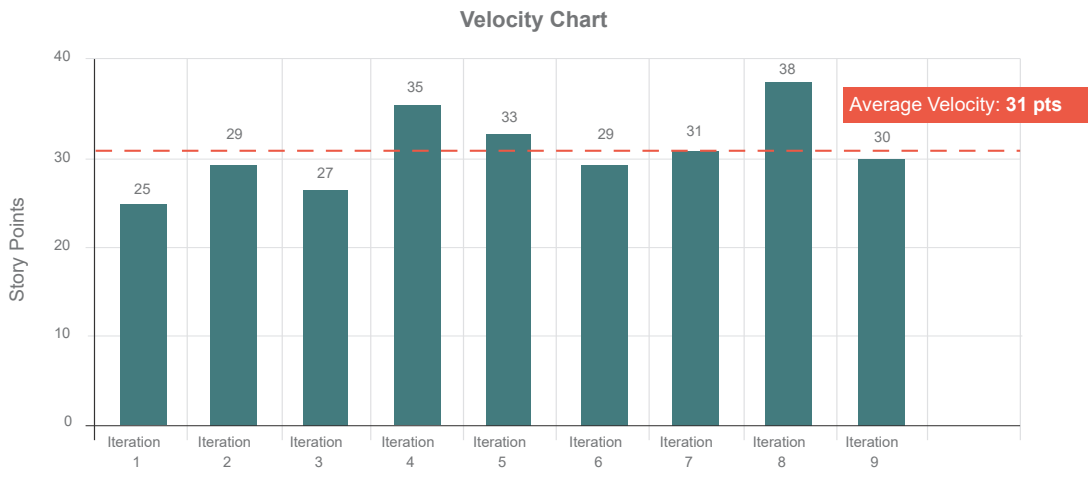


Visualizing flow distribution over time

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Flow velocity

Flow velocity measures the number of completed work items over a time period.



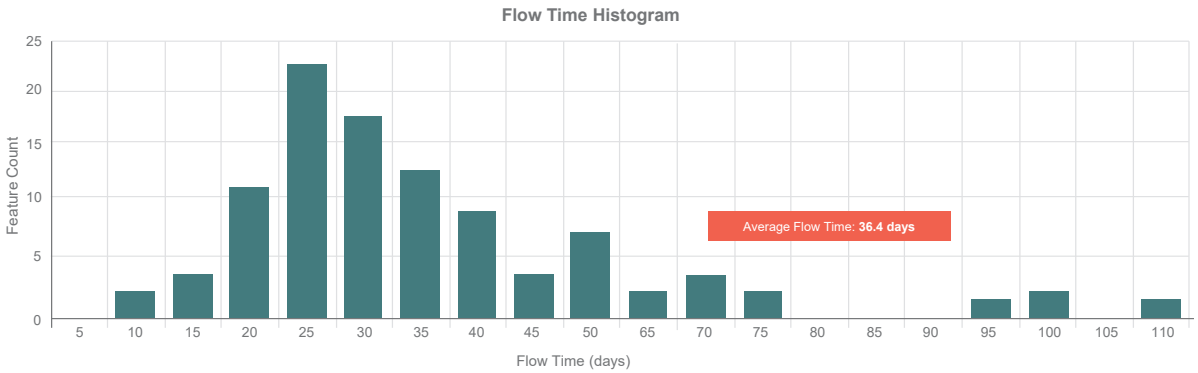
Understanding an Agile Team's flow velocity in Story points per Iteration

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Flow time

Flow time measures the time elapsed from start to completion for a given work item.

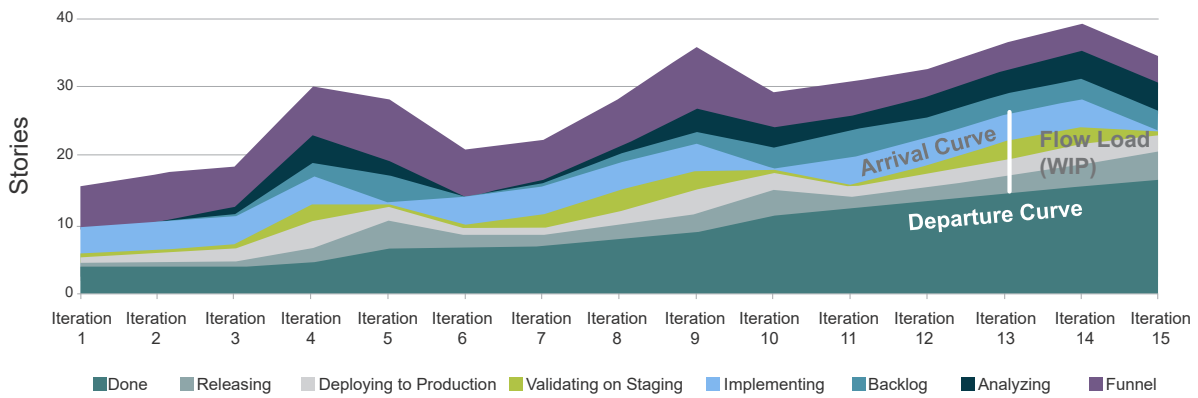


Measuring Feature flow time with a histogram

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Flow load

Flow load measures the number of work items currently in progress (active or waiting).



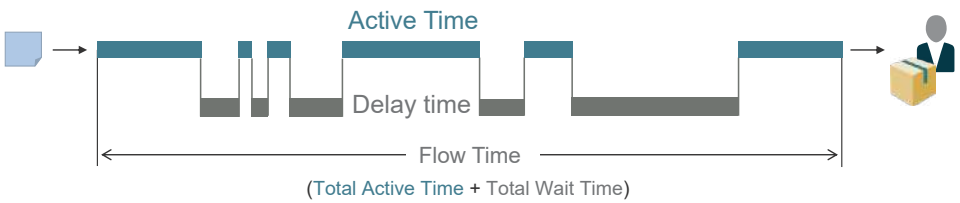
Visualizing flow load with a cumulative flow diagram

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Flow efficiency

Flow efficiency measures the ratio of the total time spent in value-added work activities divided by the total flow time.

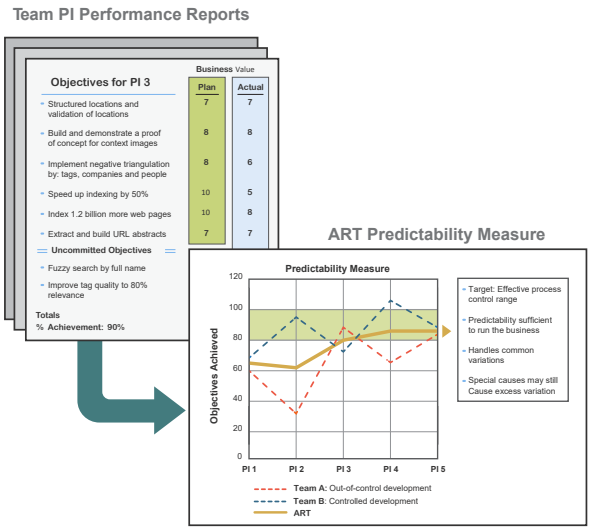


$$\text{Flow Efficiency} = \frac{\text{Total Active Time}}{\text{Flow Time}}$$

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Flow predictability

Flow predictability measures how consistently teams, ARTs, and portfolios are able to meet their commitments.



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Discussion: Measuring the flow of value

Prepare 7 min

Share 5 min

- **Step 1:** In your groups, brainstorm the Metrics you currently use to measure success on your ART. Identify each Metric and what it intends to measure.
- **Step 2:** Discuss how you would integrate Flow Metrics into your ART. Consider the following:
  - Is your ART Kanban optimized to measure Flow?
  - If you're already using Flow Metrics, how do you apply them in your role?
- **Step 3:** Be prepared to share with the class.

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5.4 Prioritizing time to innovate and plan

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Continuous Learning Culture

- Everyone in the organization learns and grows together
- Exploration and creativity are part of the organization's DNA
- Continuously improving Solutions, services, and processes is everyone's responsibility

Learning Organization



Innovation Culture



Relentless Improvement



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Innovation and Planning (IP) Iteration

"Provide sufficient capacity margin to enable cadence."

— Don Reinertsen, *Principles of Product Development Flow*

Definitions
Innovation: Opportunity for innovation spikes, hackathons, and infrastructure improvements
Planning: Provides for cadence-based planning and is an estimating guard band for cadence-based delivery

Common anti-patterns
Planning work for the IP Iteration in PI Planning
Leaving testing or bug fixing to the IP Iteration
Leaving integration of the whole system to the IP Iteration

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Innovation culture

Leaders must create an environment that supports creative thinking, curiosity, and challenging the status quo. Keys to building an innovation culture are:

- Innovative people
- Time and space for innovation
- “Go see” activities
- Experimentation and feedback
- Pivoting without mercy or guilt
- Innovation riptides



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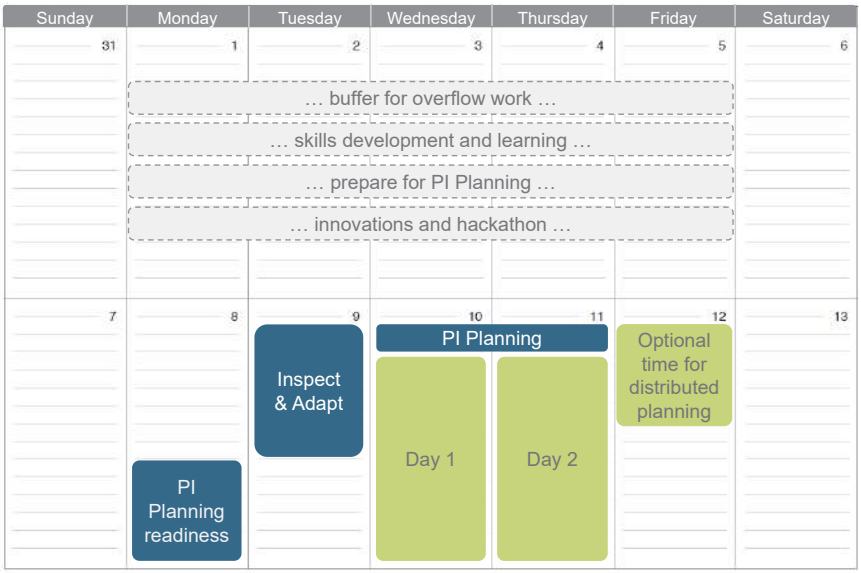
Enable continuous learning

- The IP Iteration is the perfect time to offer learning opportunities that build common skills across the ART
- The move to Lean-Agile requires many new skills, including:
  - Feature/Story writing
  - Automated testing
  - Collective ownership
  - Understanding and applying flow
  - Team building
  - ART role mastery

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Example IP Iteration calendar



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Activity: The IP Iteration

Prepare 15 min

Share 10 min

- **Step 1:** Working in your groups, use the example calendar and your own context to identify a list of activities for the IP Iteration
- **Step 2:** Working in your groups, discuss and capture answers to the following questions:
  - How would you rank the different activities in priority, with 1 being most important and 10 being least important?
  - What preparation does the RTE need to do before the IP Iteration?
  - What could happen if there was no IP Iteration?
- **Step 3:** Be prepared to share with the class

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Without the IP Iteration...

- Lack of delivery capacity buffer impacts predictability
- Little innovation; the tyranny of the urgent
- Technical debt grows uncontrollably
- People burn out
- No time for teams to plan, demo, or improve together



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5.5 Building a Continuous Delivery Pipeline with DevOps

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Video: What is DevOps?

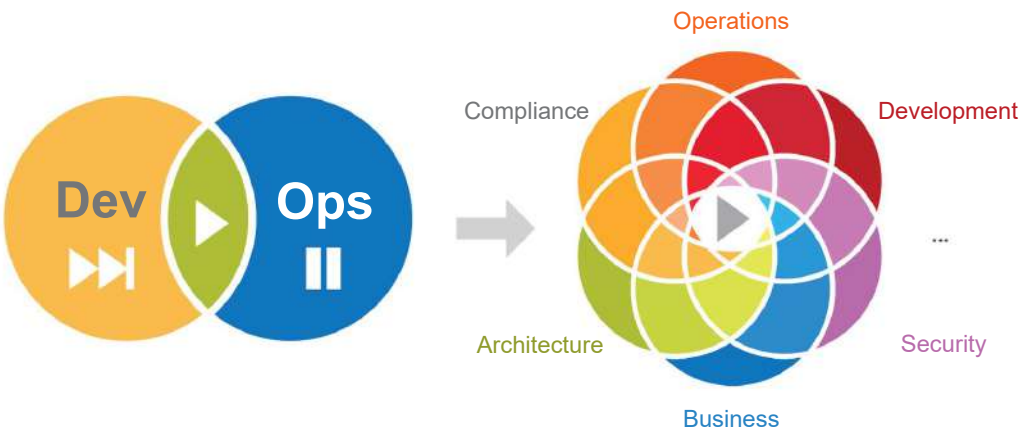
Duration 1 min



https://bit.ly/Video-WhatisDevOps

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Maximize speed and stability



Optimized for development speed

Optimized for stability

Working together for speed and stability

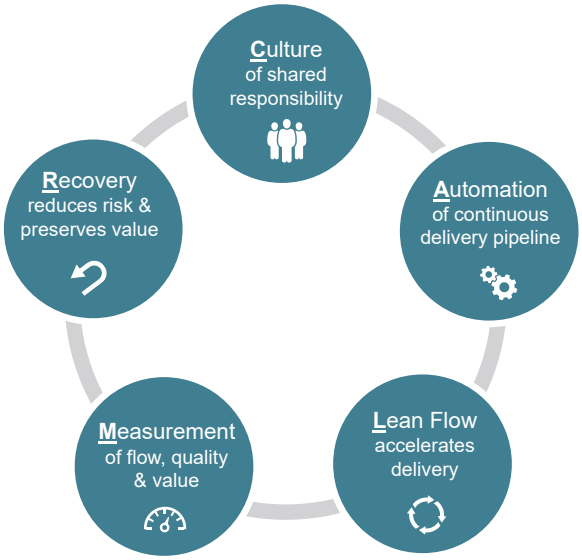
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A CALMR approach to DevOps

C	**Culture**	Establish a culture of shared responsibility for development, deployment, and operations.
A	**Automation**	Automate the Continuous Delivery Pipeline (CDP).
L	**Lean flow**	Keep batch sizes small, limit WIP, and provide extreme visibility.
M	**Measurement**	Measure the flow through the pipeline. Implement full-stack telemetry.
R	**Recovery**	Architect and enable low-risk releases. Establish fast recovery, fast reversion, and fast fix-forward.

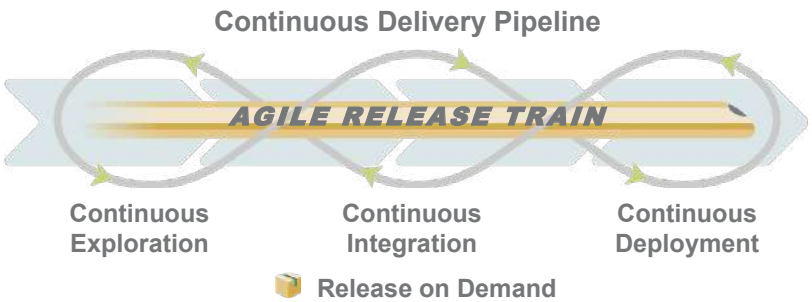


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5-47

Building the Continuous Delivery Pipeline with DevOps

- The Continuous Delivery Pipeline (CDP) represents the workflows, activities, and automation needed to deliver new functionality more frequently
- Each ART builds and maintains (or shares) a pipeline
- Organizations map their current pipeline into this new structure, remove delays, and improve the efficiency of each step

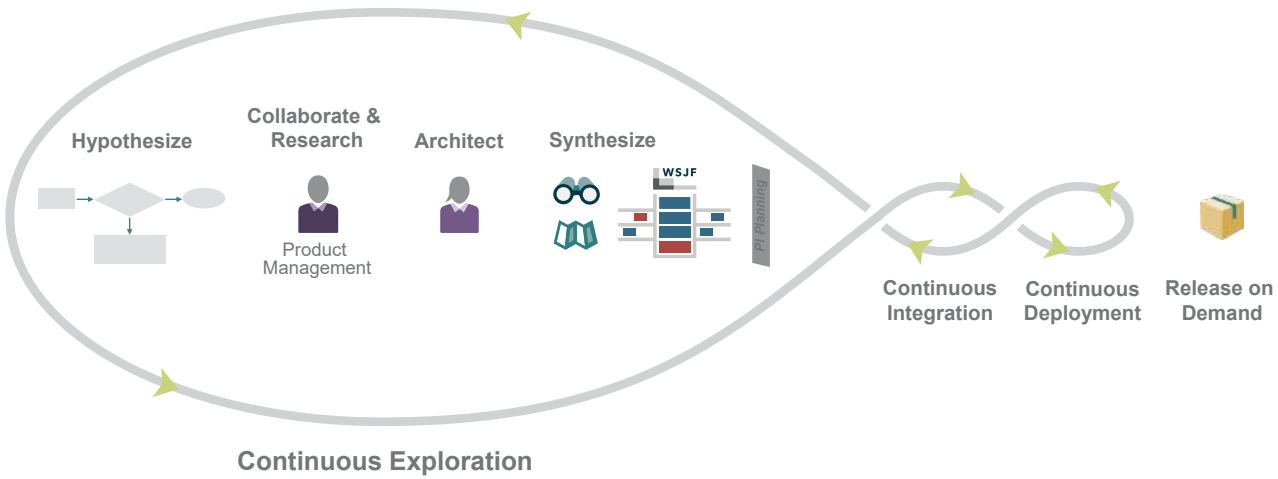


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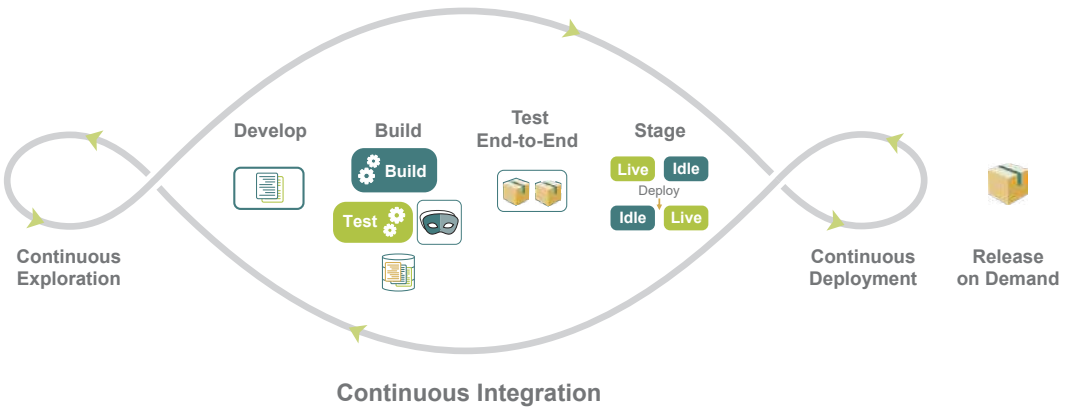
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Continuous Exploration: Understand Customer needs



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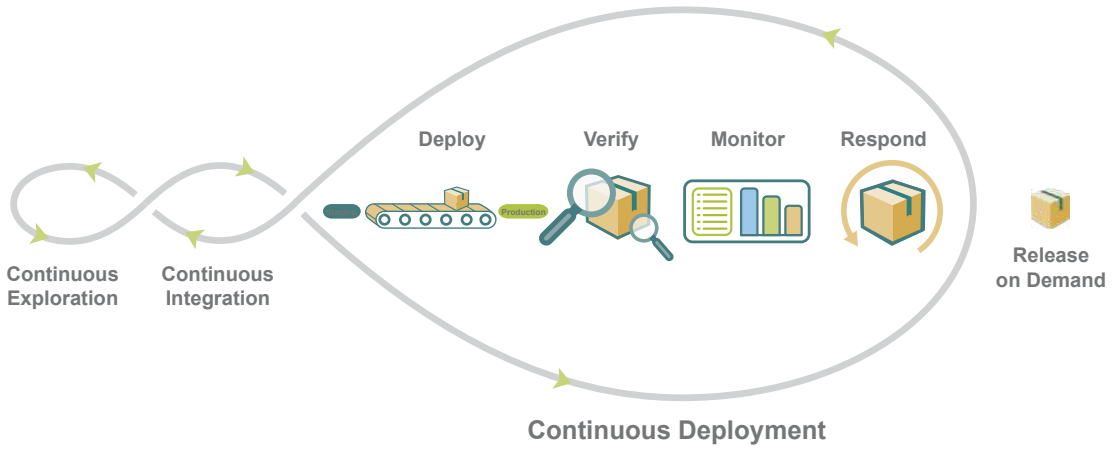
Continuous Integration: A critical technical practice of the ART



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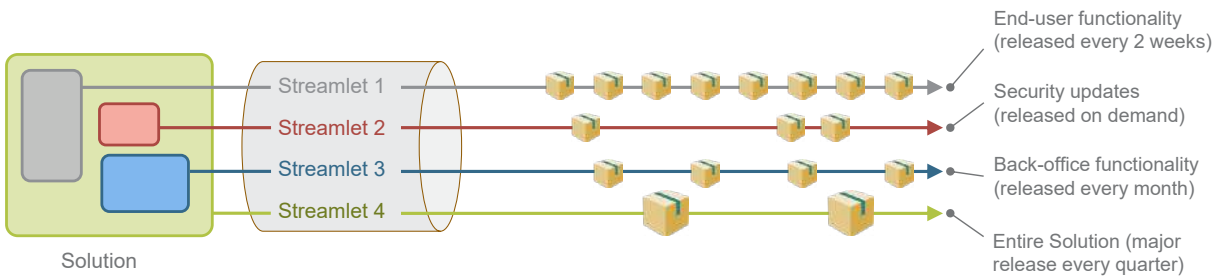
Continuous Deployment: Getting to production early



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Decouple release elements from the total Solution



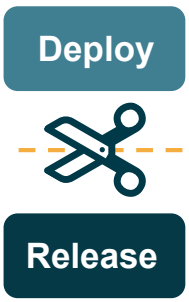
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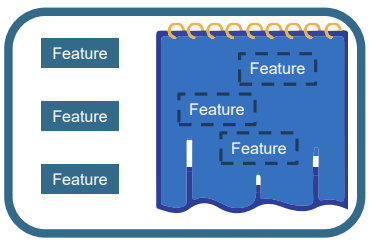
5-52

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Separate deploy from release

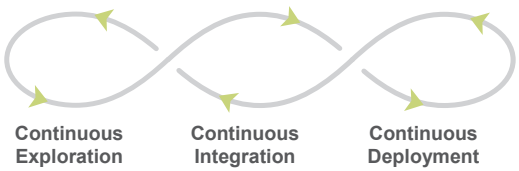
- Separate deployment from release
- Hide new functionality under Feature toggles
- Enable testing background and foreground processes in the actual production environment before exposing new functionality to users
- Consider the timing of the release a business decision

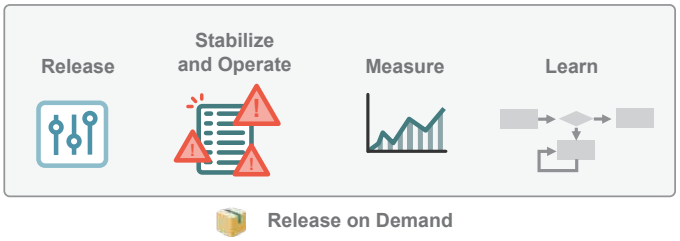




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Release on Demand: Making value available when it's needed





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Activity: Create a two-minute pitch

Prepare: 10 min
Share: 5 min

- **Step 1:** Create a three-minute pitch to the Business Owners and Product Management about the need to build DevOps capabilities. Be sure to include the following:
  - A clear statement about what is likely to happen if the practice is not implemented
  - A statement about how the RTE can support this practice
  - Metrics to support your pitch
- **Step 2:** Be ready to deliver your pitch to the class.

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Action Plan: RTE Action Plan

Duration: 5 min

- **Step 1:** Locate the RTE Action Plan section in your workbooks
- **Step 2:** Add more tools and techniques to the Action Plan by reflecting on the following:
  - What tools and practices will you implement first to enable flow in your ART?
- **Step 3:** Share one of your insights with the class



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Action Plan

Executing a PI

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Lesson review

In this lesson you:

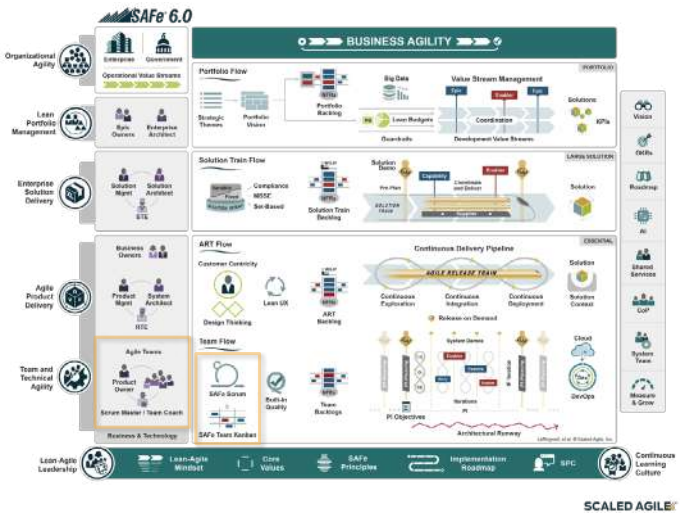
- Identified key team and ART events
- Summarized basic Agile quality practices
- Described the tools needed to visualize and measure flow
- Prepared to facilitate IP Iteration activities
- Summarized the Continuous Delivery Pipeline with DevOps

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Articles used in this lesson

Read these Framework articles to learn more about topics covered in this lesson

- "Agile Teams" https://www.scaledagileframework.com/agile-teams/
- "SAFe Team Kanban" https://www.scaledagileframework.com/safe-team-kanban/
- "SAFe Scrum" https://www.scaledagileframework.com/safe-scrum



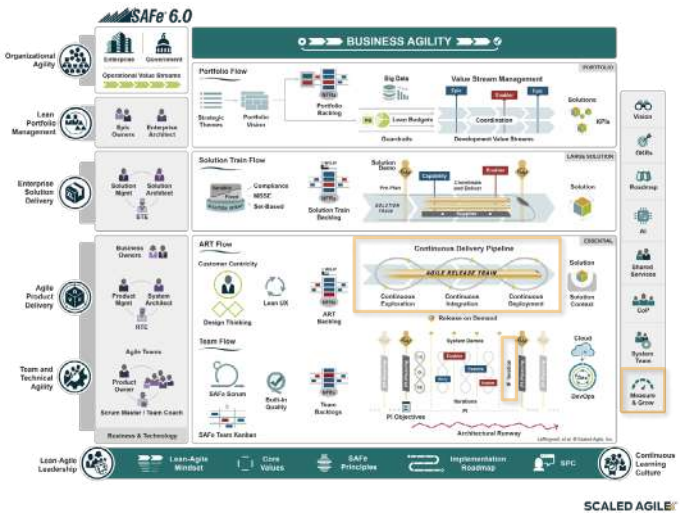
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Articles used in this lesson

Read these Framework articles to learn more about topics covered in this lesson

- "Measure and Grow" https://www.scaledagileframework.com/measure-and-grow/
- "Innovation and Planning Iteration" https://www.scaledagileframework.com/innovation-and-planning-iteration/
- "Continuous Delivery Pipeline" https://www.scaledagileframework.com/continuous-delivery-pipeline/

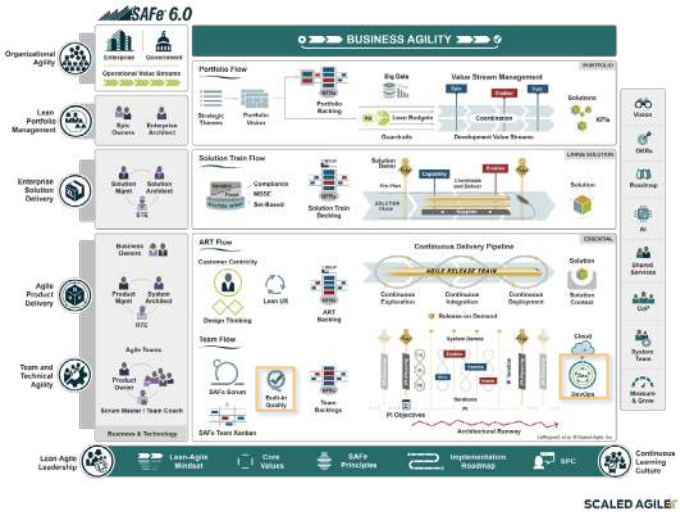


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Read these Framework articles to learn more about topics covered in this lesson

- "DevOps" https://www.scaledagileframework.com/devops/
- "Built-in Quality" https://www.scaledagileframework.com/built-in-quality/
- "Applying Kanban in SAFe" https://www.scaledagileframework.com/advanced-topic-applying-kanban-in-safe/



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Continue your SAFe journey with the following resources:

Download “The Facilitator’s Guide to SAFe: Team Sync” for support facilitating your Team Sync and ideas for overcoming potential issues you might encounter. https://bit.ly/Studio-FGTeamSync	Download “The Facilitator’s Guide to SAFe: Iteration Planning” to learn more about the purpose of Iteration Planning and for preparation checklists to get ready to facilitate your Iteration Planning event. https://bit.ly/Community-FGIterationPlanning
Download “The Facilitator’s Guide to SAFe: Iteration Review and Demo” for guidance in preparing to facilitate Iteration Review and Demo events. https://bit.ly/Community-FGIterationReview	Download “The Facilitator’s Guide to SAFe: Iteration Retrospectives” for guidance in preparing to facilitate Iteration Retrospectives. https://bit.ly/Community-FGRetrospective
Download “The Facilitator’s Guide to SAFe: Backlog Refinement” for support with agenda setting, preparation checklists, and tips and tricks for facilitating great backlog refinement events. https://bit.ly/Community-FGBacklogRefinement	Download “The Facilitator’s Guide to SAFe: Coach Sync” for a suggested agenda, talking points, and ideas to keep the event engaging. https://bit.ly/Studio-FGCoachSync

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Continue your SAFe journey with the following resources:

Download “The Facilitator’s Guide to SAFe: PO Sync” for guidance on setting the agenda and coaching for participation. https://bit.ly/Community-FGPOSync	Download “The Facilitator’s Guide to SAFe: ART Sync” to learn more about the purpose of the event and for checklists on how to combine PO Sync and Coach Sync. https://bit.ly/Community-FGARTSync
Download “The Facilitator’s Guide to SAFe: System Demo” for guidance in preparing to facilitate System Demo. https://bit.ly/Community-FGSystemDemo	Access the collaboration template “Customizing your Kanban” to build a Kanban that will work for your team. https://bit.ly/Template-CustomizeKanban
Watch this three-part video series, *Kanban for Teams*, to walk through the processes of setting up, implementing, and incrementally improving a team Kanban. https://bit.ly/Playlist-KanbanTeams	Watch this nine-minute video, *The Lost Art of the Demo: Tips from the Field for Running Your Most Important Meeting*, to explore how to make your System Demos really matter to the stakeholders who attend it. https://bit.ly/Video-LostArtofDemo

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References

Kersten, Mik. *Project to Product: How to Survive and Thrive in the Age of Digital Disruption with the Flow Framework*. IT Revolution: Portland, 2018.

Reinertsen, Donald G. *The Principles of Product Development Flow: Second Generation of Lean Product Development*. Redondo Beach: Celeritas 2009. 178.

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Lesson notes

Enter your notes below. If using a digital workbook, save your PDF often so you don't lose any of your notes.

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Lesson 6

Fostering Relentless Improvement

SAFe® Course - Attending this course gives learners access to the SAFe Release Train Engineer exam and related preparation materials.

Lesson Topics

6.1 Explore the three components of an Inspect & Adapt event

6.2 Measure and Grow the ART

6.3 Systems Thinking and Value Stream Mapping

6.4 Optimizing flow



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Learning objectives

At the end of this lesson, you should be able to:

- Define the three components of an Inspect & Adapt event
- Describe how to Measure and Grow the ART
- Interpret the concepts of Systems Thinking and Value Stream Mapping
- Summarize practices to optimize flow

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Relentless Improvement in SAFe

A commitment to relentless improvement is foundational to the success of SAFe.

- Relentless Improvement is one of the four SAFe Core Values
- Relentless Improvement is a dimension of the Continuous Learning Culture core competency



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6.1 Explore the three components of an Inspect & Adapt event

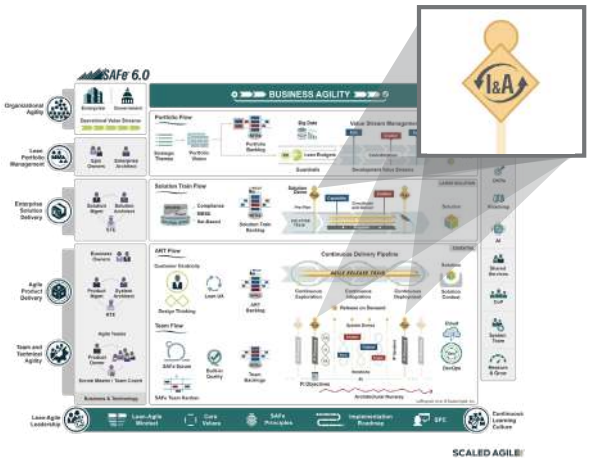
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Improving results with the Inspect & Adapt event

Three parts of Inspect & Adapt (I&A):

1. **The PI System Demo**
2. **Quantitative and Qualitative Measurement**
3. **Problem-Solving Workshop**

- **Timebox:** 3 – 4 hours per PI
- **Attendees:** Teams and stakeholders



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PI System Demo

At the end of the PI, teams demonstrate the current state of the Solution to the appropriate stakeholders.

- Often led by Product Management, POs, and the System Team
- Attended by Business Owners, ART stakeholders, Product Management, RTE, SM/TCs, and Agile Teams



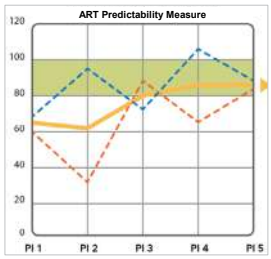


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6-7

Quantitative and qualitative measurements

- The teams review Metrics they have agreed to collect
- Facilitated by the RTE
- Timebox: 45 – 60 minutes



Objectives for PI 3

	Business Value Plan	Business Value Actual
- Structured locations and validation of locations	7	7
- Build and demonstrate a proof of concept for context images	8	8
- Implement negative triangulation by: tags, companies, and people	8	6
- Speed up indexing by 50%	10	5
- Index 1.2 billion more web pages	10	8
- Extract and build URL abstracts	7	7
Uncommitted Objectives		
	7	0
	4	4
	50	45

Functionality	PI 1	PI 2	PI 3
ART velocity			
Predictability measure			
# Features planned			
# Features accepted			
# Enablers planned			
# Enablers accepted			
# Stories planned			
# Stories accepted			
Quality			
Unit test coverage %			
Defects			
Total tests			
% automated			
# NFR tests			

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6-8

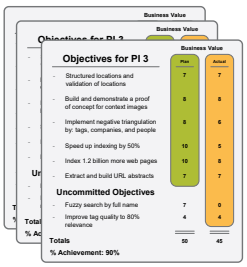
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Assessing team performance

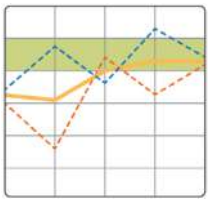
During the IP Iteration, the PI Objectives for all teams are assigned an actual business value from 1 to 10.

- The RTE and SM/TCs facilitate the business value assignment events between the teams and the Business Owners
- Business Owners evaluate each PI Objective against the stated objectives, including timeliness, content, and quality against the planned value
- The RTE aggregates and averages the scores across all objectives for the ART

Team PI Performance Report



ART Predictability Measure



Team PI performance report

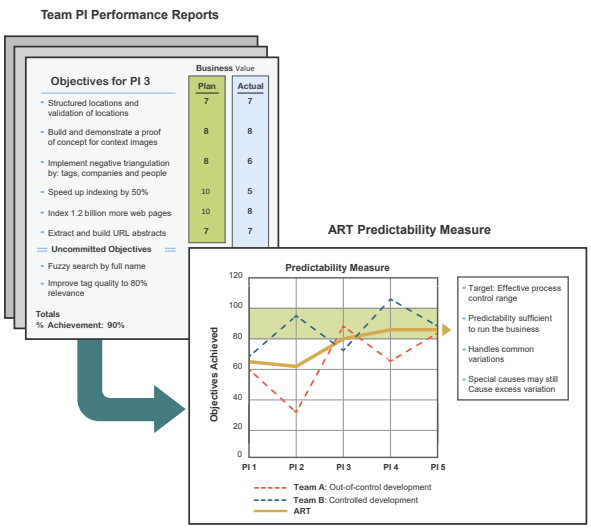
- The planned total does not include uncommitted objectives
- Actual total includes uncommitted objectives
- Percent achievement equals actual total divided by the planned total
- A team can achieve greater than 100% (as a result of achieved uncommitted objectives)
- Effort required for uncommitted objectives is included in the load (this is not extra work the team does on weekends)
- Individual team totals are rolled up into the ART Predictability Measure

Objectives for PI 3	Business Value: Plan	Business Value: Actual
- Structured locations and validation of locations	7	7
- Build and demonstrate a proof of concept for context images	8	8
- Implement negative triangulation by: tags, companies, and people	8	6
- Speed up indexing by 50%	10	5
- Index 1.2 billion more web pages	10	8
- Extract and build URL abstracts	7	7
Uncommitted Objectives		
- Fuzzy search by full name	7	0
- Improve tag quality to 80% relevance	4	4
Totals	50	45
% Achievement: 90%		

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ART Performance Metrics

- During the I&A, the RTE presents and discusses the ART Predictability Measure as well as any other agreed-upon ART Metrics being collected.
- Discussions should always focus on trends in the Metrics rather than static data snapshots
- Other example Metrics:
  - Deployments and releases per PI
  - Average cycle time
  - PI Feature throughput
  - Enabler throughput
  - Flow Metrics



Activity: Preparing for the PI System Demo and quantitative metrics

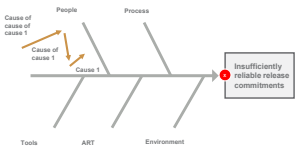
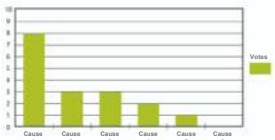
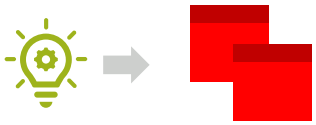
Prepare 7 min | Share 3 min

- **Step 1:** Working in your groups, discuss and capture the answers to the following questions:
  - What preparations does the RTE need to do before the I&A for the PI System Demo?
  - What preparations does the RTE need to do before the I&A for the quantitative and qualitative measurements?
  - Who can assist the RTE with these preparations?
  - What can the RTE do to ensure key ART stakeholders attend the I&A?
- **Step 2:** Be prepared to share with the class

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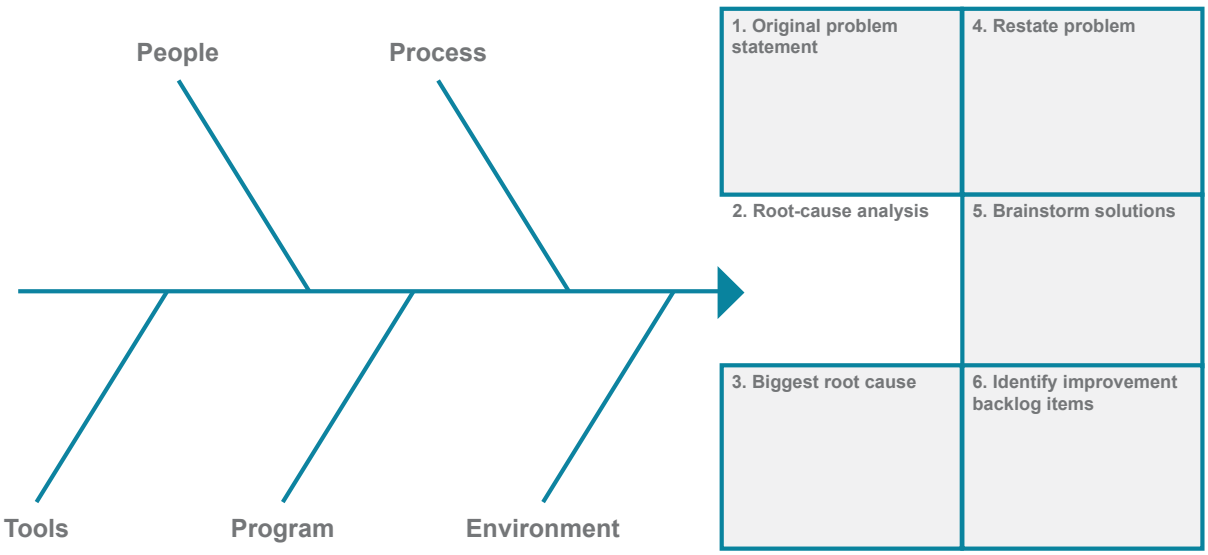
The Problem-Solving Workshop

After a retrospective, teams use root cause analysis to address the larger impediments that are limiting velocity.

Agree on the problem to solve	Apply root cause analysis and five whys	Identify the biggest root cause using Pareto analysis
X Insufficiently reliable release commitments		
Restate the new problem for the biggest root cause	**Brainstorm solutions**	**Identify improvement backlog items**
X Insufficient Architectural Runway		

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The root-cause analysis board



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Agree on the problem to solve

Clearly stating the problem is key to problem identification and correction.

- You must define the problem or situation so that everyone involved in the workshop understands it
- A clearly defined problem focuses your investigation efforts and saves time
- A problem that is not well-defined may result in failure to reach the proper countermeasure



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Anatomy of a well-defined problem

Think about the *What, When, Where,* and *Impact:*

What

When

We discovered three significant design problems in the October deployment of the new EMV vehicles at the Thrills Amusement Park.

Impact

Where

The design flaws caused us to recall the vehicles and invest three months in materials, redesign, and testing. We delivered late, paid substantial penalties, and lost credibility with the customer.

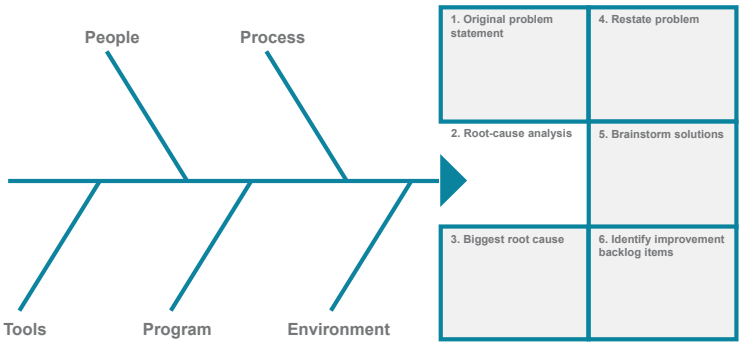
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Activity: Agree on the problem

Duration: 10 min

- **Step 1:** Working in your groups, build this problem-solving board
- **Step 2:** Choose one person's context to define the problem and create a clear problem statement
- **Step 3:** Capture the statement in the *Original problem statement* section



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Finding the root cause: the 5 Whys

"By repeating why five times, the nature of the problem, as well as its solution, becomes clear."
—Taiichi Ohno, *Toyota Production System*

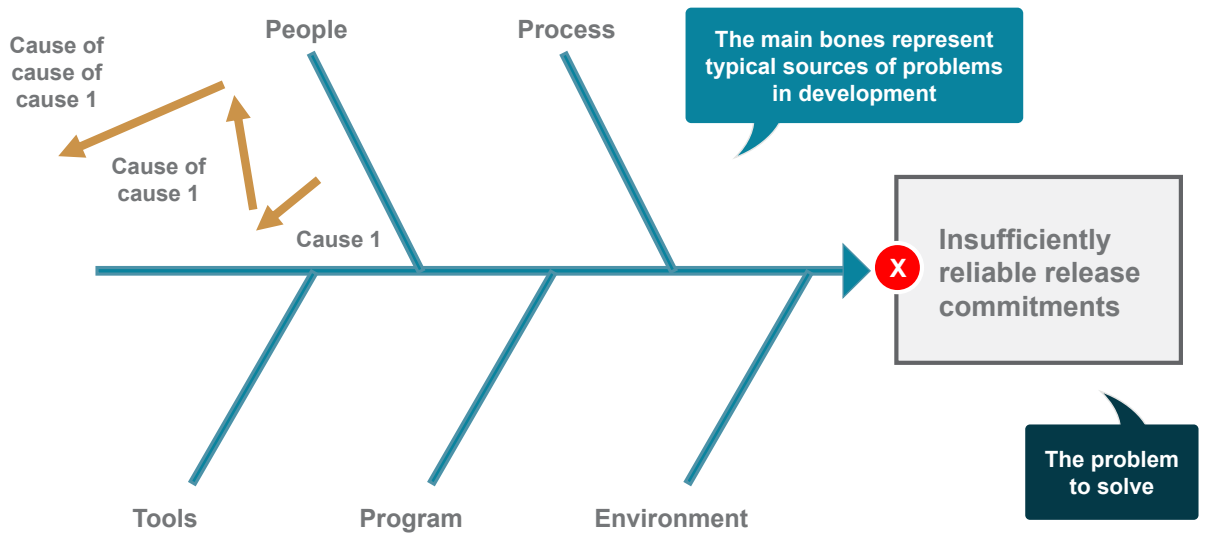
- The 5 Whys is a proven problem-solving technique used to explore cause-and-effect relationships
- The key is to avoid assumptions and logic traps
- Instead, trace the chain of causality in direct increments from the effect to a root cause

The problem: My car will not start.	
Why?	The battery is dead.
Why?	The alternator is not functioning.
Why?	The alternator belt has broken.
Why?	The alternator belt was well beyond its useful service life.
Why?	I have not been maintaining my car according to the recommended service schedule **(root cause)**

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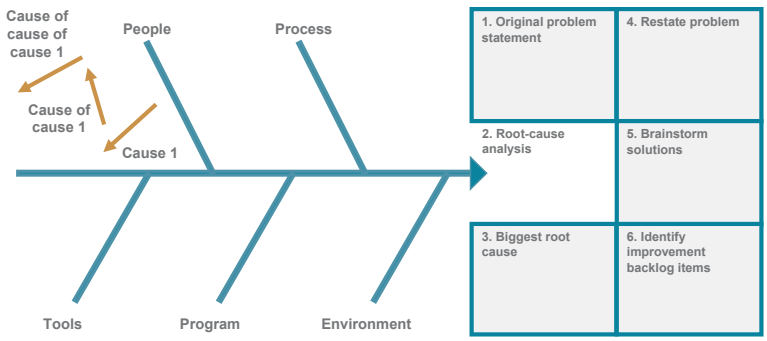
Root cause analysis diagram



Activity: Root cause analysis

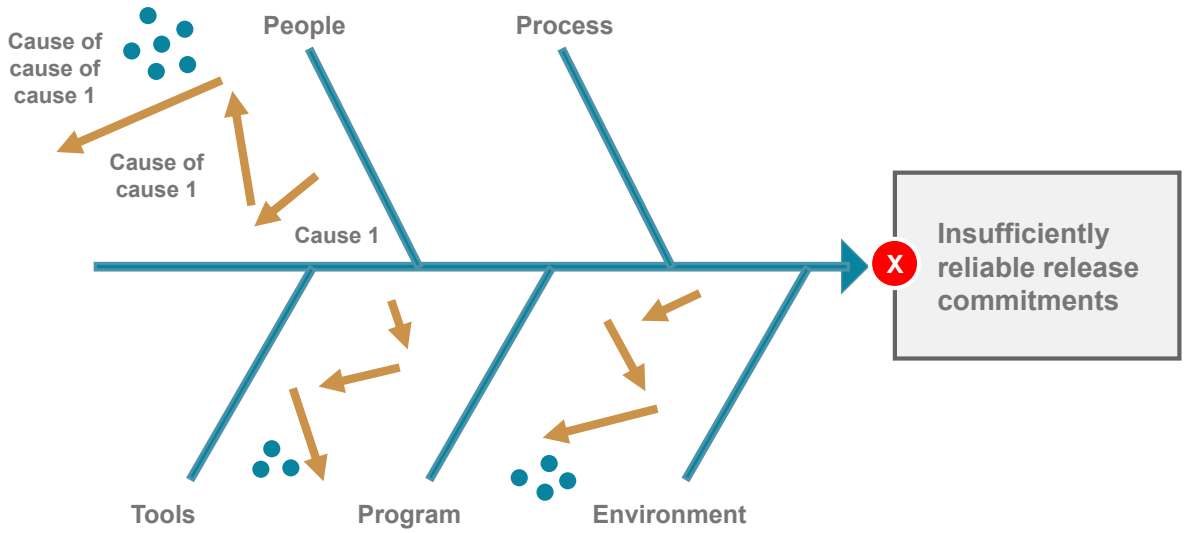
Duration 25 min

- **Step 1:** Working in your groups, brainstorm potential causes of the problem. Write the causes down in the *Root-cause analysis* section of the problem-solving board.
- **Step 2:** For each cause identified, use the 5 Whys technique to get to a potential root cause.



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Dot vote on root causes

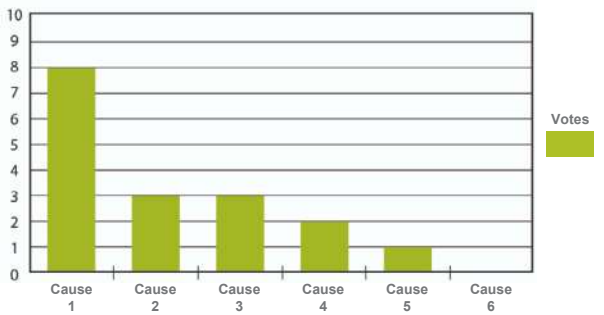


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6-21

Use Pareto analysis to identify the biggest root cause

- Pareto analysis is a statistical decision technique that uses the 80/20 rule to narrow down the number of actions that produce the most significant overall effect
- It uses the principle that 20% of root causes can cause 80% of problems
- It is useful where many possible sources and actions are competing



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Activity: Restate the new problem

Duration: 10 min

- **Step 1:** Working in your groups, dot vote to identify the biggest root cause on your board.
- **Step 2:** Use Pareto analysis to identify the biggest root cause. Put it under the *Biggest Root Cause* heading on the problem-solving board.
- **Step 3:** Restate the problem to address the identified root cause. Be sure to include the economic impact of the problem. Place the restated problem in the *Restate problem* section on the problem-solving board.

Example:

We did not have the ability to measure or test the full electrical load on vehicles in real operating conditions.

Impact:

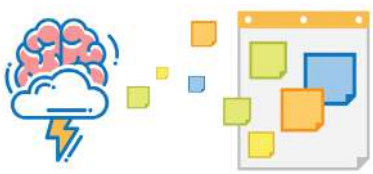
We had to upgrade the deployed power distribution system beyond what was specified. Major cost and schedule overrun.

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Activity: Brainstorm solutions

Duration: 8 min

- **Step 1:** Individually brainstorm and write ideas on sticky notes and put them up on the board
- **Step 2:** Working in your groups, discuss each idea
- **Step 3:** Organize ideas into affinity groups
- **Step 4:** Dot vote to identify the top contenders to put in the *Identify improvement backlog items* section on the problem-solving board



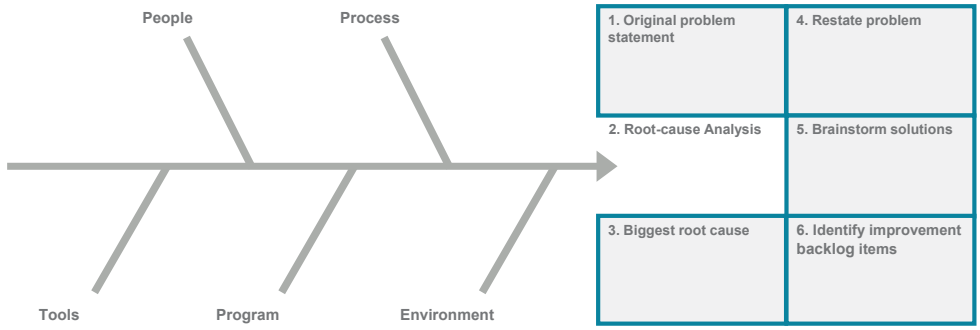
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Discussion: Problem-solving workshop readout

Duration: 10 min

- **Step 1:** Select a person from your group to do a readout
- **Step 2:** Start the readout by stating the original problem and conclude with the proposed improvement backlog items



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6-25

6.2 Measure and Grow the ART

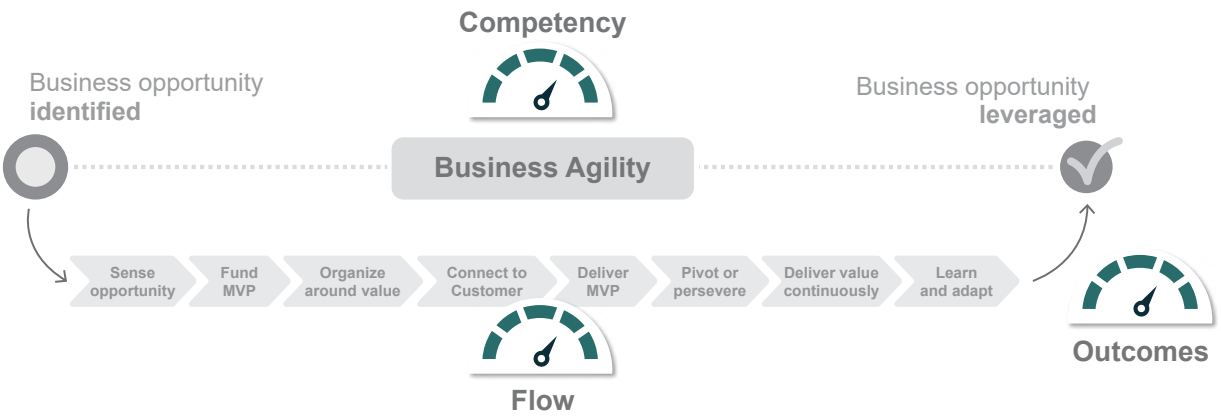
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Measure outcomes, competency, and flow

Below is a comprehensive and simple model for measuring progress toward Business Agility.

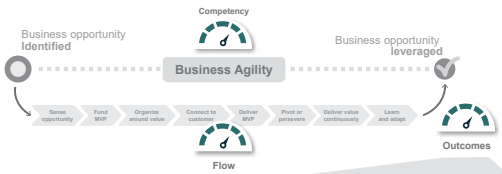


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Outcome Metrics

Do our solutions deliver the expected benefits to the business?

- KPIs represent ongoing 'health' Metrics that can be used to measure overall business performance
- Objectives and Key Results (OKRs) define the specific objectives that we are working toward in order to achieve future success



Metrics
Key Performance Indicators (KPIs)
Objectives and Key Results (OKRs)
PI Objectives
Iteration Goals
Employee Engagement

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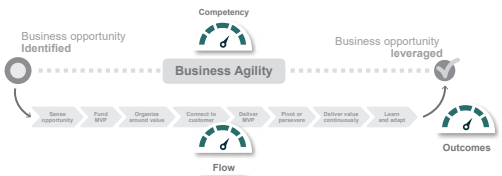
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Flow Metrics

How efficient is the organization at delivering value to the Customer?

Flow Metrics:

- Provide insights into the efficiency of the delivery pipeline
- Highlight opportunities for removing delays and increasing speed and agility



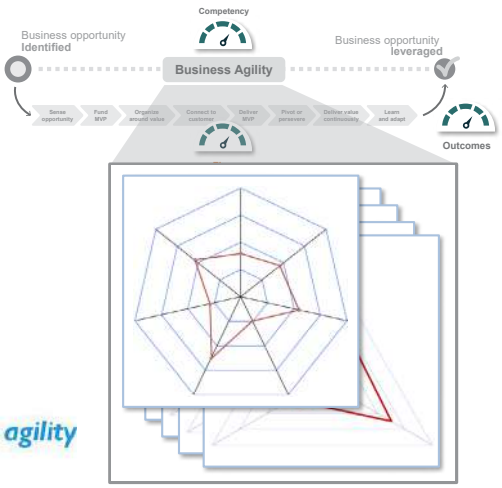
Metric
Flow Distribution
Flow Velocity
Flow Time
Flow Load
Flow Efficiency
Flow Predictability

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Competency Metrics

How proficient is the organization in the practices that enable Business Agility?

- Organizations use the Seven Core Competency assessments to improve technical and business practices that support Business Agility
- Levels of proficiency can be compared against the 'worldwide index' and other organizations in your industry



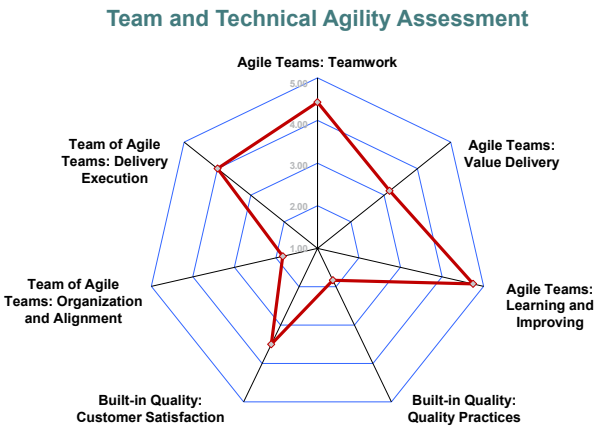
comparative agility

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Measure competency: core competency assessments

- One assessment for each one of the Seven Core Competencies
- Assess at a greater level of detail to generate deeper insights
- Measure the progress being made toward a specific core competency
- Identify specific practices for potential improvement



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Using SAFe assessments

- Perform assessments to quantify the Lean-Agile journey of the ART using the collective intelligence on the train
- Get help from the SM/TCs and the Agile Teams to perform regular assessments
- Run regular assessments with the key ART roles

Team and Technical Agility

Agile Product Delivery

DevOps Health Radar

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Where do we go next?

Growth recommendations are activities designed to increase levels of competency.

1. Identify growth recommendations
   - Two sources of growth recommendations:
     - Create growth recommendations from the assessment data
     - Review the provided growth recommendations
2. Create backlog items of improvement opportunities
3. Prioritize backlog items

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Example growth recommendation from TTA Assessment

Statement	The ART includes the necessary people to develop and deliver the Solution.
Growth Recommendation	**The ART does not include business teams.** Identify the additional teams needed to build the entire business Solution.
Activity	1. Watch the video *Organizational Agility: Lean-Agile Ways of Working Across the Enterprise*, paying close attention to the broad definition of a business Solution 2. Using this definition, identify the scope of the business Solution your ART is developing 3. Identify additional business functions that need to join the ART to develop the complete business Solution
Outcome	On completion, you should be able to: • Describe the full business Solution that the ART is building • Identify the right business roles/teams to be added to the ART

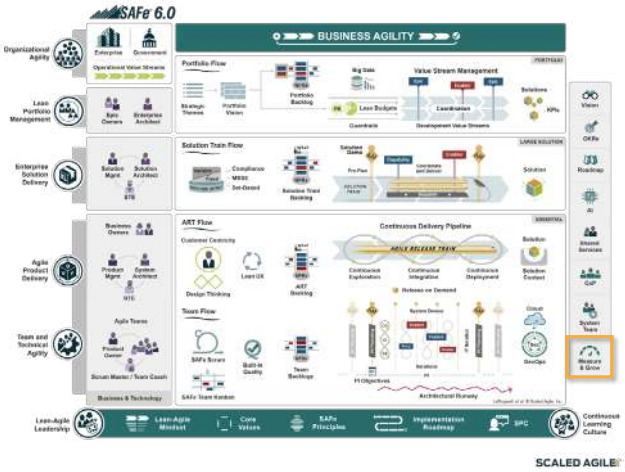
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Activity: APD and TTA Assessment tools

Prepare 10 min | Share 5 min

- **Step 1:** Form pairs in your groups and select either the Agile Product Delivery (APD) or Team and Technical Agility (TTA) assessments to review
- **Step 2:** Navigate to https://www.scaledagileframework.com/measure-and-grow/ and download the assessment
- **Step 3**: Complete the assessment
  - Identify assessment areas you want to improve or change
  - Discuss how often you would coach periodic health checks for the team and the ART
- **Step 4:** Be prepared to share with the class



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Critical success factors for effective measurement

1. Use measurement in conjunction with other discovery tools
2. Apply Metrics where they support improved decision-making and help to answer questions
3. Understand the effect of Metrics on behavior
4. Interpret Metrics carefully

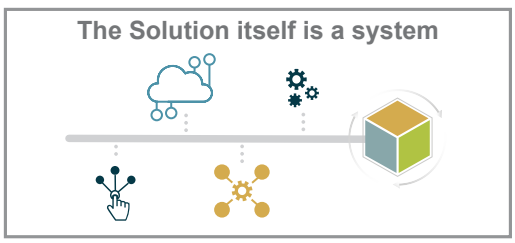
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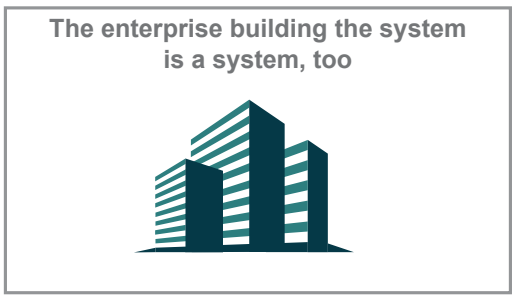
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6.3 Systems Thinking and Value Stream Mapping

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Systems Thinking





Attributes of Systems Thinking

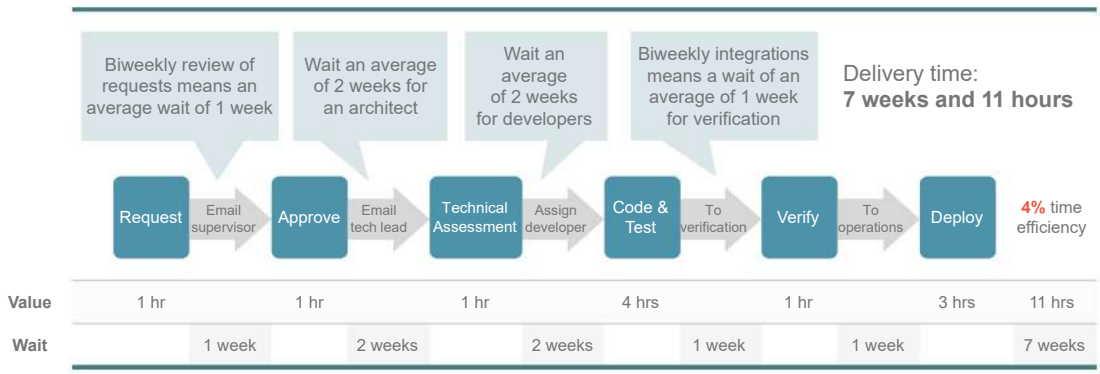
- Optimizing a component does not optimize the system
- For the system to behave well as a system, a higher-level understanding of behavior and architecture is required
- The value of a system passes through its interconnections
- A system can evolve no faster than its slowest integration point

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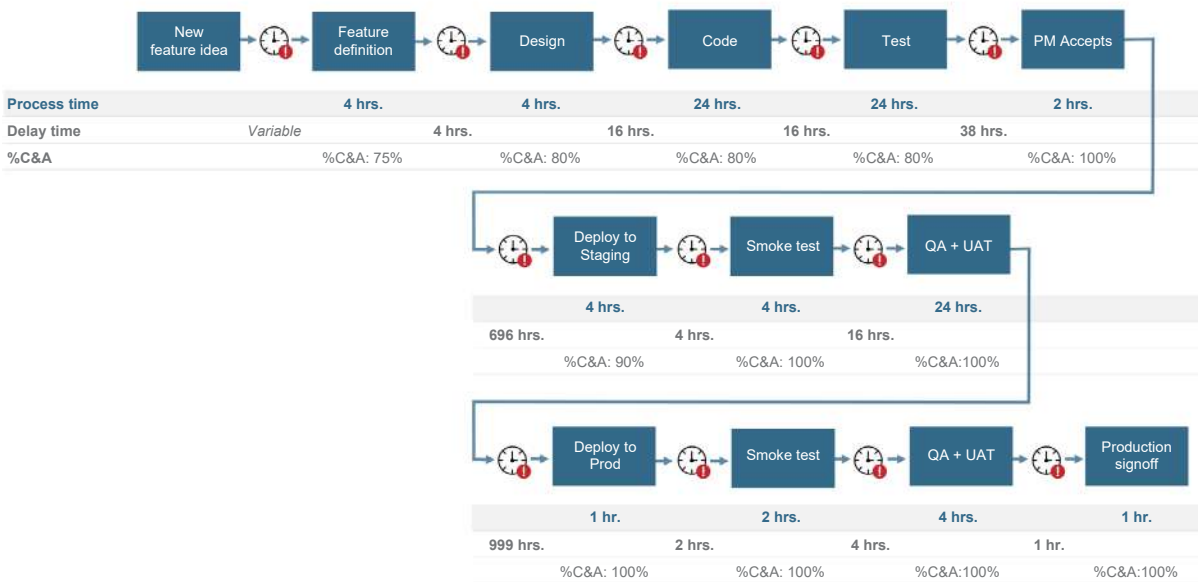
The RTE optimizes the full Value Stream

- Most problems with your process will surface as delays
- Most of the time spent getting to market is a result of these delays
- Reducing delays is the fastest way to reduce time-to-market



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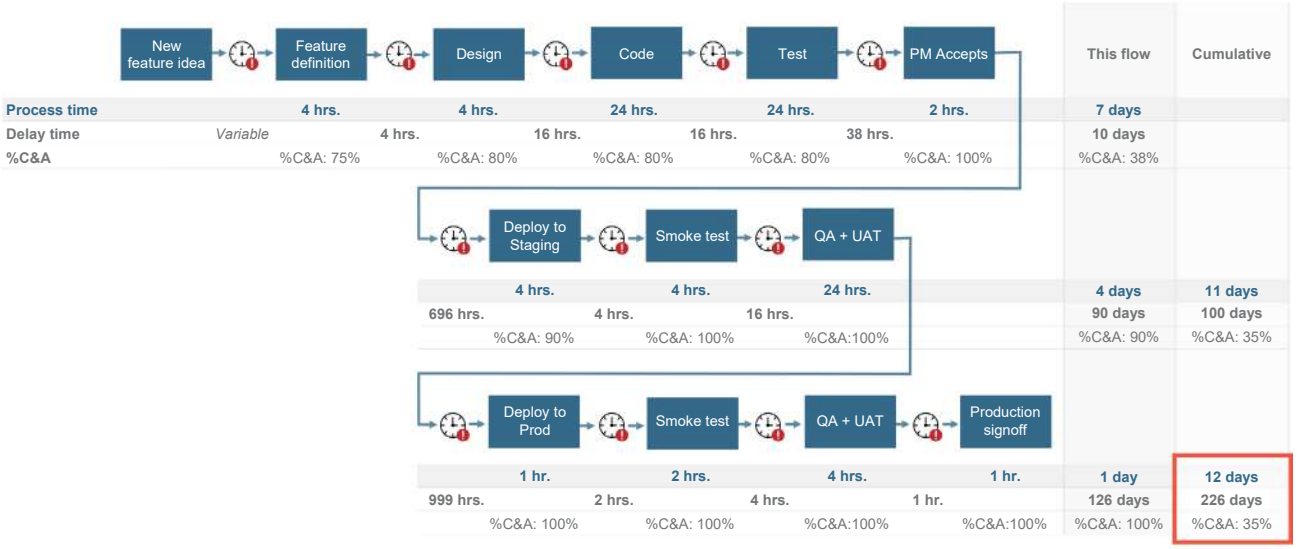
Value Stream Mapping: Measure the steps



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Value Stream Mapping: Calculate total Metrics



Flow efficiency = 5%

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Activity: Development Value Stream Mapping – current state

Prepare 15 min

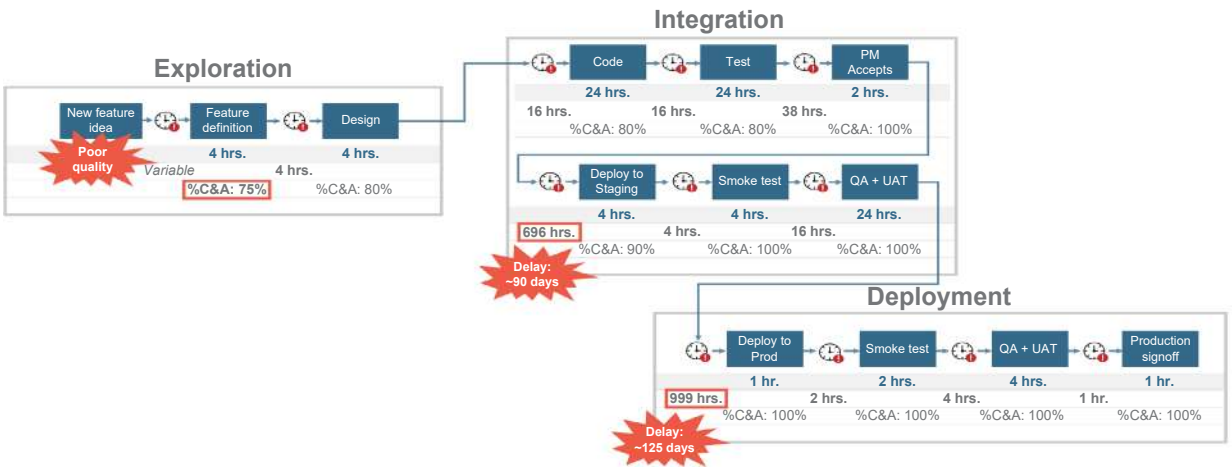
Share 5 min

- **Step 1:** Working in your groups, map a current development process from one person's real context
- **Step 2:** Identify the delays between the steps
- **Step 3:** Be prepared to share your results with the class

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Value Stream Mapping: Think about the future state



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Activity: Development Value Stream Mapping – future state

Prepare 15 min

Share 5 min

- **Step 1:** Working in your groups, use the same work context as in the previous activity.
- **Step 2:** Map the desired future state of the development process. Use the principles, practices, and tools found in SAFe.
- **Step 3:** Be prepared to share how you identified bottlenecks and what you would do to remove them with the class.

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The RTE moves from bottleneck to bottleneck

- The Value Stream Mapping of your Continuous Delivery Pipeline is an Enabler that helps the RTE identify and remove bottlenecks
- Every system has only one or a few bottlenecks that significantly constrain performance
- Once you have identified and removed the current bottleneck, there will be another bottleneck



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6.4 Optimizing flow

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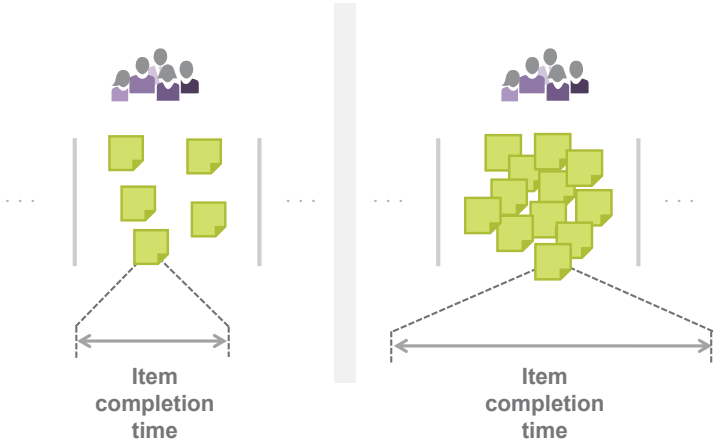
Accelerating flow with the Eight Flow Accelerators

1. Visualize and limit WIP
2. Address bottlenecks
3. Minimize handoffs and dependencies
4. Get faster feedback
5. Work in smaller batches
6. Reduce queue lengths
7. Optimize time ‘in the zone’
8. Remediate legacy policies and practices

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1. Visualize and limit WIP

- **Why it matters:**
  - Excessive WIP decreases team productivity and impedes the flow of value
  - Excessive WIP confuses individual and team priorities, causes frequent context switching, and increases waste and overhead
- **What to do about it:**
  - Make current WIP visible
  - Set WIP limits to balance WIP against available capacity

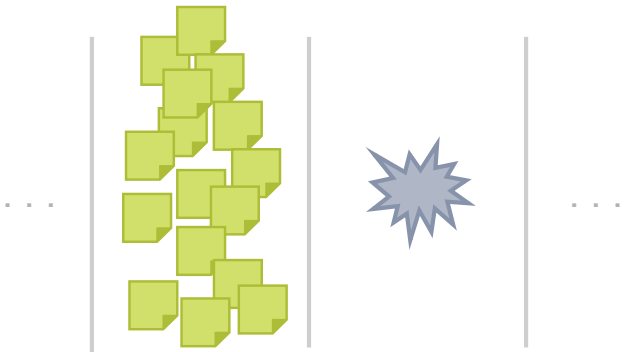


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2. Address bottlenecks

- **Why it matters:**
  - A bottleneck constrains the productivity of the entire ART
  - Teams must address them to improve flow and continuously optimize
- **What to do about it:**
  - Identify bottlenecks and understand their impact
  - Identify the resolution: bypass? Increase skills towards the bottleneck? Prioritize enablers?
  - Assess the improvement impact on the system



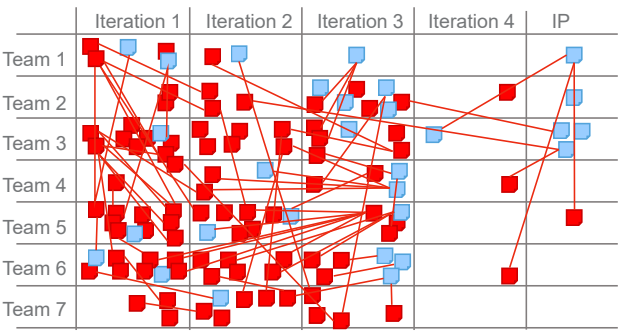
Work piles up in front of the bottleneck

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3. Minimize handoffs and dependencies

- **Why it matters:**

  Excessive and unnecessary dependencies and handoffs disrupt Team Flow, create delays, and increase context-switching overhead.
- **What to do about it:**
  - Use the ART Planning Board to visualize dependencies within and external to the ART
  - Foster incremental execution of dependencies
  - Organize around value, and reorganize when necessary



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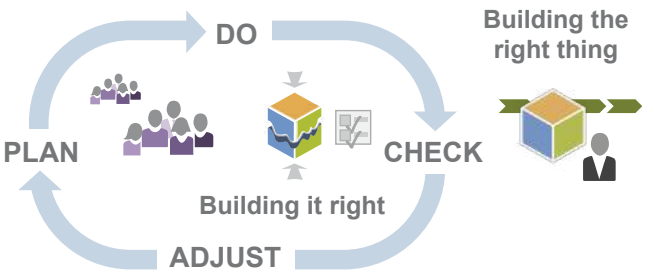
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4. Get faster feedback

- **Why it matters:**

  When feedback is delayed or missing, mistakes pile up quickly, leading to substantial rework for multiple teams, slow delivery, and unsatisfied Customers.

- **What to do about it:**
  - Determine which types of feedback are missing or inadequate
  - Shift reviews left
  - Provide Solution telemetry
  - Frequently engage with the Customer and Business Owner



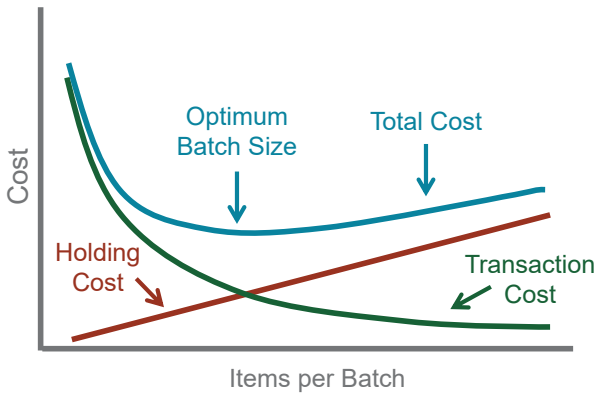
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5. Work in smaller batches

- **Why it matters:**

  Operating in large batches leads to delayed feedback, significant rework, and high variability. Teams have many different types of batches in play, including feedback batches, integration batches, and deployment batches.

- **What to do about It:**
  - Use recommended cadence and team size
  - Ensure enablers support automating the delivery pipeline
  - Provide Solution telemetry
  - Explicitly plan for small batches



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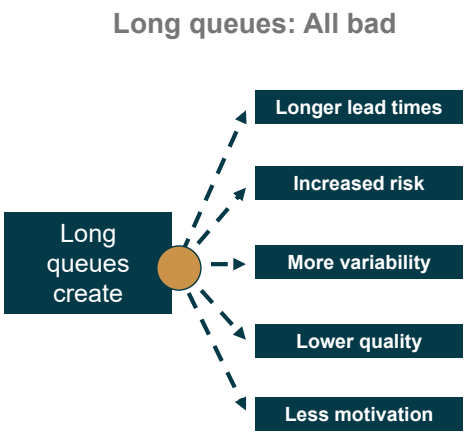
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6. Reduce queue lengths

- **Why it matters:**

  Queues represent committed work. The longer the queue, the longer the wait time for new functionality to be delivered to the Customer.

- **What to do about it:**
  - Short Iteration time boxes and crisp Iteration and PI Objectives bring focus
  - Ensure all work goes through the backlog
  - Enable Product Managers and POs to exert positive yet firm ways to prioritize
  - Leave capacity for emergent priorities



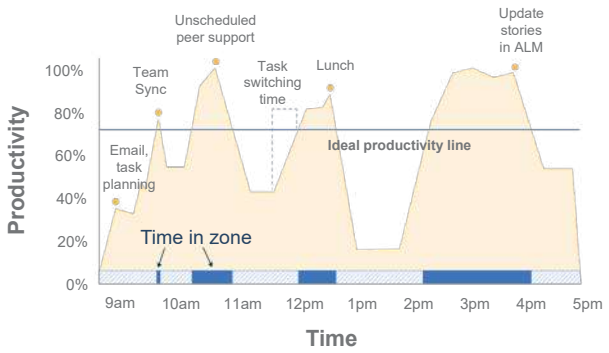
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7. Optimize time 'in the zone'

- **Why it matters:**

  Solution development relies heavily on creativity, focus, and deep, intellectual effort. It may take up to 20 minutes to fully immerse in work, and a simple external factor may instantly interrupt it.

- **What to do about it:**
  - Continually optimize the efficiency of all meetings
  - Refine what productive collaboration patterns mean to the team
  - Maintain Solution health
  - Make WIP visible, and update limits across teams as needed



2.4 hrs (time in zone) / **8 hrs** (full day) = **30%** of workday spent **in zone**

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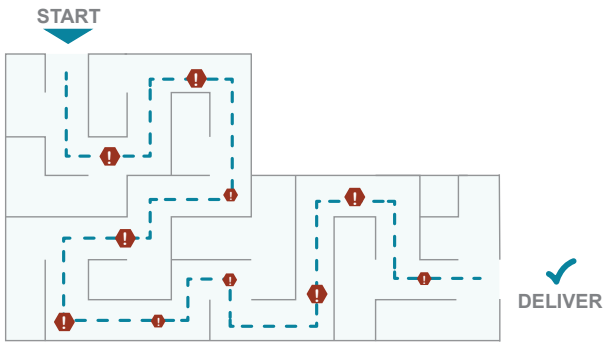
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8. Remediate legacy policies and practices

- **Why it matters:**

  Teams find themselves in a difficult predicament: pretend to conform to conflicting policies and reduce transparency or fight the system while development is in process, slowing progress and creating friction. We want to be fully Agile, not half.

- **What to do about it:**
  - Know that these impediments likely exist
  - Grant time and space to raise issues
  - Actively solve the issues you can within the teams and ART
  - Raise the visibility of the effects of those out of ART control



- Extraneous meetings
- Extra reporting
- Timesheet reporting
- Legacy CCB
- Waterfall quality management system
- Waterfall mindset
- Iron triangle
- Stage-gate milestones
- Legacy compensation
- Obsolete standards

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Measure flow with six flow Metrics

Metric	Description
Flow Distribution	Proportion of work items by type in a system
Flow Velocity	Number of completed work items over a time period
Flow Time	Time elapsed from start to completion for a given work item
Flow Load	Number of work items currently in progress (active or waiting)
Flow Efficiency	Ratio of the total time spent in value-added work activities divided by the total flow time
Flow Predictability	How consistently teams, ARTs, and portfolios can meet their commitments

Reference: Kersten, *Project to Product*

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Activity: Flow accelerators and Flow Metrics

Prepare 10 min | Share 5 min

- **Step 1:** Working in your groups, review the example problems on the following slide and identify which flow accelerator(s) would help to resolve the issue
- **Step 2:** Identify which Flow Metric(s) will be useful in tracking progress on the Solution
- **Step 3:** Be prepared to share with the class

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Problems Surfaced
Too much focus on business features leads to Solution health degradation, slowing development
Lack of stable velocity from one Iteration to the next
Slow time to market causes Customers to wait and incurs a cost of delay
Excess work in process leads to increased flow time as queues build up in the system
Large amounts of waste in the system, along with bottlenecks and delays that need addressing
Low or erratic predictability highlights underlying problems in technology, planning, or organization performance that need addressing

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RTE Action Plan

Duration: 5 min

- **Step 1:** Locate the RTE Action Plan section in your workbooks
- **Step 2:** Add more tools and techniques to the Action Plan by reflecting on the following:
  - Identify which three flow accelerators would be most useful to apply to your ART
- **Step 3:** Share one of your insights with the class



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Lesson review

In this lesson, you:

- Defined the three components of an Inspect & Adapt event
- Described how to Measure and Grow the ART
- Interpreted the concepts of Systems Thinking and Value Stream Mapping
- Reviewed practices to optimize flow

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Action Plan

Fostering Relentless Improvement

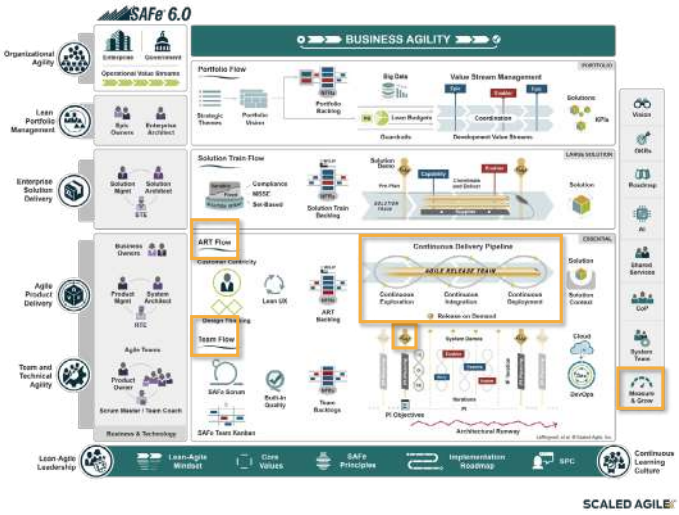
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Articles used in this lesson

Read these Framework articles to learn more about topics covered in this lesson

- "Inspect & Adapt" https://www.scaledagileframework.com/inspect-and-adapt/
- "Measure and Grow" https://www.scaledagileframework.com/measure-and-grow/
- "Continuous Delivery Pipeline" https://scaledagileframework.com/continuous-delivery-pipeline/
- "ART Flow" https://www.scaledagileframework.com/art-flow/

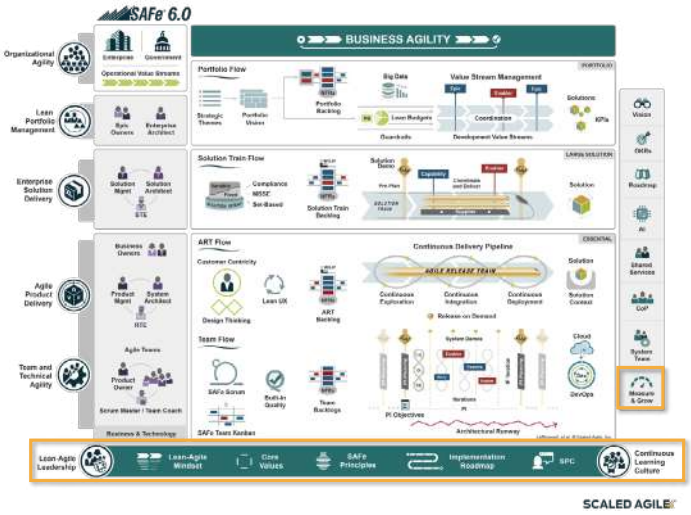


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Articles used in this lesson

Read these Framework articles to learn more about topics covered in this lesson

- "Facilitating SAFe Assessments" https://scaledagileframework.com/facilitating-safe-assessments/
- "Accelerate Business Agility" https://scaledagileframework.com/accelerate/
- "SAFe Core Values" https://scaledagileframework.com/safe-core-values/
- "Continuous Learning Culture" https://scaledagileframework.com/continuous-learning-culture/



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Continue your SAFe journey with the following resources:

Download and use the resources found in the "SAFe PI Execution Toolkit" to support an ART in successful PI Execution. https://bit.ly/Community-ToolkitsandTemplates	Use the "SAFe Assessments Workshop Toolkit" to run a workshop to identify growth opportunities for your team as you work toward mastering SAFe and Business Agility. https://bit.ly/Community-ToolkitsandTemplates
Download the "Facilitator's Guide to SAFe: Inspect & Adapt" for support with agenda setting, preparation guidance, and tips and tricks for facilitating Inspect & Adapt events. https://bit.ly/Community-FGInspectandAdapt	Use the resources available on the "Measure and Grow" SAFe Studio page to determine your progress towards Business Agility. https://bit.ly/Community-MeasureAndGrow
Watch this 60-minute webinar, *Driving Relentless Improvement: Applying the SAFe Assessments and Identifying Growth Recommendation, to explore* different use cases for applying the SAFe assessments and techniques for analyzing the data to help identify the next steps to take. https://bit.ly/Video-ApplyingSAFeAssessments	Use the resources available on the "Value Stream Mapping" SAFe Studio page to visualize value streams, measure flow, identify bottlenecks, and create an actionable improvement plan. https://bit.ly/SAFeValueStreamMapping

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Lesson notes

Enter your notes below. If using a digital workbook, save your PDF often so you don't lose any of your notes.

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Lesson 7

Serving the ART

SAFe® Course - Attending this course gives learners access to the SAFe Release Train Engineer exam and related preparation materials.

Lesson Topics

7.1 Explore the characteristics of a servant leader

7.2 Evolving group dynamics

7.3 Examine coaching techniques

7.4 Explore group facilitation techniques

7.5 Identify the steps to create an ART with a one-team culture



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Learning objectives

At the end of this lesson, you should be able to:

- Define the characteristics of a servant leader
- Identify patterns in the evolution of group dynamics
- Describe coaching techniques
- Apply group facilitation techniques
- Define the steps to create an ART with a one-team culture

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7.1 Explore the characteristics of a servant leader

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Servant leadership

A servant leader knows that their own growth comes from facilitating the growth of others who deliver the results.

"The great leader is seen as servant first, and that simple fact is the key to his greatness."

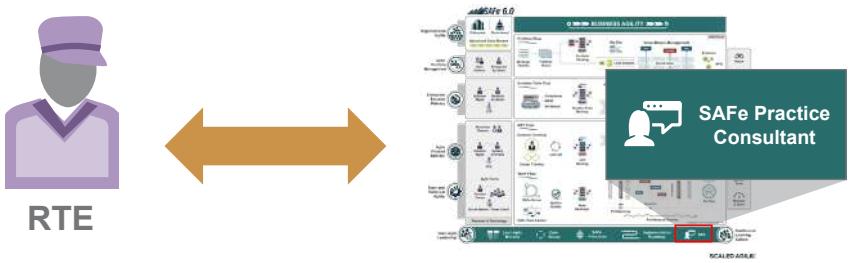
—Robert K. Greenleaf, *Servant Leadership*



The RTE is a servant leader

The RTE focuses on optimizing the flow of value through growing and maturing the teams and roles on the ART.

- This is a challenging, large, and complex task
- The RTE will need coaching support from an on-site SAFe Practice Consultant (SPC)



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Activity: The servant leader's characteristics

Prepare 10 min | Share 10 min

- **Step 1:** Read the Eight Behaviors of Servant Leaders on the following slides
- **Step 2:** Working in your groups, select one characteristic you find challenging
- **Step 3:** Brainstorm and capture specific activities, professional development, and tools that you can use to improve upon that characteristic
- **Step 4:** Be prepared to share with the class

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Eight Behaviors of Servant Leaders (part one)

Behavior	In the context of SAFe
Listens and supports team members in decision identification	• Encourages everyone to express their opinions • Notices hesitant behavior and body language during SAFe events and meetings • Helps teams identify positive and negative changes during the Inspect & Adapt event
Creates an environment of mutual influence	• Facilitates PI Planning and shared team events for all ART members and stakeholders • Asks for opinions/input and carefully considers the response
Understands and empathizes with others	• Shares in celebrating every successful System Demo; feels bad about impediments, failures, etc.
Persuades rather than uses authority	• Asks questions to encourage teams to look at decisions from new perspectives • Articulates facts; helps the teams see things they may have overlooked; helps rethink

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Eight Behaviors of Servant Leaders (part two)

Behavior	In the context of SAFe
Encourages and supports the personal development of each individual	• Encourages team learning • Fosters collaborative practices like teamwork, Continuous Integration, collective code ownership, short design sessions, and specification workshops • Encourages rotation in technical areas of concern, such as functionality, components/layers, and role aspects • Facilitates team decision-making rather than making decisions for the teams
Thinks beyond day-to-day activities; applies Systems Thinking	• Sets long-term operating goals for the team, such as Lean-Agile practices to master, new skills to acquire, etc. • Examines what is missing in order to make the environment better for everyone; prioritizes improvement activities and makes them happen

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Eight Behaviors of Servant Leaders (part three)

Behavior	In the context of SAFe
Supports the teams' commitments	• Facilitates ad-hoc meetings if needed • Helps teams access external sources of information like SMEs and shared resources (architects, UX designers, tech writers) • Helps clarify and articulate the rationale behind priorities, Milestones, and commitments • Helps teams prepare for System Demo • Helps teams find techniques to be more collaborative
Is open and appreciates openness	• Shows appreciation for team members who raise serious issues • Encourages and facilitates open communication among team members • Encourages healthy conflict during team meetings • Gives open, honest opinions

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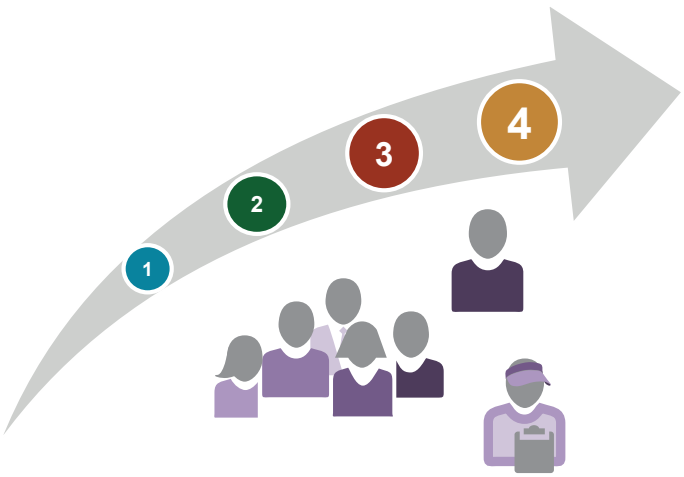
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7.2 Evolving group dynamics

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Stages of high-performing teams

1. Forming
2. Storming
3. Norming
4. Performing



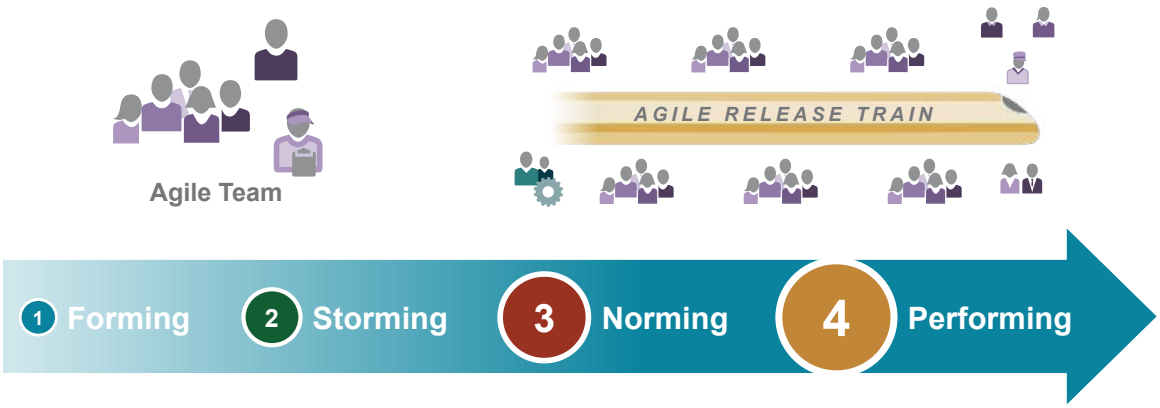
Reference: Tuckman, *Developmental sequence in small groups*

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The growth of a team of teams

The growth of individual teams in the ART progresses through each Tuckman stage, and so does the ART.

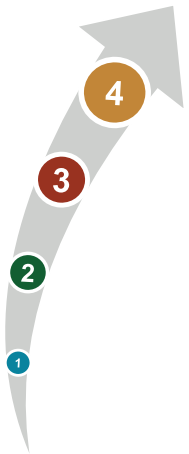


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Activity: The RTE and ART group dynamics

Prepare 7 min | Share 5 min

- **Step 1:** Working in your groups, consider each Tuckman stage and identify the following:
  - What can you observe that would indicate that the ART is at this stage?
  - What RTE behaviors or activities can help the ART grow as a team of teams and progress to the next stage?
- **Step 2:** Be prepared to share with the class



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The RTE facilitates and accelerates ART growth

FORMING
- Forming teams
- Establishing ART Vision
- Establishing ART events
- Fostering collaboration

STORMING
- Surfacing and resolving team conflict
- Dealing with individual and team performance
- Fostering relentless improvement

NORMING
- Establishing the ART as a community
- Improving engineering practices
- Fostering effective communication

PERFORMING
- Creating space for spontaneous leadership and self-organization across the ART
- Creating a flow of knowledge across the teams and the ART



AGILE RELEASE TRAIN

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7-15

7.3 Examine coaching techniques

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Coaching sometimes requires a shift from old behaviors to new ones

From…	To …
Coordinating individual contributions	Coaching the whole team to collaborate
Acting as a subject matter expert	Being a facilitator
Talking about deadlines and technical options	Focusing on business value delivery
Driving 'the right' (your) decisions	Doing the right thing for the business right now
Fixing problems for others	Facilitating team problem-solving

From fixing problems… → …to helping others fix them!

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Video: The power of empathy

Duration 4 min



https://bit.ly/The-power-of-empathy

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Why are questions powerful?

- They provoke thought
- They generate curiosity in the listener
- They channel focus
- They generate energy and forward movement
- They stimulate reflective conversation
- They surface underlying assumptions
- They invite creativity and new possibilities
- They inspire more questions
- They help reach for deep meaning



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Powerful questions you can ask

Powerful questions like these can help connect ideas and generate deeper insights.

- What new connections did you make?
- What had real meaning to you?
- What surprised you?
- What challenged you?
- What's missing from this picture so far?
- What is it we're not seeing?
- Where do we need more clarity?
- What has been your major learning, insight, or discovery so far?
- What is the next level of thinking we need to do?
- What hasn't been said that would help us reach a deeper level of understanding and clarity?
- What would you do if success were guaranteed?

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Activity: Coaching with powerful questions

Prepare 8 min | Share 2 min

- **Step 1:** Working in your group, select one person to play the role of the coach.
- **Step 2:** As a group, brainstorm an issue with a deadline that you are currently facing and present it to the coach.
- **Step 3:** As the coach, coach the team to resolve the issue by only responding in one of two ways:
  - Reflective listening, such as “I hear you saying …”
  - Asking a powerful question
- **Step 4:** Be prepared to share your experience with the class. Were the coaches able to guide the team with powerful questioning rather than by telling them what to do?

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The five dysfunctions of a team

- Teamwork is the ultimate competitive advantage. However, many teams are dysfunctional.
- Absence of trust is the key problem that leads to the other four dysfunctions.



Reference: Lencioni, *The Five Dysfunctions of a Team*

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Activity: The ART can't run without trust

Prepare 10 min | Share 5 min

- **Step 1:** Working in your groups, discuss which SAFe events and activities can help establish trust among the members of the ART and ART stakeholders.
- **Step 2:** Identify these activities on the Big Picture and note why they would help establish trust.
- **Step 3:** Choose one of the SAFe activities you identified. Discuss and capture how the RTE can act to further increase the amount of trust built in that activity.
- **Step 4:** Be prepared to share your ideas with the class.

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7-23

The RTE can mitigate the five dysfunctions using SAFe

Inattention to Results	Results are empirically reviewed at the end of every Iteration and release. Iteration Retrospectives drive continuous improvement.
Avoidance of Accountability	Stakeholders, peers, and review of results drive accountability.
Lack of Commitment	Teams make shared commitments to each other and to external stakeholders.
Fear of Conflict	Scrum creates a safe environment for conflict; the SM/TC encourages discussion of disagreements. Shared commitment avoids individual conflict that occurs when objectives are not aligned.
Absence of Trust	The environment is safe. The team shares commitment and goals, displays hyper-transparency, and engages in retrospectives.

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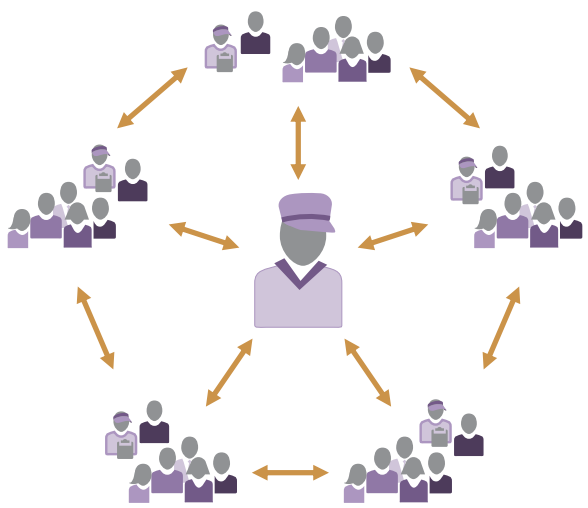
7.4 Explore group facilitation techniques

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Facilitating large groups

Often during a PI, the RTE will be the facilitator for meetings and events with many people attending. Possible events are:

- Inspect & Adapt (I&A)
- Management review and problem-solving
- PI Planning
- ART Backlog refinement
- ART Sync
- Coach Sync



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Large group facilitation preparation and techniques

- Allow social networks to form. Everyone should be in a mindset of safety, and this happens through the natural formation of social networks.
- Provide clear instructions in writing, but offer no more than three steps at a time. Handouts are extremely helpful.
- Design, test, and leverage your audiovisuals.
- Optimize the physical spaces for maximum participation. Round tables and pods are best. Leverage the wall space as well.

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Large group facilitation preparation and techniques (continued)

- Use video conferencing tool features like polls that can simplify participation in virtual events.
- Timebox everything and use your tools to manage timeboxes.
- Call upon coaches in your organizations to help. You'll need all the coaching power you can get!
- Create energy through fun!

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Discussion: Group facilitation techniques

Prepare 5 min

Share 10 min

- **Step 1:** Working in your groups, select one event from the Big Picture where you will be facilitating a large group of people
- **Step 2:** Discuss techniques you will use to facilitate that event
- **Step 3:** Be prepared to share your techniques with the class

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7.5 Identify the steps to create an ART with a one-team culture

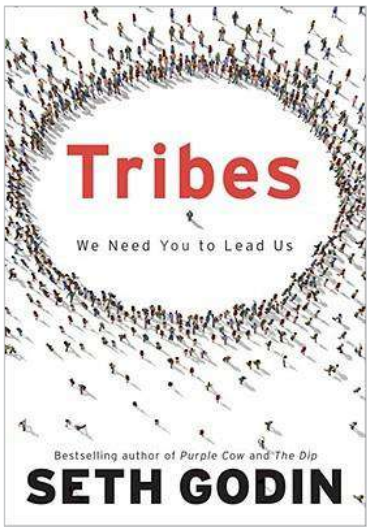
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The ART as a tribe

"A tribe is a group of people connected to one another, connected to a leader, and connected to an idea."

— Seth Godin, *Tribes: We Need You to Lead Us*



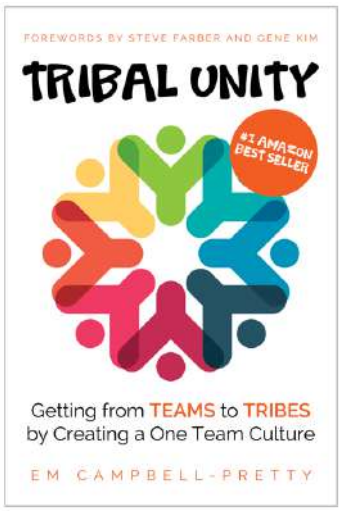
Reference: Godin, *Tribes: We Need You to Lead Us*

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7-31

Tribal unity: Steps to creating a one-team culture

1. Create a great team
   - Compose a SAFe Agile Team of ten or fewer individuals
   - Form via self-selection
   - Use Agile practices
2. Connect the teams and create a tribe
   - Share a collective identity
   - Share experiences
   - Celebrate as a tribe



Reference: Campbell-Pretty, *Tribal Unity*

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Tribal unity: Steps to creating a one-team culture (continued)

3. Connect the tribe to its leader
   - Connect everyone to the leader during Gemba
   - Create an Agile Team of Agile leaders
   - Encourage vulnerability in front of the tribe
4. Connect the tribe to an idea
   - Have a vision
   - Communicate the vision
   - Learn together

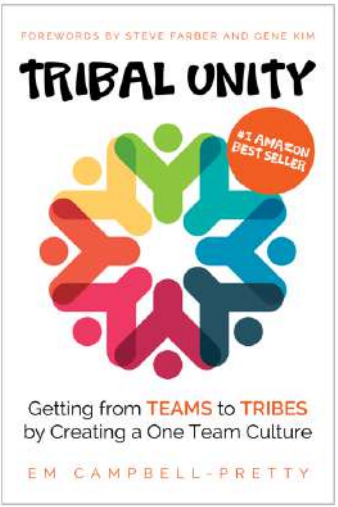


Reference: Campbell-Pretty, *Tribal Unity*

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Tribal Unity: Steps to creating a one-team culture (continued)

5. Sustaining tribal unity
   - Quantify culture with an employee Net Promoter Survey (eNPS)
   - Record and share tribal legends
   - Set up successors for success
6. Engaging management in tribal unity
   - Tap into your empathy
   - Shine a light on a bright spot
   - Invite them into Gemba



Reference: Campbell-Pretty, *Tribal Unity*

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Video: Quick tips: Sustaining the victory

Duration 2 min



https://bit.ly/Video-SustainingVictory

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Discussion: Sustaining the victory

Prepare 10 min | Share 10 min

- **Step 1:** As an RTE, consider how you will sustain the victory:
  - How are you connecting team victories to the Vision?
  - How are you creating connections between yourself and the teams?
- **Step 2:** Working in your groups, select one person's context. Discuss an upcoming PI event and how the RTE could sustain the victory. Build on the facilitation skills to practice what you will say during the event.
- **Step 3:** Be prepared share with the class.

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RTE Action Plan

Duration: 5 min

- **Step 1:** Locate the RTE Action Plan section in your workbooks
- **Step 2:** Begin adding tools to the Action Plan by brainstorming the following:
  - What behaviors will you shift away from and move into as you develop your RTE coaching and facilitation skills?
- **Step 3:** Share one of your insights with the class



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Lesson review

In this lesson you:

- Defined the characteristics of a servant leader
- Identified patterns in the evolution of group dynamics
- Described coaching techniques
- Applied group facilitation techniques
- Defined the steps to create an ART with a one-team culture

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Action Plan

Serving the ART

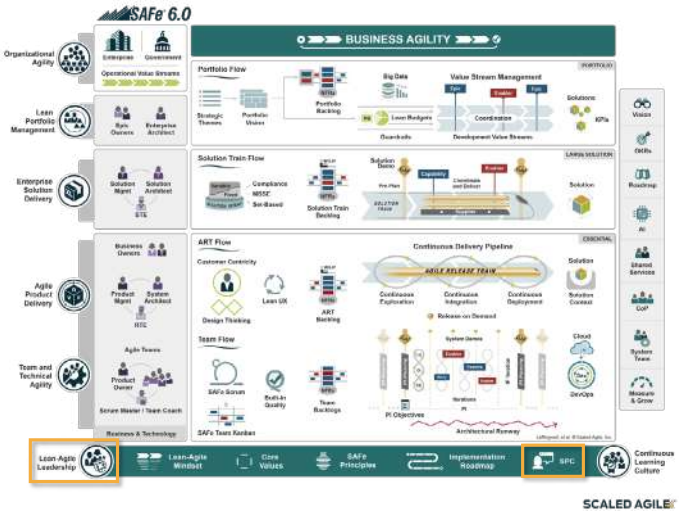
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Articles used in this lesson

Read these Framework articles to learn more about topics covered in this lesson

- "SAFe Practice Consultant" https://scaledagileframework.com/SPC/
- "Lean-Agile Leadership" https://www.scaledagileframework.com/lean-agile-leadership/
- "Release Train Engineer" https://www.scaledagileframework.com/release-train-engineer/



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Continue your SAFe journey with the following resources:

Download and use the resources found in the "SAFe PI Execution Toolkit" to support an ART in successful PI Execution. https://bit.ly/Community-ToolkitsandTemplates	Access the Collaborate template "SAFe Agile Team Charter" to support your team in defining their purpose, responsibilities, and other critical elements. https://bit.ly/Template-SAFeAgileTeamCharter
Download and review the "Facilitation Guidance Brief" for a collection of content designed by the Scrum Master Research Cohort to help improve your facilitation skills. https://bit.ly/Community-FacilitationGuidanceBrief	Download the "Team Formation Toolkit" to help your team clearly define their purpose, responsibilities, and other elements. https://bit.ly/Community-ToolkitsandTemplates

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Lencioni, Patrick. *The Five Dysfunctions of a Team: A Leadership Fable.* San Francisco: Jossey-Bass, 2011. Kindle edition.

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Tuckman, Bruce and Mary Ann Jensen. “Stages of Small-Group Development Revisited.” *Group & Organization Management* 2, no. 4 (1977): 419-427.

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Lesson notes

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Lesson 8

Continuing Your Learning Journey

SAFe® Course - Attending this course gives learners access to the SAFe Release Train Engineer exam and related preparation materials.

Lesson Topics

8.1 Create a personal RTE action plan

8.2 Create an ART Improvement Roadmap

8.3 Commit to the plans



8-2

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Learning objectives

At the end of this lesson, you should be able to:

- Create a personal RTE action plan
- Create an ART Improvement Roadmap
- Commit to the plans

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8.1 Create a personal RTE action plan

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Activity: Personal RTE Action Plan

Prepare 7 min

Share 5 min

- **Step 1:** Reflect on your goal for this course from Lesson 1. Has your goal been achieved? What is your next learning goal?
- **Step 2:** Review the ideas in the Action Plan you wrote after each lesson in your workbook. Consider each action based on:
  - Time to implement
  - Degree of difficulty
  - Potential impact on the ART
- **Step 3:** Priority rank each action and build an implementation plan.
- **Step 4:** Be prepared to share with the class.

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8-5

8.2 Create an ART Improvement Roadmap

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8-6

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Personal RTE Action Plan

Instructions: Reflect on your goal for this course from Lesson 1. Has your goal been achieved? What is your next learning goal? Review the ideas in the Action Plan you wrote after each lesson in your workbook. Consider each action based on: Time to implement; Degree of difficulty; Potential impact on the ART. Priority rank each action and build an implementation plan. Be prepared to share with the class.

Reflections

Action Plans

Lesson 1:
Exploring the RTE role and Responsibilities

Lesson 2:
Applying SAFe Principles

Lesson 3:
Organizing the ART

Lesson 4:
Planning a PI

Lesson 5:
Executing a PI

Lesson 6:
Fostering Relentless Improvement

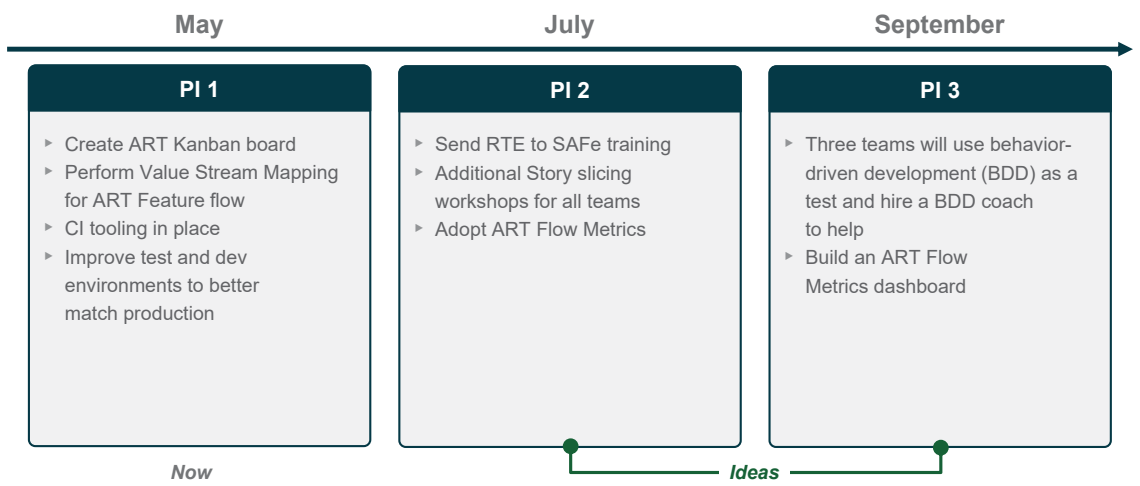
Lesson 7:
Serving the ART

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Improvement Roadmap

An ART Improvement Roadmap guides ART evolution over time.



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Activity: Define your ART Improvement Roadmap

Prepare 6 min Share 6 min

- **Step 1:** Select areas where you see the most opportunity for improvement
- **Step 2:** Identify three action items that you believe can immediately improve your ART
- **Step 3:** Write these items in the ART improvement list in your workbook
- **Step 4:** Be prepared to share with the class

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Define your ART Improvement Roadmap

Instructions: Select areas where you see the most opportunity for improvement. Identify three action items that you believe can immediately improve your ART. Write these items in the ART improvement list in your workbook. Be prepared to share with the class.

Action item 1

Action item 2

Action item 3

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8.3 Commit to the plans

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Activity: Commit to the plans

Duration



- **Step 1:** In your groups, take turns reviewing the personal Action Plan and the ART Improvement Roadmap you created.
- **Step 2:** Commit to each other that you will do your best to act on the items in the list.
- **Step 3:** Agree on an action to keep each other accountable. For example, you might agree to check in with each other in two to eight weeks to discuss the progress of your plans.

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> Lean-Agile Leaders are lifelong learners who help teams build better systems through understanding and exhibiting the values, principles and practices of Lean, Systems Thinking, and Agile development.”
>
> —Dean Leffingwell, Creator of SAFe



Dean Leffingwell, Creator of SAFe.

8-11

Lesson review

In this lesson you:

- Created a personal RTE action plan
- Created an ART Improvement Roadmap
- Committed to the plans

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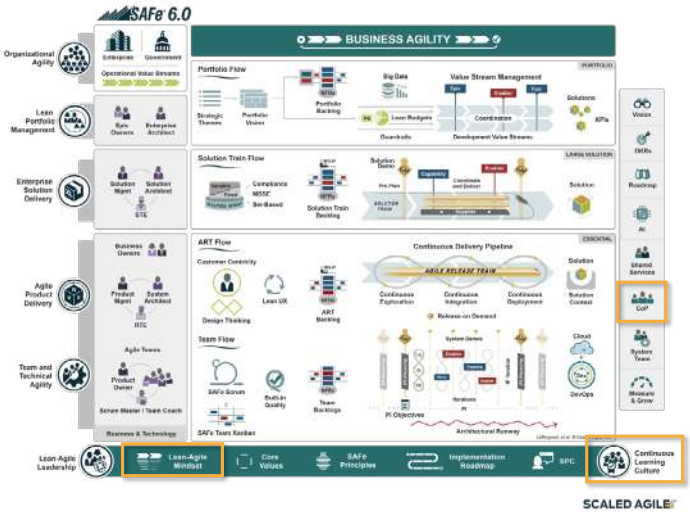
8-12

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Articles used in this lesson

Read these Framework articles to learn more about topics covered in this lesson

- "Communities of Practice" https://www.scaledagileframework.com/communities-of-practice/
- "Lean-Agile Leadership" https://www.scaledagileframework.com/lean-agile-leadership/
- "Continuous Learning Culture" https://www.scaledagileframework.com/continuous-learning-culture/



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RTE reading list

- *Principles of Product Development Flow* by Donald G. Reinertsen
- *Agile Software Requirements: Lean Requirements Practices for Teams, Programs, and the Enterprise* by Dean Leffingwell
- *Lean Product and Process Development* by Allen C. Ward and Durward Sobek II
- *Lean Thinking* by James P. Womack and Daniel T. Jones
- *Out of the Crisis* by W. Edwards Deming
- *Unicorn Project* by Gene Kim
- *Kanban: Successful Evolutionary Change for Your Technology Business* by David J. Anderson

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Lesson 9
Practicing SAFe

SAFe® Course - Attending this course gives learners access to the SAFe Release Train Engineer exam and related preparation materials.

RTE SAFe 6

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Video: SAFe certification benefits

Duration 3 min



https://bit.ly/Video-SAFeCertificationBenefits

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The Learning Plan is YOUR path to certification

Next Steps:



Download the workbook and access the **exam study guide** and **practice test**



Take the **Certification Exam** and showcase your **Digital Badge** to get recognized as a Certified SAFe Release Train Engineer

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9-3

Feedback is a gift

Help us improve by completing the Course Feedback Survey



9-4

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Activity: My Learning Plan

Duration 7 min

- **Step 1:** Individually, navigate to your learning plan via:
  - https://safe.scaledagile.com/
- **Step 2:** In your Learning Plan, navigate to and complete the course feedback survey. The survey results will remain anonymous to the instructor.
- **Step 3:** After completing the survey, review the rest of the certification prep materials. Take special note of your exam deadline and plan your timeline to prepare for and complete the exam!

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Video: Welcome to SAFe Studio

Duration 3 min



https://bit.ly/Video-WelcomeSAFeStudio

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SAFe Studio Resources

- Online Learning
- Videos
- Toolkits
- My SAFe Events
- SAFe FAQs
- SAFe Assessments
- SAFe Forums

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Activity: SAFe Studio Exploration

Duration: 7 min

- **Step 1:** Using the sidebar navigation, explore each area of SAFe Studio to locate the following:
  - Identify a video about a SAFe event you're curious to learn more about.
  - Select an online learning of interest.
- **Step 2:** Navigate to the SAFe Forum and make a post. You could introduce yourself, ask a question from the class parking lot, or respond to a post from another SAFe Studio user.

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SAFe Glossary

SAFe Glossary:
Visit the Scaled Agile Framework site (*www.scaledagileframework.com/glossary/*) to download glossaries translated into other languages.

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